Simple local trace formulas for unramified p-adic groups*

Draft 2. Corrected and expanded version

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Abstract

This revision was written with the help of AI systems: Claude (Anthropic) refereed the 1997 version and drafted the revision as a co-author, and ChatGPT (OpenAI) refereed the first revised draft. See the Acknowledgments for details. The paper revises and updates the first author's earlier paper [Jo92]. Let G be a connected unramified semisimple group over a p-adic field F, K a hyperspecial maximal compact subgroup, f a spherical function with Satake transform $f^\vee$, and ϕ a smooth compactly supported function on the regular elliptic set. Following Waldspurger's method for $GL(N)$, we study the spectral expansion

$$\int_{G(F)\times G(F)} f(h)\phi(g^{-1}hg)\,dg\,dh = \int f^\vee(\chi)\,dI_\phi(\chi)$$

over the compact torus of unitary unramified characters, and with it the spectral expansion of elliptic orbital integrals of spherical functions. We prove a truncated form of the identity, with an explicit kernel $I^T(\chi,\phi)$. By Brion's theorem, I^T is a finite sum of vertex terms whose T-independent part $I(\chi,\phi)$ is a rational function of χ. Under a regularity hypothesis on the vertex terms (and, in rank ≥ 2, a uniform form of Casselman's asymptotics) we prove that $dI_\phi = I(\chi,\phi)\,d\mu(\chi)$, with $d\mu$ the Plancherel measure.

Three examples are computed in closed form.

- For $PGL(2)$ the hypothesis holds.
- For $SL(2)$ it fails, and dI_ϕ has an atom at the unramified quadratic character, given by the κ-orbital integral, as the fundamental lemma predicts.
- For $PGL(3)$, a computation on the Bruhat–Tits building shows that the hypothesis also fails for an unramified elliptic element of depth 0. Numerically, the spectral measure is nevertheless absolutely continuous, with an explicit density.

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We propose a weaker regularity hypothesis that separates these cases. We also conjecture that the singular part of dI_ϕ is supported where the R-groups of the unitary unramified principal series are non-trivial, and that for $SL(n)$ it is given by endoscopy. We give the κ-twisted version for $SL(n)$ and a sketch of the derivation from Arthur's local trace formula.

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1 Introduction

This section states the problem and the main results. We define the elliptic distributions T_ϕ and the spectral identity (1.2) that the paper is about. We then explain how the identity relates to Arthur's local trace formula and to Waldspurger's work for $GL(N)$, and what it says about elliptic orbital integrals of spherical functions. The section ends with a road map of the proof and a list of what is proved, what is conditional and what is conjectural.

Readers who prefer to see the identity at work first can go directly to the examples: $PGL(2)$ and $SL(2)$ in §6, and $PGL(3)$ in Appendix A.

Let G be a connected unramified group over a p-adic field F, let G_{reg} denote the set of regular semisimple elements of $G(F)$, and let G_{ell} denote the set of regular elliptic elements of $G(F)$. Let $C_c^\infty(G(F))$ denote the algebra of locally constant compactly supported functions on $G(F)$ and let $\mathscr{H}(G,K)$ denote the commutative subalgebra of spherical (bi-K-invariant) functions, where K is a hyperspecial maximal compact subgroup of $G(F)$.

Standing hypothesis (H). Throughout, G is semisimple, or more generally reductive with anisotropic center (equivalently, the split component A_G of its center is trivial). Then $G(F)^1 = G(F)$, the spherical transform (2.9) is an integral over all of $G(F)$, and $X_*(A_G) = 0$ in (3.3). Levi subgroups M of G are of course reductive with non-compact center; they enter only through Macdonald's formula and Casselman's asymptotics for M, which hold for all reductive groups. The endoscopic groups H_m of §5 satisfy (H).

Remark 2026.1 (scope). Hypothesis (H) cannot simply be dropped, with $G(F)^1$ in place of $G(F)$. For $G = GL(2)$ and $f = \mathbf{1}_{\pi I_2 K}$ the support of f misses $G(F)^1$, so $f^\vee \equiv 0$ while $f(\pi I_2) = 1$, and Corollary 2.3 fails. An extension to general reductive G would have to keep track of central characters (equivalently, of the variable in $\mathfrak{a}_G^*$) in the Plancherel formula and in the truncation; this is not attempted here. Waldspurger's $GL(N)$ results [Wa89a] are of this kind.

Let $\Phi \subset C_c^\infty(G(F))$ denote the subspace of functions supported in G_{ell}. Each $\phi \in \Phi$ has compactly generated support modulo conjugation in the sense of Kazhdan [Ka86, §4]. For each $\phi \in \Phi$,

define

$$T_\phi : f \longmapsto \int_{G(F)\times G(F)} f(h)\phi(g^{-1}hg)\,dg\,dh. \tag{1.1}$$

It is not hard to show that T_ϕ is an "elliptic" invariant distribution with compactly generated support (Lemma 4.4). In this note we restrict T_ϕ to $\mathscr{H}(G,K)$ and compute, in §§3–4, a (Macdonald–)Plancherel-type formula for T_ϕ:

$$T_\phi(f) = \int f^\vee(\chi)\,dI_\phi(\chi), \tag{1.2}$$

where χ runs over the compact torus $\widehat{T}^u$ of unitary unramified characters (modulo W) and dI_ϕ is a measure; see Theorem 4.5 below. For this, we use the geometric lemma of Bernstein and Zelevinsky, Macdonald's Plancherel formula, Macdonald's formula for the spherical function, results of Casselman on intertwining operators and asymptotics of the unramified principal series, and Brion's decomposition of lattice-point generating functions. This derivation follows the procedure of Waldspurger [Wa89a] rather closely, where the case of $GL(N)$ was worked out in detail. In §5, we examine briefly the "κ-stable" version of (1.1) for $SL(n)$:

$$T_\phi^\kappa : f \longmapsto I^\kappa(f,\phi) := \int_{SL(n,F)\times PGL(n,F)} \kappa(g) f(g^{-1}hg)\phi(h)\,dg\,dh, \tag{1.3}$$

where κ is an unramified character of $PGL(n,F)$ of order dividing n. After using the Weyl integration formula to obtain a "geometric" expansion for T_ϕ^κ, we use the fundamental lemma (now a theorem) to rewrite the result (see (5.11), (5.8) and Proposition 5.2 below).

Of course, the distribution T_ϕ also occurs in the context of Arthur's local trace formula [Ar89, Ar91]. Let R denote the unitary representation of $G(F)\times G(F)$ on $L^2(G(F))$ given by $(R(x_1,x_2)\psi)(y) := \psi(x_1^{-1}yx_2)$. Given $f = (f_1,f_2)$ in $C_c^\infty(G(F)) \times C_c^\infty(G(F)) \hookrightarrow C_c^\infty(G(F)\times G(F))$, the kernel of the integral operator $R(f)$ is

$$K_f(x_1,x_2) = \int_{G(F)} f_1(x_1y)f_2(yx_2)\,dy = \int_{G(F)} f_1(y)f_2(x_1^{-1}yx_2)\,dy. \tag{1.4}$$

As in the global trace formula, one wants to find both a "geometric" and a "spectral" formula for a truncated version of the integral of $K_f(x,x)$. It should be emphasized that this is done below only for a very restricted class of $f = (f_1,f_2)$: f_1 spherical and f_2 supported on the elliptic set. For such f no truncation is needed on the geometric side (Lemma 4.4), and §7 explains how (1.2) follows from Arthur's formula.

Thus part of this paper could be viewed as a special case of Arthur's local trace formula or as a generalization of part of Waldspurger's work [Wa89a]. Another way one might interpret the distributions $\phi \mapsto I(\chi,\phi)$ is as follows. Proposition 4.2 below shows that $\phi \longmapsto I(\chi,\phi)$ is linear and invariant under conjugation by K. When Hypothesis (R) holds it is G-invariant (Corollary 4.5A), and it is expected, but not proved here, to be G-admissible in the sense of [HC78]. Regarding this invariant distribution as a function (as [HC78, Theorem 19] would then allow), one may rewrite (1.2) as

$$\int_{G(F)} f(g^{-1}\gamma g)\,dg = \int f^\vee(\chi)\,dI_\gamma(\chi), \tag{1.5}$$

for $\gamma \in G_{\mathrm{ell}}$ and f spherical. Under (R) this holds with $dI_\gamma = I(\chi,\gamma)\,d\mu(\chi)$ (Corollary 4.5A); in general, that the right side is given by a measure at all is part of Conjecture 4.6. Thus (1.5) gives

a relatively explicit formula for the unramified part of the "spectral support" of an elliptic orbital integral on $G(F)$.

One might expect that always $dI_\gamma = I(\chi,\gamma)\,d\mu(\chi)$, with $I(\cdot,\gamma)$ a rational function on the complex torus of unramified characters (whose compact real form carries the Plancherel measure [Ma71]). This is so for $PGL(2)$ (§6.1), where $I(\cdot,\gamma)$ has double poles at the non-regular characters that are cancelled by the double zeros of the Plancherel density. It *fails* for $SL(2)$ (§6.2): there dI_γ has, besides an absolutely continuous part, a point mass at the unramified quadratic character χ_0. This is a non-regular unitary character at which the Plancherel density does not vanish, and $i_{G,A}(\chi_0)$ is reducible. The atom is half the κ-orbital integral, $\frac{1}{2}f^\vee(\chi_0)$, as the fundamental lemma predicts. Theorem 4.5 and Conjecture 4.6 state the general form.

A somewhat analogous formula to (1.5), for stable unipotent orbits, was conjectured in [As91]. Assem's conjecture is a theorem for $GL(n)$, and for $GL(n)$ the germ expansion and Assem's formula yield a relatively explicit formula for $I(\chi,\gamma)$ (this idea can essentially be found in [Wa89a]). Finally, in the case of $SL(n)$ the fundamental lemma of Waldspurger [Wa91] may be reformulated as a functorial property of the spectral measures dI_γ^κ (Proposition 5.2).

Road map. The proof of (1.2) has three steps.

1. *Truncated inner products* (§3). For a truncation parameter T, the integral J^T of a product of two matrix coefficients over $G(T) = \bigcup_a KaK$ (the finitely many double cosets "below T") is computed from Casselman's asymptotics on the regions where they apply. Brion's theorem turns the resulting lattice-point sums into vertex terms $\psi(T_Q)R_Q$, one for each standard parabolic Q; the vertex $T_G = 0$ gives a T-independent term (Proposition 3.6).

2. *The truncated kernel* (§4.2). Expanding the spherical function in matrix coefficients expresses $I^T(\chi,\phi) = \int_{G(T)\times G}\Gamma_\chi(g^{-1}hg)\phi(h)$ through J^T. So I^T is a sum of vertex terms, and its T-independent part defines $I(\chi,\phi)$ (Proposition 4.2).

3. *Removing the truncation* (§4.3). For T large, $T_\phi(f) = \int f^\vee I^T\,d\mu$ exactly (Fubini on a compact set). If the vertex terms are regular enough (Hypothesis (R)), their contributions vanish as $T\to\infty$ and $T_\phi(f) = \int f^\vee I\,d\mu$ (Theorem 4.5). Where (R) fails they can leave atoms (Conjecture 4.6, §6.2), or they can cancel in the limit (Appendix A).

What is proved, what is conditional, what is conjectural.

- *Proved for all G satisfying (H):* the truncated identity, Theorem 4.5(a); the combinatorial Lemmas 3.5A–3.5B (also checked by exact computation for $SL(3)$ and $PGL(3)$); Lemma 2.8 when the coroot lattice is saturated in $X_*(A)$.

- *Proved modulo one input from Casselman's theory:* Propositions 3.6 and 4.2 use Casselman's asymptotics in the form (C) of §3.1, on the regions $A_P^-(\varepsilon)$. In rank one, (C) is Casselman's theorem itself, and these propositions are unconditional.

- *Proved under Hypothesis (R):* the spectral identity, Theorem 4.5(b), and Corollary 4.5A. Hypothesis (R) is a strong condition with representation-theoretic content, not a routine regularity assumption. It is not necessary (Appendix A).

- *Examples, proved:* for $PGL(2)$, (R) holds and $I(\chi,\gamma)$ is given in closed form (Theorem 6.1, Lemma 6.1A). For $SL(2)$, (R) fails and the spectral measure has an atom (Proposition 6.4, for every q).
- *Example, numerical:* for $PGL(3)$ and an unramified elliptic γ of depth 0, $I(\chi,\gamma)$ is given in closed form, (R) fails, and the spectral identity holds with no singular part (Appendix A; one combinatorial input is observed, not proved).
- *Endoscopy:* Proposition 5.1 follows from the fundamental lemma, which is a theorem; its normalizations were checked for $SL(2)$. Proposition 5.2 is conditional on a stable form of Theorem 4.5 for the endoscopic group.
- *Conjectural:* Conjecture 4.6 (shape of the spectral measure when (R) fails) and Conjecture 7.3 (comparison with Arthur's weighted characters). Section 7 describes a proposed route to Conjecture 4.6, not a proof.

Readers who want a concrete picture first may read §6 (the examples $PGL(2)$ and $SL(2)$) before §§3–4; it uses only §2 and the statements of §4.

A note on numbering. We apologize for the unusual numbering of this paper. Sections, results and equations keep the numbers that they had in the 1997 version. This is so that readers who want to can compare the revision with the original. Items added in 2026 carry letters, as in Lemma 3.5A, §2.1A and (2.13A), or new numbers, as in Section 6 and the sections after it. Some numbers do not occur, because the 1997 items that carried them were removed. Examples are Section 4.1 and Remarks 6–8 of the 1997 version. A concordance with the 1997 numbering is in the revision log (the file `revision_log.md` that accompanies the source).

Literature review

The paper was published in 1992, and its corrected and extended version was circulated in October 1997. We divide the literature at 1997. The first part describes the background available when the 1997 version was written, including work that appeared in 1997. The second part describes what came after.

Work up to 1997

This part describes the work available up to 1997: spherical harmonic analysis (Satake, Macdonald, Casselman), Arthur's program and its elliptic form, explicit spherical orbital integrals, endoscopy and transfer, the building, and lattice-point counting.

Spherical harmonic analysis. The spherical function and the Hecke algebra of a p-adic group go back to Satake [Sa63], who identified $\mathscr{H}(G,K)$ with W-invariant functions on the torus of unramified characters (the Satake isomorphism). Macdonald [Ma71] found the explicit formula for zonal spherical functions (Lemma 2.1) and the spherical Plancherel measure (Lemma 2.2). Casselman [Ca80] re-derived Macdonald's formula from intertwining operators and the asymptotics of matrix coefficients, which are developed in his notes [Ca95] (first circulated in 1974). Cartier's

survey [Ca79] collects the unramified theory used throughout this paper. The geometric lemma of Bernstein and Zelevinsky [BZ77] describes Jacquet modules of induced representations. Harish-Chandra's theory of admissible invariant distributions [HC78] supplies the local integrability of characters and the G-admissibility used in §1. Keys [Ke82] determined the R-groups of unramified principal series of Chevalley groups.

Arthur's program. Arthur's trace formula [Ar78] and its invariant form [Ar81, Ar88] introduced truncation, (G,M)-families, and weighted orbital integrals and characters. Arthur [Ar87] expressed characters of supercuspidal representations as weighted orbital integrals, a precursor of the cuspidal specialization in §7.3. "Towards a local trace formula" [Ar89] and "A local trace formula" [Ar91] apply these ideas to the regular representation of $G(F)\times G(F)$ on $L^2(G(F))$, producing the identity used in §7. The elliptic part of harmonic analysis was developed by Kazhdan [Ka86] (cuspidal geometry, the elliptic pairing and density) and Bernstein–Deligne–Kazhdan [BDK86] (trace Paley–Wiener theorem).

Elliptic harmonic analysis. Arthur's paper on elliptic tempered characters [Ar93] introduced the virtual characters $T_{\mathrm{ell}}(M)$ and put the local trace formula in its invariant, elliptic form. This is the form used in §7.3, and it explains the singular part of Conjecture 4.6. Waldspurger's local trace formula for p-adic Lie algebras [Wa95a] is the Lie-algebra analogue of [Ar91]. His ICM address [Wa95b] surveys the comparison of orbital integrals as it stood in 1994.

Spherical orbital integrals. Explicit spherical orbital integrals appear in Langlands' base change for $GL(2)$ [La80] and in Kottwitz's computation for $GL(3)$ [Ko80]. Waldspurger [Wa89a] computed the spectral expansion of spherical orbital integrals for $GL(N)$, by the method followed here. His Shalika germ paper [Wa89b] and Assem's work on unipotent orbital integrals [As91] give complementary information near the identity.

Endoscopy. Labesse and Langlands [LL79] worked out stable conjugacy, L-indistinguishability and κ-orbital integrals for $SL(2)$; the atom of §6.2 is a shadow of their theory. Kottwitz developed rational and stable conjugacy [Ko82] and the elliptic singular terms of the stable trace formula [Ko86]. Langlands and Shelstad defined transfer factors [LS87]. The fundamental lemma for the unit element of $SL(n)$ was proved by Kazhdan [Ka84] for κ of order n and by Waldspurger [Wa91] in general. Clozel [Cl90] and Labesse [La90] treated stable base change. Henniart's work on the cyclic case of the local Langlands conjecture [He86] uses the κ-orbital integrals of $SL(n)$ considered in §5.

Transfer and the fundamental lemma. Hales [Ha95] reduced the fundamental lemma for the whole Hecke algebra to the unit element. Waldspurger showed that the fundamental lemma implies transfer [Wa97], and Courtès proved transfer for $SL(n)$-type groups [Co97].

Elliptic theory via the building. Schneider and Stuhler [SS97] constructed Euler–Poincaré functions from the Bruhat–Tits building and identified their orbital integrals with characters on the elliptic set.

Lattice points. The combinatorial input of §3 is Brion's theorem [Br88] on generating functions of lattice points in rational polytopes.

Work since 1997

This part describes later work that bears on the paper: the proof of the fundamental lemma, elliptic representations and the building, new expositions of the Plancherel formula, and relative local trace formulas.

Transfer and the fundamental lemma. Laumon and Ngô proved the fundamental lemma for unitary groups [LN08], and Ngô proved it in general for Lie algebras [Ng10]. Combined with Waldspurger's transfer from Lie algebras to groups and change of characteristic, and with [Ha95], this makes (5.11) a theorem. For statements and surveys, see Hales [Ha05] and Kottwitz [Ko05]. For twisted endoscopy, Lemaire, Mœglin and Waldspurger reduced the fundamental lemma to unit elements [LMW18], and Mœglin and Waldspurger developed the twisted local trace formula [MW18].

Elliptic theory via the building. Reeder [Re01] related the Euler–Poincaré pairing of Iwahori-spherical representations to elliptic representations of Weyl groups. Opdam and Solleveld [OS13] computed Ext groups of tempered representations of affine Hecke algebras and recovered Arthur's formula for the elliptic pairing. These results bear directly on the singular part in Conjecture 4.6: the elliptic tempered virtual characters with Iwahori-fixed (in particular K-fixed) vectors are exactly what Reeder and Opdam–Solleveld classify.

The Plancherel formula. Waldspurger's exposition of Harish-Chandra's Plancherel formula [Wa03] gives complete proofs of the harmonic analysis behind §7. A new edition of Macdonald's notes, with A.-M. Aubert, has appeared [MA26].

Local relative trace formulas. The local trace formula has been generalized to relative settings: Feigon [Fe12] and Delorme–Harinck [DH17] for $PGL(2)$, and Beuzart-Plessis [BP20] for the Gan–Gross–Prasad conjecture, following Waldspurger's work on orthogonal groups.

Lattice points. Barvinok's book [Ba08] is a modern account of Brion's theorem and its algorithmic uses.

Remark 2026.18 (literature). All bibliographic entries below were checked against publisher, Numdam/EuDML or library records in November 2026. The review is not exhaustive. In particular, we have not surveyed systematically the later explicit computations of spherical orbital integrals of $GL(n)$ or $SL(n)$ in terms of the building, which would be the natural comparison for §6.

2 Notation and background

This section collects the background from the harmonic analysis of unramified groups that the rest of the paper uses. Nothing in it is new. §2.1 sets up root data, lattices and the torus of unrami-

fied characters. §2.1A recalls the representation theory used throughout: smooth and admissible representations, matrix coefficients, the unramified principal series, the spherical function, and Casselman's description of matrix coefficients at infinity. §2.2 recalls Casselman's computation of intertwining operators on the spherical vector, which produces the c-function. §2.3 gives Macdonald's formula for the spherical function and his Plancherel formula; together they turn spherical functions into Laurent polynomials on a compact torus, and this is the setting of the spectral side of (1.2). §2.3A collects the Fourier analysis on that torus that the proofs use, including the Dirichlet-kernel mechanism behind atoms in the spectral measure. §2.4 recalls the Bernstein–Zelevinsky geometric lemma, which describes Jacquet modules of principal series and is used in §3.

Each subsection ends with an example, worked out for $PGL(2)$ and $SL(2)$; the same examples run through §§3–5 and are collected in §6. A reader who knows [Ca79] can skim this section and come back to it as needed. §2.5, on (G,M)-families, is included for comparison with Arthur's and Waldspurger's methods and plays no role in the proofs.

2.0 Conventions

Here we fix notation for the field, the group, its root data, Haar measures and unramified characters.

- F is a p-adic field with ring of integers $\mathcal{O}$, uniformizer π, residue field of order q, $|\pi|_F = q^{-1}$, and normalized valuation ν_F.
- A is a maximal F-split torus, $M_0 = Z_G(A)$ its centralizer (a maximal torus, since G is quasi-split), $B = M_0 N_0 \supset M_0$ a Borel subgroup over F, and K a hyperspecial maximal compact subgroup in good position relative to A. $I \subset K$ is the Iwahori subgroup attached to B.
- *Roots and weights.* Δ is the set of simple (reduced) roots of A in B, $\Delta^\vee = \{\alpha^\vee\}$ the simple coroots, $\hat{\Delta} = \{\varpi_\alpha\}$ the fundamental weights (in $(\mathfrak{a}_0^G)^*$), so that $\langle \alpha^\vee, \varpi_\beta \rangle = \delta_{\alpha\beta}$. This is Arthur's notation.
- $\mathfrak{a}_0 = X_*(A) \otimes \mathbb{R}$. For a standard parabolic $P = MN$, $\mathfrak{a}_0 = \mathfrak{a}_P \oplus \mathfrak{a}_0^P$, and we write $X = X_P + X^P$.
- *Chamber.* $A^- = \{a \in A(F) : |\alpha(a)|_F \le 1 \ \forall \alpha \in \Delta\}$. Since $|\alpha(a)|_F = q^{-\langle \nu_A(a), \alpha \rangle}$ (see (2.4)), $\nu_A(A^-)$ is the closed dominant cone $X_*(A)^+$. Elements of A^- contract N_0. For a standard parabolic $P = MN$ and $0 < \varepsilon \le 1$ put

$$A_P^-(\varepsilon) = \{a \in A^- : \ |\alpha(a)|_F \le \varepsilon \ \ \forall \alpha \in \Delta \smallsetminus \Delta^M\}, \qquad A^-(\varepsilon) = A_B^-(\varepsilon).$$

- *Induction and Jacquet modules* are normalized: $i_{G,M}(\sigma) = \{f : G \to V_\sigma : \ f(mng) = \delta_P^{1/2}(m)\sigma(m)f(g)\}$ with G acting by right translation, and $r_{M,G}(\pi) = \delta_P^{-1/2} \otimes \pi_N$. Here $\delta_P(m) = |\det \mathrm{Ad}(m)|_{\mathrm{Lie}\, N}|_F$, so $\delta_B(a) = q^{-\langle \nu_A(a), 2\rho \rangle} \le 1$ on A^-.
- *Weyl group.* $W = N_G(A)/M_0$, with representatives in K; $(w\chi)(a) = \chi(w^{-1}aw)$. For a standard Levi M, $[W_M \backslash W]$ denotes the set of minimal-length representatives w of the cosets $W_M w$ (equivalently $w^{-1}\Delta^M > 0$).
- *Unramified characters.* For $\lambda \in \mathfrak{a}_{0,\mathbb{C}}^*$ put $\chi_\lambda(a) = q^{-\langle \nu_A(a), \lambda \rangle}$. This is the one definition of χ_λ used throughout.

- *Measures.* $\mathrm{vol}(K) = \mathrm{vol}(K \cap A(F)) = \mathrm{vol}(K \cap N(F)) = 1$, and $\mathrm{vol}(K \cap M(F)) = 1$ for every standard Levi M. The Haar measure on $\widehat{T}^u$ (below) has total mass 1.
- *Symbols.* T always denotes the truncation parameter. Maximal tori are M_0 or (for elliptic tori) S. Characteristic functions are $\mathbf{1}_E$. Q_G is Macdonald's constant. The letters P, Q denote parabolic subgroups. We avoid $\chi(M)$ for a characteristic function.

2.1 Root spaces

We describe the root data of an unramified group: the maximal split torus A, the relative roots, the lattices $X^*(A)$ and $X_*(A)$, and the identification of $X_*(A)$ with $A(F)/(A(F) \cap K)$, which lets us write a_λ for the element attached to a coweight λ. The last paragraph introduces the complex torus $\widehat{T}$ of unramified characters and its compact form $\widehat{T}^u$, on which the spectral side of (1.2) lives.

Let G be a connected unramified reductive group of semi-simple rank ℓ over F, split over a finite unramified extension E/F. (Recall that a reductive group G over F is unramified if it is quasi-split over F and split over a finite unramified extension [Ca79, p. 135].) Let $M_0 = Z_G(A)$, B and A be as in §2.0. Let $\Gamma = \mathrm{Gal}(E/F)$, let $X^* = X^*(M_0)$ denote the character lattice of M_0, and let $\Sigma_{M_0} \subset X^*$ denote the (absolute) root system. The Γ-module structure of X^* leaves Σ_{M_0} invariant. Let Σ denote the set of reduced roots of G relative to A and Δ the corresponding reduced fundamental system. The character lattice

$$X^*(A) = \mathrm{Hom}_{F\text{-grps}}(A, GL(1)) \tag{2.1}$$

may be regarded as a quotient of X^* containing Σ. Let

$$X_*(A) := \mathrm{Hom}_{\mathbb{Z}}(X^*(A), \mathbb{Z}) \tag{2.2}$$

denote the co-character lattice. Let $\Sigma^+ \subset \Sigma$ be the set of positive roots containing Δ. For parabolic subgroups $P \subset Q$ containing A, let Δ_P^Q denote the set of simple roots of $(P \cap M_Q, A_P)$, where $Q = M_Q N_Q$ is the Levi decomposition and A_P is the split component of the center of M_P [Ar81]; $\Delta_P^G = \Delta_P$. Each $w \in W$ has a representative in K via $N_G(A)/Z_G(A) \cong (K \cap N_G(A))/(K \cap Z_G(A))$ [Ca79, p. 140]. W is generated by the simple reflections $\{w_\alpha : \alpha \in \Delta\}$ ([Ca95], §1). Let R denote the root lattice generated by Δ and L the coroot lattice generated by $\Delta^\vee$. If G is semi-simple these lattices and $X^*(A)$, $X_*(A)$ all have rank ℓ, and $X^*(A)/R$, $X_*(A)/L$ are finite. If G is split, G is adjoint iff $X^*(A) = R$ and simply connected iff $X_*(A) = L$; for non-split G these criteria must be phrased with the absolute root datum (quasi-split $SU(3)$ and $PU(3)$ both have $X^*(A) = R$).

For each $\theta \subset \Delta$, let $P_\theta = M_\theta N_\theta \supset B$ be the corresponding standard parabolic, so $P_\emptyset = B$ and $M_\emptyset = M_0$; let $A_M = A_\theta$ be the split component of the center of $M = M_\theta$. A Levi component $M = M_P$ of a parabolic $P = M_P N_P$ is called a Levi subgroup of G; all Levi and parabolic subgroups are taken to contain A. $\mathscr{P}(M)$ is the set of parabolics with Levi component M. For $P \subset Q$ there are orthogonal decompositions $\mathfrak{a}_P = \mathfrak{a}_Q \oplus \mathfrak{a}_P^Q$ and $\mathfrak{a}_P^* = \mathfrak{a}_Q^* \oplus (\mathfrak{a}_P^Q)^*$ [Ar78]. The projections of $(\Delta_P^Q)^\vee$ form a basis of $\mathfrak{a}_P^Q$ and $\hat{\Delta}_P^Q$ a basis of $(\mathfrak{a}_P^Q)^*$. Let τ_P^Q and $\hat{\tau}_P^Q$ denote the characteristic functions on $\mathfrak{a}_0$ of

$$\{X :\ \alpha(X) > 0,\ \alpha \in \Delta_P^Q\} \qquad \text{and} \qquad \{X :\ \varpi(X) > 0,\ \varpi \in \hat{\Delta}_P^Q\},$$

respectively, as in [Ar78].

The kernel of the map $H_M : M(F) \to \mathfrak{a}_P$ of [Ar78] is denoted $M(F)^1$; it is the intersection of the kernels of the absolute values of the rational characters of $M(F)$. The Haar measure on $M(F)^1$ is the one determined by those on $M(F)$ and $\mathfrak{a}_P$.

The pairing $\langle \cdot, \cdot \rangle : X_*(A) \times X^*(A) \to \mathbb{Z}$, $\langle \lambda_*, \lambda^* \rangle := \lambda_*(\lambda^*)$, identifies $X^*(A)$ with the dual lattice of $X_*(A)$. It gives an isomorphism

$$\mathrm{Hom}_{F\text{-grps}}(GL(1), A) \cong X_*(A), \tag{2.3}$$

associating to λ the unique λ_* with $\lambda^*(\lambda(t)) = t^{\langle \lambda_*, \lambda^* \rangle}$ for all $t \in F^\times$, $\lambda^* \in X^*(A)$. Let $a_\alpha \in A(F)$ denote $\alpha^\vee(\pi)$ (for non-split G, the element of [Ca79, §3] used in Casselman's formula (2.8)). If G is split then: (a) $\delta_B(a_\alpha) = q^{-2}$ for $\alpha \in \Delta$; (b) the a_α generate the image of the coroot lattice L in $A(F)/(A(F) \cap K)$, freely, and this image is everything iff G is simply connected (in general it has finite index $[X_*(A) : L]$); (c) $w_\alpha^{-1} a w_\alpha = a\, a_\alpha^{-\langle \nu_A(a), \alpha \rangle}$ modulo $A(F) \cap K$, where ν_A is as in (2.4) ([Ca79], pp. 141–142, [Ma71], pp. 42–43). To each $\beta \in \Sigma \cup \frac{1}{2}\Sigma$ there is attached a number $q_\beta > 0$ as in [Ca79]; if G is split and $\alpha \in \Sigma$ then $q_\alpha = q$ and $q_{\alpha/2} = 1$.

Let $\mathfrak{a}_0 = X_*(A) \otimes \mathbb{R}$ and $\mathfrak{a}_0^* = X^*(A) \otimes \mathbb{R}$, and extend $\langle \cdot, \cdot \rangle$ to $\mathfrak{a}_0 \times \mathfrak{a}_0^*$. Then $\Delta^\vee \subset \mathfrak{a}_0$ and $\hat{\Delta} \subset \mathfrak{a}_0^*$ are dual bases (for semisimple G): $\langle \alpha^\vee, \varpi_\beta \rangle = \delta_{\alpha\beta}$ for $\alpha, \beta \in \Delta$.

There is a surjection $\nu_A : A(F) \to X_*(A)$ characterized by

$$\langle \nu_A(a), \lambda^* \rangle = \nu_F(\lambda^*(a)), \qquad \forall \lambda^* \in X^*(A),\ a \in A(F). \tag{2.4}$$

Thus we obtain an isomorphism

$$\nu_A^{-1} : X_*(A) \xrightarrow{\sim} A(F)/(A(F) \cap K) \ \cong\ M_0(F)/(M_0(F) \cap K), \tag{2.5}$$

and every unramified character of $M_0(F)$ may be identified with a character of the discrete group $X_*(A)$. We write $a_\lambda = \nu_A^{-1}(\lambda)$ and $\chi(\lambda) = \chi(a_\lambda)$. For each Levi M, $X_*(A_M) = X_*(A) \cap \mathfrak{a}_M$.

Tori of characters. Let $L^\vee = \frac{2\pi i}{\log q} X^*(A) \subset \mathfrak{a}_{0,\mathbb{C}}^*$. The map $\lambda \mapsto \chi_\lambda$ gives isomorphisms

$$\widehat{T} := \mathrm{Hom}(X_*(A), \mathbb{C}^\times) \cong \mathfrak{a}_{0,\mathbb{C}}^*/L^\vee, \qquad \widehat{T}^u := \mathrm{Hom}(X_*(A), \mathbb{C}^{\times 1}) \cong i\mathfrak{a}_0^*/L^\vee, \tag{2.6}$$

where $\mathbb{C}^{\times 1} = \{z : |z| = 1\}$. Here $\widehat{T}$ is the complex torus of all unramified characters and $\widehat{T}^u$, the Pontryagin dual of $X_*(A)$, is its compact real form: the unitary unramified characters. Thus

$$\mathrm{Hom}_{unr}(A(F), \mathbb{C}^{\times 1}) \cong i\mathfrak{a}_0^*/L^\vee, \tag{2.7}$$

where $\mathrm{Hom}_{unr}(H(F), \mathrm{End}\, V)$ denotes the continuous homomorphisms with a non-zero $H(F) \cap K$-fixed vector.

Example 2.1 (rank one, and $PGL(3)$). For $G = PGL(2)$ take $A = \{\mathrm{diag}(x, 1)\}$. Then $X_*(A) = \mathbb{Z}\varpi^\vee$ with $a_n := \varpi^\vee(\pi)^n = \mathrm{diag}(\pi^n, 1)$, and $\alpha^\vee = 2\varpi^\vee$. An unramified character is a number $z = \chi(a_1) \in \widehat{T} = \mathbb{C}^\times$, unitary if $|z| = 1$, and $\chi(a_\alpha) = z^2 =: y$. The Weyl group $W = \{1, s\}$ acts by $z \mapsto z^{-1}$, and $A^- = \{a_n : n \geq 0\}$.

For $G = SL(2)$ take $A = \{\mathrm{diag}(x, x^{-1})\}$. Then $X_*(A) = \mathbb{Z}\alpha^\vee$, $a_n = \mathrm{diag}(\pi^n, \pi^{-n})$ and $z = \chi(a_1) = \chi(a_\alpha)$. The two groups differ in one respect that matters later: for $PGL(2)$ the coroot is twice a generator of $X_*(A)$, so the root hyperplane $y = 1$ in $\widehat{T}^u$ consists of the two points $z = \pm 1$, while for $SL(2)$ only $z = 1$ lies on it, and $z = -1$ is a non-regular character ($s\chi = \chi$) off the root hyperplane.

For $G = PGL(3)$, $X_*(A) = \mathbb{Z}^3/\mathbb{Z}(1,1,1)$, an unramified character is $z = (z_1, z_2, z_3)$ with $z_1 z_2 z_3 = 1$, and $\chi(a_{\alpha^\vee}) = z_i/z_j$ for $\alpha = e_i - e_j$ (Appendix A).

2.1A Representations and matrix coefficients

This subsection recalls the representation theory behind the rest of the paper: smooth and admissible representations, contragredients and matrix coefficients, the unramified principal series with its spherical vector and spherical function, and the behavior of matrix coefficients at infinity. Standard references are Cartier's survey [Ca79], Casselman's notes [Ca95] and Bernstein–Zelevinsky [BZ77].

Smooth and admissible representations. A representation (π, V) of $G(F)$ on a complex vector space is *smooth* if every vector is fixed by some open compact subgroup, and *admissible* if moreover $V^{K'}$ is finite-dimensional for every open compact subgroup K'. For $n \geq 1$ let $K_n \subset K$ be the n-th principal congruence subgroup, and say that $v \in V$ has *level* n if it is fixed by K_n. For $\phi \in C_c^\infty(G(F))$ put $\pi(\phi)v = \int_{G(F)} \phi(g)\pi(g)v\,dg$. If ϕ is bi-K_n-invariant, then $\pi(\phi) = e_n\pi(\phi)e_n$, where e_n is the projection onto V^{K_n}. So for admissible π the operator $\pi(\phi)$ has finite rank, and it has a trace. This is how the test function ϕ enters in §4.2.

Contragredient and matrix coefficients. The contragredient $(\pi^\vee, V^\vee)$ acts on the smooth linear forms on V. A *matrix coefficient* of π is a function $g \mapsto \langle \pi(g)v, v^\vee \rangle$ with $v \in V$ and $v^\vee \in V^\vee$. It is locally constant, and bi-invariant under any open compact subgroup fixing v and $v^\vee$. Matrix coefficients encode the behavior of π "at infinity" in $G(F)$. For example, an irreducible π is supercuspidal exactly when its matrix coefficients are compactly supported. For the principal series below they are not, and their behavior at infinity is governed by Jacquet modules.

The unramified principal series. For a character χ of $M_0(F)$, $V_B(\chi) = i_{G,A}(\chi)$ is the space of locally constant $f : G(F) \to \mathbb{C}$ with $f(mng) = \chi(m)\delta_B^{1/2}(m)f(g)$ for $m \in M_0(F)$, $n \in N_0(F)$, with $G(F)$ acting by right translation. By the Iwasawa decomposition $G(F) = B(F)K$, such an f is determined by its restriction to K. So all the $V_B(\chi)$ are realized on one space of functions on $(B(F) \cap K)\backslash K$ (the "compact picture"), and $V_B(\chi)^{K_n}$ is the finite-dimensional space of functions on $(B(F) \cap K)\backslash K / K_n$. In particular $V_B(\chi)$ is admissible. The pairing

$$\langle f, f' \rangle_G = \int_K f(k)f'(k)\,dk \qquad (f \in V_B(\chi),\ f' \in V_B(\chi^{-1}))$$

is $G(F)$-invariant, because ff' transforms under $B(F)$ by δ_B and $\int_K$ is the invariant functional on such functions. Hence $V_B(\chi)^\vee \cong V_B(\chi^{-1})$. If χ is unitary, $(f, f') \mapsto \int_K f\overline{f'}\,dk$ makes $V_B(\chi)$ a unitary representation.

The character χ is *unramified* if it is trivial on $M_0(F) \cap K$. Then, and only then, $V_B(\chi)^K \neq 0$; it is the line spanned by the *spherical vector* $\Phi_{K,\chi}$, with $\Phi_{K,\chi}(mnk) = \chi(m)\delta_B^{1/2}(m)$.

The spherical function. The matrix coefficient of the spherical vectors,

$$\Gamma_\chi(g) = \langle \pi_\chi(g)\Phi_{K,\chi}, \Phi_{K,\chi^{-1}} \rangle_G = \int_K (\chi\delta_B^{1/2})(m(kg))\,dk \qquad (kg \in m(kg)N_0(F)K),$$

is the *zonal spherical function*. It is bi-K-invariant with $\Gamma_\chi(1) = 1$. It depends only on the W-orbit of χ, and it is an eigenfunction of convolution by every $f \in \mathscr{H}(G, K)$. Every zonal spherical

function is of this form [Ma71, Thm. 3.3.12]. By the Cartan decomposition $G(F) = \bigsqcup K a_\lambda K$ ($a_\lambda \in A^-$), Γ_χ is determined by its values on A^-, which Macdonald's formula (Lemma 2.1) computes. A second identity is the bridge from (1.1) to the rest of the paper: for $g, h \in G(F)$,

$$\Gamma_\chi(g^{-1}hg) = \langle \pi_\chi(h)\, \pi_\chi(g)\Phi_{K,\chi},\ \pi_{\chi^{-1}}(g)\Phi_{K,\chi^{-1}} \rangle_G.$$

So integrating Γ_χ over a conjugacy class amounts to integrating products of matrix coefficients of $\pi_\chi(g)\Phi_{K,\chi}$ over g. That is the content of §3 and §4.2.

Behavior at infinity. Let $v \in V$ and $v^\vee \in V^\vee$. Casselman's theorem [Ca95, §4] says that there is $\varepsilon > 0$, depending on the levels of v and $v^\vee$, such that for $a \in A^-$ with $|\alpha(a)| \le \varepsilon$ for all $\alpha \in \Delta$ (that is, a deep in the negative chamber),

$$\langle \pi(a)v, v^\vee \rangle = \delta_B^{1/2}(a)\, \langle \pi_N(a) v_N,\ v^\vee_{\bar{N}} \rangle_N,$$

where $v \mapsto v_N$ is the projection to the normalized Jacquet module along B, and $\langle \cdot, \cdot \rangle_N$ is Casselman's canonical pairing between the Jacquet modules along B and along the opposite Borel subgroup. For $V = V_B(\chi)$ with χ regular, the Jacquet module along B is $\bigoplus_{w \in W} w\chi$ (Corollary 2.5 and Remark 1). So deep in the chamber every matrix coefficient is a finite sum of characters:

$$\langle \pi_\chi(a)v, v^\vee \rangle = \delta_B^{1/2}(a) \sum_{w \in W} C_w(v, v^\vee)\, (w\chi)(a).$$

Macdonald's formula is the case $v = \Phi_{K,\chi}$, $v^\vee = \Phi_{K,\chi^{-1}}$, where the expansion holds on all of A^- and $C_w = Q_G^{-1} c((w\chi)^{-1})$. Near the walls of A^- the expansion along B fails, and one must use the Jacquet module along a larger parabolic P. That refinement is the input (C) of §3.1.

Example 2.1A ($PGL(2)$ and $SL(2)$). For $G = PGL(2)$, $(B(F) \cap K)\backslash K \cong \mathbb{P}^1(\mathscr{O})$, so $V_B(\chi)^K$ consists of the constant functions in the compact picture, and $V_B(\chi)^{K_1}$ of the functions on $\mathbb{P}^1(\mathbb{F}_q)$, of dimension $q+1$. For regular χ ($y = z^2 \ne 1$) the Jacquet module along B is $\chi \oplus s\chi$, and the spherical function is

$$\Gamma_\chi(a_n) = \delta_B^{1/2}(a_n)\big(C_1 z^n + C_s z^{-n}\big) = q^{-n/2} Q^{-1}\big(c_\alpha(\chi^{-1}) z^n + c_\alpha(\chi) z^{-n}\big) \qquad (n \ge 0),$$

exactly, by Lemma 2.1 (Example 2.3). For $PGL(2)$ every unitary unramified principal series is irreducible. For $SL(2)$ and the unramified quadratic character χ_0 ($z = -1$), $s\chi_0 = \chi_0$ is not regular, and $V_B(\chi_0) = \pi^+ \oplus \pi^-$ splits into two irreducible pieces, only one of which (π^+) has a K-fixed vector. This reducibility is behind the atom of §6.2.

2.2 Intertwining operators

The standard intertwining operators T_w between unramified principal series act on the spherical vector by a scalar, the Harish-Chandra c-function. We record this, and the factorization of c_w over positive roots, since c appears in Macdonald's formula and in the Plancherel density.

For the unramified principal series representations $i_{G,A}(\chi) = V_B(\chi)$ of $G(F)$ attached to an unramified character χ of $M_0(F)$ we refer to [Ca79]. The pairing $\langle \cdot, \cdot \rangle_G$ on $V_B(\chi) \times V_B(\chi^{-1})$, $\langle f, f' \rangle_G = \int_K f(k) f'(k)\, dk$, identifies the contragredient of $i_{G,A}(\chi)$ with $i_{G,A}(\chi^{-1})$.

Let χ be a regular unramified character (so the $w\chi$, $w \in W$, are distinct), and let $T_w : V_B(\chi) \to V_B(w\chi)$ denote the intertwining operator of [Ca79]. If $\Phi_{K,\chi} \in V_B(\chi)^K$ is the K-fixed vector with $\Phi_{K,\chi}(1) = 1$ then Casselman [Ca80, Thm. 3.1], [Ca79, Thm. 3.9] has shown that

$$T_w(\Phi_{K,\chi}) = c_w(\chi)\Phi_{K,w\chi}, \qquad c_w(\chi) = \prod_{\alpha\in\Sigma^+,\ w\alpha<0} c_\alpha(\chi), \tag{2.8}$$

$$c_\alpha(\chi) = \frac{\left(1 - q_{\alpha/2}^{-1/2} q_\alpha^{-1}\chi(a_\alpha)\right)\left(1 + q_{\alpha/2}^{-1/2}\chi(a_\alpha)\right)}{1 - \chi(a_\alpha)^2}.$$

In the split case $c_\alpha(\chi) = (1 - q^{-1}\chi(a_\alpha))/(1 - \chi(a_\alpha))$. Put $c(\chi) = c_{w_\ell}(\chi) = \prod_{\alpha>0} c_\alpha(\chi)$, w_ℓ the longest element of W. Also $T_{w_1 w_2} = T_{w_1} T_{w_2}$ if $\ell(w_1 w_2) = \ell(w_1) + \ell(w_2)$; here $\ell(w)$ is the length of w, which is also the number of walls in $\mathfrak{a}_0$ separating the positive chambers of B^w and B [Ca95, ch. 1].

Remark 2026.2 (exponent in c_α). We use Casselman's exponent $q_{\alpha/2}^{-1/2}$ in the first factor of c_α; for split G the factor $q_{\alpha/2}$ does not occur. The non-split formula has not been re-derived here; all numerical checks in this paper are for split groups.

Example 2.2 (rank one). For $PGL(2)$ the simple reflection s acts on the spherical vector by $T_s\Phi_{K,\chi} = c_\alpha(\chi)\Phi_{K,s\chi}$ with $c_\alpha(\chi) = (1 - q^{-1}y)/(1 - y)$, $y = \chi(a_\alpha) = z^2$. For $SL(2)$ the same formula holds with $y = z$. The pole of c_α is on the root hyperplane $y = 1$, and its zero is at $y = q$, off the unitary torus.

2.3 Plancherel's formula and Macdonald's formula

This subsection contains the two formulas on which the spectral side rests. Macdonald's formula (Lemma 2.1) expresses the zonal spherical function Γ_χ on the negative chamber as a sum over W of characters weighted by c-functions. Macdonald's Plancherel theorem (Lemma 2.2) identifies the Plancherel measure for $\mathscr{H}(G,K)$ as $\rho(\chi)\,d\chi$ on $\widehat{T}^u/W$, with ρ proportional to $|c(\chi)|^{-2}$. Corollary 2.3 is the resulting inversion formula, which is how a spherical function is recovered from its Satake transform.

Let $\mathscr{H}(G,K)$ denote the algebra of bi-K-invariant functions in $C_c^\infty(G(F))$ and $C^\infty(G,K)$ the analogous space of locally constant bi-K-invariant functions. For $f \in V_B(\chi)$ define $f^\natural(g) = \int_K f(kg)\,dk$, and let $\Gamma_\chi = \Phi_{K,\chi}^\natural$ be the zonal spherical function; $\Gamma_\chi(g) = \langle i_{G,A}(\chi)(g)\Phi_{K,\chi}, \Phi_{K,\chi^{-1}}\rangle_G$. Macdonald's formula states:

Lemma 2.1 (Macdonald)**.** *If χ is an unramified regular character of $M_0(F)$ then, for $a \in A^-$,*

$$\Gamma_\chi(a) = Q_G^{-1}\,\delta_B(a)^{1/2} \sum_{w\in W} c\big((w\chi)^{-1}\big)\,(w\chi)(a), \qquad Q_G := \sum_{w\in W} (IwI : I)^{-1},$$

which is $\sum_w q^{-\ell(w)}$ in the split case.

Remark 2026.3 (Lemma 2.1). Two points about the normalization.

1. The constant is $Q_G = \sum_w (IwI : I)^{-1}$; for $PGL(2)$ and $SL(2)$, $\Gamma_\chi(1) = 1$ forces $Q_G = 1 + q^{-1}$ (Example 2.3), and the Plancherel inversion (Corollary 2.3) holds with this constant (checked numerically, §6).

2. The formula holds only on the closed chamber A^- (the right side is not W-invariant in a). On the chamber A^- of §2.0 the coefficient of $(w\chi)(a)$ is $c((w\chi)^{-1})$; with $c(w\chi)$ one gets the formula for the opposite chamber $A^+ = (A^-)^{-1}$. Verified to machine precision for $PGL(2)$ and $SL(2)$. In general the pairing "chamber versus c-function" depends on the convention for T_w; we have checked it only in rank one.

For $f \in \mathscr{H}(G,K)$ and χ unramified, define the Fourier (Satake) transform of f at χ by

$$f^\vee(\chi) := \int_{G(F)} f(g)\Gamma_\chi(g^{-1})\,dg. \tag{2.9}$$

Let $\Omega_K(G)$ denote the set of zonal spherical functions of $G(F)$ relative to K:

$$\Omega_K(G) := \{\omega \in C^\infty(G,K) :\ \omega(1) = 1 \text{ and } f * \omega = \lambda_f \omega \text{ for all } f \in \mathscr{H}(G,K), \text{ some } \lambda_f \in \mathbb{C}\}. \tag{2.10}$$

Let $\Omega_K^+(G)$ denote the subset of positive definite zonal spherical functions. If χ is unitary then $\Gamma_\chi \in \Omega_K^+(G)$ [Ma71, Thm. 3.3.12]. The Fourier transform of $f \in \mathscr{H}(G,K)$ at $\omega \in \Omega_K^+(G)$ is

$$f^\vee(\omega) := \int_{G(F)} f(g)\omega(g^{-1})\,dg. \tag{2.11}$$

The relation between the Fourier transform and the Satake transform is given in [Ma71, p. 47]. The Plancherel measure $d\mu(\omega)$ is the positive measure on $\Omega_K^+(G)$ such that, for all $f \in \mathscr{H}(G,K)$,

$$f^\vee(\omega) \in L^2(\Omega_K^+(G), d\mu) \tag{2.12}$$

and

$$\int_{G(F)} |f(g)|^2\,dg = \int_{\Omega_K^+(G)} |f^\vee(\omega)|^2\,d\mu(\omega). \tag{2.13}$$

By a theorem of Godement, such a measure exists and is unique.

Lemma 2.2 (Macdonald [Ma71, Thm. 5.1.2])**.** *The support of the Plancherel measure is the image of the compact torus $\widehat{T}^u$ of (2.6) (modulo W). Let $d\chi$ be the Haar probability measure on $\widehat{T}^u \cong i\mathfrak{a}_0^*/L^\vee$. The Plancherel measure of $G(F)$ with respect to K is*

$$d\mu(\chi) = \frac{Q_G}{|W|}\,|c(\chi)|^{-2}\,d\chi.$$

Corollary 2.3. *For all $f \in \mathscr{H}(G,K)$,*

$$f(g) = \int_{\Omega_K^+(G)} f^\vee(\omega)\overline{\omega(g^{-1})}\,d\mu(\omega) = \int_{\widehat{T}^u} f^\vee(\chi)\Gamma_\chi(g)\,d\mu(\chi).$$

For all $f_1, f_2 \in \mathscr{H}(G,K)$,

$$\int_{G(F)} f_1(g)\overline{f_2(g)}\,dg = \int_{\Omega_K^+(G)} f_1^\vee(\omega)\overline{f_2^\vee(\omega)}\,d\mu(\omega) = \int_{\widehat{T}^u} f_1^\vee(\chi)\overline{f_2^\vee(\chi)}\,d\mu(\chi).$$

Proof. First, $d\mu$ is supported on $\widehat{T}^u/W$ [Ma71, Thm. 5.1.2], every ω is a Γ_χ [Ma71, Thm. 3.3.12], $\Gamma_\chi(g^{-1}) = \Gamma_{\chi^{-1}}(g)$ [Ma71, Prop. 3.3.2], and $\overline{\Gamma_{\chi^{-1}}(g)} = \Gamma_\chi(g)$ for unitary χ [Ca79, p. 150]. So in each statement the second equality follows from the first. The second statement follows from (2.13) by polarization: replace f by f_1+f_2 and by f_1+if_2 in (2.13), expand, and combine; the first gives the real part and the second the imaginary part of the cross term. The first statement follows from the second by taking f_2 to be the characteristic function of a double coset KgK and using the definition (2.11). □

Example 2.3 (Macdonald's formulas in rank one). For $PGL(2)$, $Q_G = 1+q^{-1} =: Q$ and $\delta_B^{1/2}(a_n) = q^{-n/2}$, and Lemma 2.1 reads

$$\Gamma_\chi(a_n) = Q^{-1}q^{-n/2}\big(c_\alpha(\chi^{-1})z^n + c_\alpha(\chi)z^{-n}\big) \qquad (n \geq 0).$$

The normalization can be checked at $n=0$: $c_\alpha(\chi)+c_\alpha(\chi^{-1}) = \frac{1-q^{-1}y}{1-y} - \frac{y(1-q^{-1}y^{-1})}{1-y} = 1+q^{-1}$, so $\Gamma_\chi(1)=1$. Lemma 2.2 gives the Plancherel density

$$\rho(\chi) = \frac{Q}{2}\,|c_\alpha(\chi)|^{-2} = \frac{Q}{2}\cdot\frac{|1-y|^2}{|1-q^{-1}y|^2} \qquad (y=z^2),$$

which has double zeros at the two non-regular unitary characters $z=\pm 1$. The double cosets have $\mathrm{vol}(Ka_nK) = (q+1)q^{n-1}$ for $n\geq 1$. For $SL(2)$ everything is the same with $q^{-n/2}$ replaced by q^{-n}, y by z and $(q+1)q^{n-1}$ by $(q+1)q^{2n-1}$. In particular ρ vanishes only at $z=1$, and $\rho(\chi_0) = \frac{Q}{2}\cdot\frac{4}{(1+q^{-1})^2} \neq 0$ at $z=-1$.

2.3A Fourier analysis

The spectral side of (1.2) is an integral over the compact torus $\widehat{T}^u$. This subsection collects the facts from Fourier analysis on $\widehat{T}^u$ and on the lattice $X_*(A)$ that are used in §§4–6. They are: Fourier series, the Satake transform viewed as a Fourier transform, exponential polynomials, and the behavior of oscillatory sums (Riemann–Lebesgue and the Dirichlet kernel). The last point decides whether the spectral measure has a singular part.

Fourier series on $\widehat{T}^u$. By (2.6), $\widehat{T}^u$ is the Pontryagin dual of the lattice $X_*(A)$. Each $\lambda \in X_*(A)$ gives a character $\chi \mapsto \chi(\lambda)$ of $\widehat{T}^u$, and these form an orthonormal basis of $L^2(\widehat{T}^u, d\chi)$. For $g \in L^1(\widehat{T}^u)$ the Fourier coefficients are $\hat{g}(\lambda) = \int_{\widehat{T}^u} g(\chi)\chi(\lambda)^{-1}\,d\chi$. We use three standard facts.

(F1) The finite Fourier series are exactly the restrictions to $\widehat{T}^u$ of the Laurent polynomials, i.e. of the regular functions on the complex torus $\widehat{T}$.

(F2) (Riemann–Lebesgue.) If $g \in L^1$, then $\hat{g}(\lambda) \to 0$ as $\lambda \to \infty$.

(F3) g is C^∞ if and only if $\hat{g}$ decreases rapidly: $|\hat{g}(\lambda)| \leq C_N(1+|\lambda|)^{-N}$ for every N.

Consequently, if g is smooth and μ is a linear map with $|\mu(T)| \to \infty$, then $\int g(\chi)\chi(\mu(T))\,d\chi = \hat{g}(-\mu(T))$ decays faster than any power of $|T|$. This is how oscillatory terms are disposed of in the proof of Theorem 4.5(b). If g is only in L^1, the decay can be arbitrarily slow. If g has a non-integrable singularity, the integral need not exist, and the limit can produce a singular measure (see below). This is why Hypothesis (R) asks for smoothness.

The Satake transform as a Fourier transform. By the Cartan decomposition, every $f \in \mathscr{H}(G,K)$ is a finite combination of the functions $f_\lambda = \mathbf{1}_{Ka_\lambda K}$, and by (2.9)

$$f_\lambda^\vee(\chi) = \mathrm{vol}(Ka_\lambda K)\,\Gamma_\chi(a_\lambda^{-1}).$$

By Macdonald's formula this is a Laurent polynomial in χ: the poles of the individual c-functions cancel in the sum over W. Satake's theorem [Sa63], [Ca79, §4] says that $f \mapsto f^\vee$ is an isomorphism of $\mathscr{H}(G,K)$ onto the W-invariant Laurent polynomials. By Lemma 2.2 and Corollary 2.3 it extends to an isometry of $L^2(K\backslash G(F)/K)$ onto $L^2(\widehat{T}^u/W, d\mu)$, with inverse $f(g) = \int f^\vee(\chi)\Gamma_\chi(g)\,d\mu(\chi)$. Thus both sides of (1.2) are linear functionals on W-invariant finite Fourier series. Whether the right side is given by a measure, and of what kind, is the question of §4.3.

Exponential polynomials. A function F on the lattice points of a cone in a lattice L is an *exponential polynomial* if $F(T) = \sum_\mu \psi_\mu(T)P_\mu(T)$, a finite sum with distinct characters ψ_μ of L and polynomials P_μ. In this paper $\psi_\mu(T) = \chi(\mu(T))$ for linear maps $\mu : L \to X_*(A)$, and χ is generic. The expansion is unique: if F vanishes on all lattice points of a translate of an open cone, then every P_μ vanishes. To see this, restrict to lines $T_0 + tT_1$ along which the $\psi_\mu(T_1)$ are distinct. In one variable, $\sum_j z_j^t p_j(t) = 0$ for all large t, with distinct z_j, forces $p_j = 0$, because $\sum_t F(t)x^t$ is a rational function whose poles at the z_j^{-1} have orders $1 + \deg p_j$. Uniqueness is what makes "the T-independent part" of the truncated kernel well defined (Proposition 4.2(c)). For example, Example 4.2 writes the truncated kernel of $PGL(2)$ as an exponential polynomial in t with the three characters 1, y^t and y^{-t} and constant coefficients. In Example 3.6 the trivial character carries a polynomial of degree one in t, with slope $1/\rho(\chi)$.

Oscillatory sums, Dirichlet kernels and atoms. Brion's theorem (Lemma 3.5A) turns sums over truncated cones into vertex terms. The basic one-variable case, on the circle $z = e^{i\theta}$ with $d\chi = d\theta/2\pi$, is

$$\sum_{m=0}^{M-1} z^m = \frac{1}{1-z} - \frac{z^M}{1-z}.$$

The first term does not depend on M, and the second oscillates, with a coefficient that has a simple pole at $z = 1$. The coefficient is not integrable, but the sum is a polynomial. As $M \to \infty$, in the sense of distributions on the circle,

$$\sum_{m=0}^{M-1} z^m \;\longrightarrow\; \tfrac{1}{2} + \tfrac{1}{2}\delta_1 + \tfrac{i}{2}\,\mathrm{p.v.}\cot(\theta/2), \tag{2.13A}$$

where δ_1 is the unit point mass at $z = 1$ for the probability measure $d\chi$. (Pair both sides with z^{-k}: the left side gives 1 for $0 \le k < M$ and 0 otherwise, and the right side gives $\mathbf{1}_{k\ge 0}$, using the Fourier series $\cot(\theta/2) = 2\sum_{k\ge 1}\sin k\theta$.) A one-sided geometric sum whose ratio passes through 1 therefore leaves half a point mass in the limit; this is the Sokhotski–Plemelj formula on the circle. A two-sided sum $\sum_{|m|<M} z^m$ tends to the full point mass δ_1. If the sum is multiplied by a smooth function b, the point mass is weighted by $b(1)$. If it is multiplied by a function that vanishes at $z = 1$, as ρ does on the root walls, no point mass survives.

Example 2.3A (the circle). For $PGL(2)$, $\widehat{T}^u = \{|z| = 1\}$, W acts by $z \mapsto z^{-1}$, and the W-invariant Laurent polynomials are $\mathbb{C}[z+z^{-1}]$. From Example 2.3,

$$f_1^\vee(\chi) = (q+1)\,Q^{-1}q^{-1/2}\,\frac{-yz+q^{-1}z+z^{-1}-q^{-1}yz^{-1}}{1-y} = q^{1/2}(z+z^{-1}),$$

the classical Hecke eigenvalue. In general $f_n^\vee$ is a W-invariant Laurent polynomial of degree n in z. For $SL(2)$ the Dirichlet kernel (2.13A) in the variable z^2, weighted by ρ, which does not vanish at $z=-1$, produces the atom $\frac{1}{2}f^\vee(\chi_0)$ of §6.2 (Example 4.5).

2.4 The Jacquet functor

We recall normalized induction and the Jacquet functor, and the special case of the Bernstein–Zelevinsky geometric lemma that describes the Jacquet module of an unramified principal series along a standard parabolic (Lemma 2.4, Corollary 2.5). For regular χ the Jacquet module is a direct sum indexed by $[W_M\backslash W]$ (Remark 1). This decomposition is what makes the asymptotics of matrix coefficients in §3.1 a finite sum of exponentials.

For any parabolic $P = MN$ of G we have $G(F) = P(F)K$, and for each Levi M and parabolic P^M of M we have $M(F) = P^M(F)(K \cap M(F))$ [Ca79, p. 140]. In this case the notion of "compactly induced" representations [BZ77, §1.8] agrees with the usual notion of unitarily induced representations. Let

$$i_{G,M} : \mathrm{Alg}\,M \to \mathrm{Alg}\,G \tag{2.14}$$

denote normalized induction, in the notation of [BZ77] (so $i_{G,M}(\tau) = V_P(\tau)$ in Arthur's notation), and let

$$r_{M,G} : \mathrm{Alg}\,G \to \mathrm{Alg}\,M \tag{2.15}$$

denote the normalized Jacquet functor [BZ77, §2.3] (called the first Jacquet functor in [Ca79, §2.2]). We sometimes write $\pi_N = r_{M,G}(\pi)$.

Let W_M denote the Weyl group of M. The special case of the geometric lemma of Bernstein–Zelevinsky which we need is the following.

Lemma 2.4 (Bernstein–Zelevinsky [BZ77, §2.12])**.** *There is an enumeration $w_1, \dots, w_k$ of $[W_M\backslash W]$ such that, for each character χ of $M_0(F)$, the M-module $r_{M,G} \circ i_{G,A}(\chi)$ has a filtration*

$$r_{M,G} \circ i_{G,A}(\chi) = V_k \supset V_{k-1} \supset \cdots \supset V_1 \supset V_0 = \{0\} \qquad \textit{with} \qquad V_j/V_{j-1} \cong i_{M,A}(w_j\chi).$$

The subquotients need not be irreducible (take $M = G$). Let $V_{(s)}$ denote the semi-simplification of an M-module V.

Corollary 2.5. *For each character χ of $M_0(F)$, $\big(r_{M,G} \circ i_{G,A}(\chi)\big)_{(s)} \cong \bigoplus_{w \in [W_M\backslash W]} i_{M,A}(w\chi)_{(s)}$.*

Proof. Follows from Lemma 2.4 and the definition of the semi-simplification. □

Remark 1. If χ is regular, then $r_{M,G} \circ i_{G,A}(\chi) \cong \bigoplus_{w \in [W_M\backslash W]} i_{M,A}(w\chi)$, a direct sum of (possibly reducible) M-modules. Indeed the cuspidal supports $W_M \cdot w\chi$ of the summands are distinct for distinct cosets $W_M w$ (if $w'\chi = vw\chi$ with $v \in W_M$ then $w' = vw$ by regularity), and a finite-length M-module splits according to the generalized eigencharacters of the Bernstein center of M, which

separate cuspidal supports. Without regularity the Jacquet module need not be semisimple: for $M = G$ and $\chi = \delta_B^{1/2}$ on $SL(2)$, $i_{G,A}(\chi)$ is a non-split extension of the trivial representation by the Steinberg representation.

Example 2.4 (rank one). For $G = SL(2)$ and $M = A$, Lemma 2.4 says that $r_{A,G}(i_{G,A}(\chi))$ has a filtration with subquotients χ and $s\chi = \chi^{-1}$. If $\chi^2 \neq 1$ it is the direct sum $\chi \oplus \chi^{-1}$ (Remark 1). For $\chi = \chi_0$ the two subquotients coincide. The Jacquet module is still $\chi_0 \oplus \chi_0$, one copy from each of π^+ and π^- (Example 2.1A). For $\chi = \delta_B^{1/2}$ the extension is non-split.

2.5 (G,M)-families

Arthur's (G,M)-families are collections of functions indexed by parabolics that agree along common walls; the point of the definition is that certain alternating sums of them are smooth. We recall the definition, a p-adic variant for families on $\widehat{T}^u$, and Waldspurger's lemmas relating them.

These results are used in §4 only to explain where the regularity Hypothesis (R) comes from; see Remark 2026.12.

For a collection $\{c_P(\lambda) : P \in \mathscr{P}(M)\}$ of smooth functions on $i\mathfrak{a}_M^*$ we say that it is a (G,M)-**family** if for any adjacent $P, P' \in \mathscr{P}(M)$, $c_P(\lambda) = c_{P'}(\lambda)$ on the hyperplane spanned by the common wall of the chambers of P and P' [Ar81, §6]. For a collection $\{\tilde{c}_P(\chi) : P \in \mathscr{P}(A)\}$ of smooth functions on $\widehat{T}^u$ we say that it is a $(G,A)^\sim$-**family** if for any adjacent $P, P' \in \mathscr{P}(A)$ we have $\tilde{c}_P(\chi_\lambda) = \tilde{c}_{P'}(\chi_\lambda)$ on the set

$$\left\{\lambda \in i\mathfrak{a}_0^* : \ \langle \alpha^\vee, \lambda \rangle \in \frac{2\pi i}{\log q}\mathbb{Z}\right\} = \{\lambda : \ \chi_\lambda(a_\alpha) = 1\},$$

where α is one of the two roots separating P and P' [Wa89a, p. 26].

Let $Q \subset R$ be parabolic subgroups and let L_Q^R denote the lattice generated by the projections of $\{\alpha^\vee : \alpha \in \Delta_Q^R\}$. Let

$$\theta_Q^R(\lambda) = \mathrm{vol}(\mathfrak{a}_Q^R / L_Q^R)^{-1} \prod_{\alpha \in \Delta_Q^R} \lambda(\alpha^\vee), \qquad \lambda \in i\mathfrak{a}_Q^*, \tag{2.16}$$

and define $\tilde{\theta}_B = \tilde{\theta}$ by

$$\tilde{\theta}(\chi) := \prod_{\alpha \in \Delta} (1 - \chi(a_\alpha)). \tag{2.17}$$

Lemma 2.6 (Arthur [Ar81, Lemma 6.2])**.** *If $\{c_P(\lambda) : P \in \mathscr{P}(M)\}$ is a (G,M)-family then $c_M(\lambda) := \sum_{P \in \mathscr{P}(M)} c_P(\lambda)\theta_P(\lambda)^{-1}$ extends to a smooth function on $i\mathfrak{a}_M^*$.*

Lemma 2.7 (Waldspurger [Wa89a, §7])**.** *If $\{\tilde{c}_P(\chi) : P \in \mathscr{P}(A)\}$ is a $(G,A)^\sim$-family then*

$$\tilde{c}_A(\chi) := \sum_{P \in \mathscr{P}(A)} \tilde{c}_P(\chi)\tilde{\theta}_P(\chi)^{-1}$$

extends to a smooth function on $\widehat{T}^u$. Here $\tilde{\theta}_P(\chi) := \tilde{\theta}_B(w_P\chi)$, where $\tilde{\theta}_B$ is defined in (2.17) and $w_P \in W$ is the unique element with $P = w_P^{-1}Bw_P$.

The factor $\tilde{\theta}$ is the periodic analogue of θ:

Lemma 2.8 (compare Waldspurger [Wa89a, II.7.1])**.** *Assume (H), and assume that the coroot lattice $\mathbb{Z}\Delta^\vee$ is saturated in $X_*(A)$, i.e. $X_*(A)/\mathbb{Z}\Delta^\vee$ is torsion-free. (Under (H) this means $\mathbb{Z}\Delta^\vee = X_*(A)$; e.g. $G = SL(n)$, or G simply connected.) Then there exists $u_B \in C_c^\infty(i\mathfrak{a}_0^*)$ such that*

(a) for all $\lambda \in i\mathfrak{a}_0^*$, $\sum_{r\in L^\vee} u_B(\lambda+r)\,\theta_B(\lambda+r)^{-1} = \tilde{\theta}_B(\chi_\lambda)^{-1}$ (an identity of meromorphic functions);

(b) if $r \in L^\vee$ lies in the support of u_B then $r = 0$.

Proof. Let $U = \{\lambda \in i\mathfrak{a}_0^* : |\langle\alpha^\vee,\lambda\rangle| < 2\pi/\log q \ \ \forall \alpha \in \Delta\}$, a bounded open set since $\Delta^\vee$ spans $\mathfrak{a}_0$. The zeros of $\tilde{\theta}_B(\chi_\lambda) = \prod_\alpha (1 - q^{-\langle\alpha^\vee,\lambda\rangle})$ are the hyperplanes $\langle\alpha^\vee,\lambda\rangle \in \frac{2\pi i}{\log q}\mathbb{Z}$. In U only the hyperplanes $\langle\alpha^\vee,\lambda\rangle = 0$ occur, and these are exactly the zeros of θ_B, each simple. So $\theta_B(\lambda)/\tilde{\theta}_B(\chi_\lambda)$ is smooth and nowhere zero on U. By saturation, $r \mapsto (\langle\alpha^\vee, r\rangle)_{\alpha\in\Delta}$ maps $X^*(A)$ onto $\mathbb{Z}^\Delta$, so every λ has a translate $\lambda + r$, $r \in L^\vee$, with $|\langle\alpha^\vee,\lambda+r\rangle| \le \pi/\log q$ for all α. Hence $U + L^\vee = i\mathfrak{a}_0^*$, and there is $\eta \in C_c^\infty(U)$, $\eta \ge 0$, with $\sum_{r\in L^\vee}\eta(\lambda+r) = 1$ (normalize any $\eta_0 \in C_c^\infty(U)$ that is positive on the closed set where all $|\langle\alpha^\vee,\lambda\rangle| \le \pi/\log q$ by its periodization). Put $u_B = \eta\,\theta_B/\tilde{\theta}_B(\chi_\cdot)$. Then u_B is smooth, (a) holds because $\tilde{\theta}_B(\chi_{\lambda+r}) = \tilde{\theta}_B(\chi_\lambda)$, and (b) holds because $r \in U \cap L^\vee$ forces $\langle\alpha^\vee, r\rangle = 0$ for all α. □

Remark 2026.4 (Lemma 2.8). Without saturation, Lemma 2.8 is false. For $G = PGL(2)$, $X_*(A) = \mathbb{Z}\varpi^\vee$ and $\alpha^\vee = 2\varpi^\vee$, so with $z = \chi_\lambda(\varpi^\vee(\pi))$ we have $\tilde{\theta}_B(\chi_\lambda) = 1 - z^2$. This has a pole at $z = -1$, i.e. at $\langle\varpi^\vee,\lambda\rangle = \pi i/\log q$. But $\langle\alpha^\vee,\lambda+r\rangle = \frac{2\pi i}{\log q}(1+2k) \neq 0$ there for every $r \in L^\vee$, so the left side of (a) is smooth near that point for any smooth u_B. The extra singular hyperplanes of $\tilde{\theta}_B$ are not $L^\vee$-translates of those of θ_B. A general version would have to replace θ_B by a function adapted to the full periodic hyperplane arrangement (or pass to the cover defined by $\mathbb{Z}\Delta^\vee$); we do not pursue this. (A compactly supported η with $\sum_r \eta(\lambda+r) = 1$ cannot be concentrated near 0, so there is no easy fix.) This phenomenon (the extra hyperplane $z = -1$) is the same one that produces the atom for $SL(2)$ in §6.2, there in the frequencies $w\chi/w'\chi$ rather than in χ itself.

Lemma 2.9 (Waldspurger [Wa89a, §7])**.** *Assume the hypotheses of Lemma 2.8, and let $\{\tilde{c}_P(\chi) : P \in \mathscr{P}(A)\}$ be a $(G,A)^\sim$-family. For $P \in \mathscr{P}(A)$ let $c_P(\lambda) = \tilde{c}_P(\chi_\lambda)u_P(\lambda)$, where $u_P(\lambda) = u_B(w_P\lambda)$ and u_B is as in Lemma 2.8. Then $\{c_P\}_P$ is a (G,A)-family, and*

$$\tilde{c}_A(\chi_\lambda) = \sum_{r\in L^\vee} c_A(\lambda+r), \qquad c_A(\lambda) = \sum_P c_P(\lambda)\theta_P(\lambda)^{-1}.$$

Remark 2. Lemma 2.9 follows from Lemmas 2.6–2.8. These results relate the $(G,A)^\sim$-families occurring implicitly in §4 to Arthur's (G,A)-families [Ar81].

Remark 2026.5 (§2.5). Lemma 2.6 is standard, and Lemma 2.8 is proved above under the saturation hypothesis. The precise form of Lemmas 2.7 and 2.9 has not been re-checked against [Wa89a, §7] (where $G = GL(N)$, for which the coroot lattice is saturated). None of Lemmas 2.6–2.9 is used in any proof below.

Example 2.5 (rank one). Let $G = SL(2)$ or $PGL(2)$ and $M = A$. Then $\mathscr{P}(A) = \{B, \bar{B}\}$, the two parabolics are adjacent, and their common wall is $\{0\}$. Write $\xi = \lambda(\alpha^\vee)$ for $\lambda \in i\mathfrak{a}_0^*$ and $v =$

$\mathrm{vol}(\mathfrak{a}_0/\mathbb{Z}\alpha^\vee)$. Since $\Delta_{\bar{B}} = \{-\alpha\}$, (2.16) gives $\theta_B(\lambda) = v^{-1}\xi$ and $\theta_{\bar{B}}(\lambda) = -v^{-1}\xi$. A (G,A)-family is a pair of smooth functions $c_B, c_{\bar{B}}$ with $c_B(0) = c_{\bar{B}}(0)$, and

$$c_A(\lambda) = v\,\frac{c_B(\lambda) - c_{\bar{B}}(\lambda)}{\xi}.$$

This is smooth because the numerator vanishes at $\xi = 0$, as Lemma 2.6 asserts. For example, $c_B = 1$ and $c_{\bar{B}}(\lambda) = e^{-m\xi}$ with $m \geq 0$ give $c_A(\lambda) = v(1 - e^{-m\xi})/\xi$, whose value at $\lambda = 0$ is vm, the length of the segment from 0 to $m\alpha^\vee$.

The periodic version is similar. Put $y = \chi(a_\alpha)$. By (2.17), $\tilde{\theta}_B(\chi) = 1 - y$, and $\tilde{\theta}_{\bar{B}}(\chi) = \tilde{\theta}_B(w_\alpha\chi) = 1 - y^{-1}$. A $(G,A)^\sim$-family is a pair of smooth functions $\tilde{c}_B, \tilde{c}_{\bar{B}}$ on $\widehat{T}^u$ that agree where $y = 1$. For example, $\tilde{c}_B = 1$ and $\tilde{c}_{\bar{B}}(\chi) = y^m$ with $m \geq 0$ give

$$\tilde{c}_A(\chi) = \frac{1}{1-y} + \frac{y^m}{1-y^{-1}} = \frac{1-y^{m+1}}{1-y} = \sum_{k=0}^{m} y^k,$$

a Laurent polynomial, hence smooth, as Lemma 2.7 asserts. It is the sum of χ over the $m+1$ lattice points $k\alpha^\vee$, $0 \leq k \leq m$, and its value at $y = 1$ is $m+1$. The real and periodic examples differ as the length of a segment differs from the number of lattice points in it. Compare the vertex decomposition in rank one (Example 3.5).

3 Inner products of some matrix coefficients

This section contains the analytic heart of the argument. Its goal is Proposition 3.6, which computes the truncated integral J^T of a product of two matrix coefficients of unramified principal series over the finite union $G(T)$ of double cosets $Ka_\lambda K$ "below T".

The computation has two ingredients. §3.1 uses Casselman's theory of Jacquet modules to write each matrix coefficient, on suitable regions of the negative chamber, as a finite sum of characters of λ (Lemmas 3.1–3.4); the precise form of this input that we need is stated as (C). §3.2 sums these characters over the truncated cone of dominant coweights using Brion's theorem on lattice-point generating functions (Lemmas 3.5A–3.5C). The result is a sum of vertex terms, one for each standard parabolic subgroup, of which exactly one is independent of T.

3.1 Matrix coefficients

We fix the level n of the vectors involved and describe matrix coefficients of the unramified principal series on the regions $A_P^-(\varepsilon)$, the parts of the negative chamber that are far from the walls not in P. The key statement is the input (C) below and its consequence, Lemma 3.4: on each such region, the product of two matrix coefficients is a finite sum of characters of λ, with coefficients given by pairings on Jacquet modules. In rank one, (C) is Casselman's theorem; in higher rank it is the one representation-theoretic input that has not been checked in full (Remark 2026.6).

Assume as before that G satisfies (H), and use the measures of §2.0. Let χ be an unramified regular character of $M_0(F)$ and $V_B(\chi) = i_{G,A}(\chi)$. For $n \geq 1$ let $K_n \subset K$ be the n-th principal congruence subgroup; it is normal in K and has an Iwahori factorization with respect to every standard parabolic. A vector has *level* n if it is fixed by K_n. Write $\pi = \pi_\chi = i_{G,A}(\chi)$; its restriction to K does not depend on χ (compact picture).

Casselman's asymptotics; the input (C). Let $P = MN$ be standard and $\bar{P} = M\bar{N}$ its opposite. By Casselman [Ca95, §4] there is a canonical M-invariant pairing $\langle \cdot, \cdot \rangle_N$ between $r_{M,G}(\pi_\chi)$ (taken with respect to P) and $r_{M,\bar{P}}(\pi_{\chi^{-1}})$ (taken with respect to $\bar{P}$). We use the following statement, which we call (C):

(C) *For every n there is $\varepsilon = \varepsilon(n) \in (0,1]$, independent of the unramified character χ, such that for every standard P and all $f \in V_B(\chi)$, $f' \in V_B(\chi^{-1})$ of level n*

$$\langle \pi_\chi(a) f, f' \rangle_G = \delta_P^{1/2}(a) \, \langle r_{M,G}(\pi_\chi)(a) f_N, f'_{\bar{N}} \rangle_N \qquad \text{for all } a \in A_P^-(\varepsilon). \tag{3.1}$$

For $P = B$ the region is $A^-(\varepsilon)$, and (C) is Casselman's theorem [Ca95, Prop. 4.3.3], [Ca80]; the independence of ε from χ follows from [Ca79, §3] (the restriction of π_χ to K does not depend on χ). In rank one only $P = B$ and $P = G$ occur, so (C) holds. For intermediate P, (C) asks for the asymptotic on all of $A_P^-(\varepsilon)$, where a is deep in the N-directions only; Casselman's theorem as usually stated gives it for a in the split center A_M of M.

Remark 2026.6 ((3.1)). In (3.1) the second space is the Jacquet module for the *opposite* parabolic, the factor $\delta_P^{1/2}(a)$ is needed, and ε depends on the level of f, f'. (C) is an essential input to Propositions 3.6 and 4.2 in rank ≥ 2, not an editorial detail. Casselman's proof (Jacquet's lemma for a contracting element, with a level subgroup having an Iwahori factorization) appears to use only that $a \in A^-$ contracts N_0 and contracts $N \cap K_n$ by a factor $\leq \varepsilon$, which holds on $A_P^-(\varepsilon)$; but we have not written out this extension, and we have not found it stated in this form in the literature. A reduction to the A_M case is not immediate: writing $a = a_e a_\nu$ with $a_\nu \in A_M$, the level of $\pi(a_e)\Phi$ grows with the size of a_e, so the constants become circular. Accordingly, Propositions 3.6 and 4.2 are stated as conditional on (C) in rank ≥ 2.

For $f \in V_B(w\chi)$ let $R_M f = \delta_P^{-1/2} \cdot f|_{M(F)} \in i_{M,A}(w\chi)$; the map R_M factors through $r_{M,G}$ and is M-equivariant [Wa89a, §I]. For $w \in [W_M \backslash W]$ the maps $R_M \circ T_w$ give the decomposition of Remark 1,

$$r_{M,G}(\pi_\chi) \xrightarrow{\sim} \bigoplus_{w \in [W_M \backslash W]} i_{M,A}(w\chi), \qquad f_N \longmapsto (f_w)_w, \quad f_w := R_M T_w f. \tag{3.2}$$

Lemma 3.1. *For χ unramified regular, $f \in V_B(\chi)$ and $a \in A(F)$, $\big(\pi_\chi(a) f\big)_w = i_{M,A}(w\chi)(a) f_w$ for every $w \in [W_M \backslash W]$.*

Proof. R_M and T_w are intertwining maps, so $R_M T_w \pi_\chi(a) f = i_{M,A}(w\chi)(a) R_M T_w f$. □

Lemma 3.2. *Let χ be regular. In terms of the decomposition (3.2) of $r_{M,G}(\pi_\chi)$ and the corresponding decomposition of $r_{M,\bar{P}}(\pi_{\chi^{-1}})$ into summands isomorphic to $i_{M,A}((w\chi)^{-1})$, $w \in [W_M \backslash W]$, the pairing $\langle \cdot, \cdot \rangle_N$ is block-diagonal, and on the w-th block it is a scalar multiple of the standard pairing*

$$\langle u, u' \rangle_M := \int_{M(F) \cap K} u(k) u'(k) \, dk$$

between $i_{M,A}(w\chi)$ and $i_{M,A}((w\chi)^{-1})$.

Remark 3. For our choice of Haar measures, if u, u' are $(M \cap K)$-fixed vectors in $i_{M,A}(w\chi)$ and $i_{M,A}((w\chi)^{-1})$, then $\langle u, u' \rangle_M = u(1) u'(1)$.

Proof of Lemma 3.2. The pairing $\langle\cdot,\cdot\rangle_N$ is M-invariant, so its restriction to the (w,w') block lies in $\mathrm{Hom}_M\big(i_{M,A}(w\chi)\otimes i_{M,A}((w'\chi)^{-1}),\mathbb{C}\big)=\mathrm{Hom}_M\big(i_{M,A}(w\chi),i_{M,A}(w'\chi)\big)$. A non-zero element forces a common cuspidal support, $W_Mw\chi=W_Mw'\chi$, hence $W_Mw=W_Mw'$ by regularity. For $w=w'$, Bruhat theory gives $\dim\mathrm{End}_M(i_{M,A}(w\chi))\le\#\{v\in W_M: vw\chi=w\chi\}=1$. □

The next result generalizes Macdonald's formula. An analogous result is [Wa89a, Lemma I.3.1]. Here we use Macdonald's formula for G and for M to evaluate the constants.

Lemma 3.3. *Let χ be unramified regular, $f'\in V_B(\chi^{-1})$ of level n, and $\varepsilon=\varepsilon(n)$. There are M-maps $\bar R_{M,w}: V_B(\chi^{-1})\to i_{M,A}((w\chi)^{-1})$, $w\in[W_M\backslash W]$, factoring through $r_{M,\bar P}$, such that for $a\in A_P^-(\varepsilon)$*

$$\langle\pi_\chi(a)\Phi_{K,\chi},f'\rangle_G=\delta_P^{1/2}(a)\sum_{w\in[W_M\backslash W]}c_w(\chi)\,\langle i_{M,A}(w\chi)(a)\Phi^M_{w\chi},\,\bar R_{M,w}f'\rangle_M,$$

where $\Phi^M_{w\chi}$ is the normalized $(K\cap M)$-fixed vector of $i_{M,A}(w\chi)$. For $f'=\Phi_{K,\chi^{-1}}$ one has $\bar R_{M,w}\Phi_{K,\chi^{-1}}=c(w,M,\chi)\,c_w(\chi^{-1})\,\Phi^M_{(w\chi)^{-1}}$ with

$$c(w,M,\chi)=Q_G^{-1}Q_M\,\frac{\prod_{\alpha\in\Sigma^+\smallsetminus\Sigma_M^+}c_\alpha\big((w\chi)^{-1}\big)}{c_w(\chi)\,c_w(\chi^{-1})},$$

and then the identity holds for all $a\in A^-$. In particular, for $M=M_0$, $c(w,M_0,\chi)=Q_G^{-1}c((w\chi)^{-1})/\big(c_w(\chi)c_w(\chi^{-1})\big)$, as in [Ma71].

Remark 4. Since K_n is normal in K, $\pi(k)f'$ has level n whenever f' does, so one $\varepsilon=\varepsilon(n)$ serves for all $\pi(k)f'$, $k\in K$; and since the vectors $u(w,k)$ of Lemma 3.4 have level $\le n$ in M, the same ε serves for the analogous identity on Levi subgroups.

Proof of Lemma 3.3. The first identity is (3.1) combined with Lemmas 3.1 and 3.2, with $(\Phi_{K,\chi})_w=R_MT_w\Phi_{K,\chi}=c_w(\chi)\Phi^M_{w\chi}$ by (2.8); the scalars of Lemma 3.2 are absorbed into $\bar R_{M,w}$.

For the constants take $f'=\Phi_{K,\chi^{-1}}$. Then the left side is $\Gamma_\chi(a)$, and by $K\cap M$-invariance $\bar R_{M,w}\Phi_{K,\chi^{-1}}$ is a multiple of $\Phi^M_{(w\chi)^{-1}}$; write the multiple as $c(w,M,\chi)c_w(\chi^{-1})$. Since $\langle i_{M,A}(w\chi)(a)\Phi^M_{w\chi},\Phi^M_{(w\chi)^{-1}}\rangle_M=\Gamma^M_{w\chi}(a)$, Macdonald's formula for M (Lemma 2.1 for M, valid on $A^-\subset A_M^-$) gives, with $\delta_P^{1/2}\delta_{B\cap M}^{1/2}=\delta_B^{1/2}$,

$$\mathrm{RHS}=Q_M^{-1}\delta_B^{1/2}(a)\sum_{w\in[W_M\backslash W]}c(w,M,\chi)c_w(\chi)c_w(\chi^{-1})\sum_{v\in W_M}c_M\big((vw\chi)^{-1}\big)(vw\chi)(a),$$

while Macdonald's formula for G gives $\Gamma_\chi(a)=Q_G^{-1}\delta_B^{1/2}(a)\sum_{\sigma\in W}c((\sigma\chi)^{-1})(\sigma\chi)(a)$. Every $\sigma\in W$ is uniquely $\sigma=vw$ with $v\in W_M$, $w\in[W_M\backslash W]$, and the characters $a\mapsto(\sigma\chi)(a)$ are linearly independent on $A_P^-(\varepsilon)$ for regular χ. Comparing coefficients,

$$c(w,M,\chi)=Q_G^{-1}Q_M\frac{c((vw\chi)^{-1})}{c_M((vw\chi)^{-1})\,c_w(\chi)c_w(\chi^{-1})}\qquad(v\in W_M).$$

The ratio $c(\eta)/c_M(\eta)=\prod_{\alpha\in\Sigma^+\smallsetminus\Sigma_M^+}c_\alpha(\eta)$ is unchanged when η is replaced by $v\eta$, because W_M permutes $\Sigma^+\smallsetminus\Sigma_M^+$; this gives the formula. Since both Macdonald formulas hold on all of A^-, so does the identity for spherical vectors. □

Remark 2026.7 (Lemma 3.3). For general f' the functional $f' \mapsto \langle \pi(a)\Phi, f' \rangle$ factors through the $\bar{P}$-Jacquet module, not the P-Jacquet module, so we have stated the lemma with $\bar{R}_{M,w}$. Identifying $\bar{R}_{M,w}$ with "$R_M T_w$" requires Waldspurger's opposite-parabolic conventions. Only the spherical case is used explicitly below.

Lemma 3.4. *Let χ, χ' be regular unramified characters, $u \in V_B(\chi^{-1})$ and $u' \in V_B(\chi'^{-1})$ of level n, and $a \in A_P^-(\varepsilon(n))$. Then*

$$\int_{KaK} \langle \pi_\chi(g)\Phi_{K,\chi}, u\rangle_G \langle \pi_{\chi'}(g)\Phi_{K,\chi'}, u'\rangle_G \, dg = \operatorname{meas}(KaK)\, \delta_P(a) \sum_{w,w' \in [W_M \backslash W]} c_w(\chi) c_{w'}(\chi')$$
$$\times \int_K \langle i_{M,A}(w\chi)(a)\Phi^M_{w\chi}, u(w,k)\rangle_M \langle i_{M,A}(w'\chi')(a)\Phi^M_{w'\chi'}, u'(w',k)\rangle_M \, dk,$$

where $u(w,k) = \bar{R}_{M,w}(\pi(k)u)$ and $u'(w',k) = \bar{R}_{M,w'}(\pi(k)u')$ (with $\bar{R}$ formed for χ, resp. χ').

Remark 5. If κ is an unramified character of $G(F)$ then κ is constant on KaK, so the integral above with an extra factor $\kappa(g)$ equals $\kappa(a)$ times the integral without it. This is used for Proposition 3.7.

Proof of Lemma 3.4. The push-forward of Haar measure on $K \times K$ under $(k_1, k_2) \mapsto k_1 a k_2$ is $\operatorname{meas}(KaK)^{-1}$ times Haar measure on KaK. Since $\Phi_{K,\chi}$ is K-fixed, $\langle \pi_\chi(k_1 a k_2)\Phi_{K,\chi}, u\rangle_G = \langle \pi_\chi(a)\Phi_{K,\chi}, \pi(k_1^{-1})u\rangle_G$. Substituting $k = k_1^{-1}$, the left side is

$$\operatorname{meas}(KaK) \int_K \langle \pi_\chi(a)\Phi_{K,\chi}, \pi(k)u\rangle \langle \pi_{\chi'}(a)\Phi_{K,\chi'}, \pi(k)u'\rangle \, dk$$

Now apply Lemma 3.3 to each factor, which is legitimate by Remark 4, and multiply out. □

Example 3.1 (rank one). For $G = PGL(2)$ there are two parabolics. For $P = G$, the region $A_G^-(\varepsilon)$ is all of A^- and Lemma 3.3 is a tautology. For $P = B$, the region $A_B^-(\varepsilon)$ is $\{a_m : m \geq m_0\}$ with m_0 depending on the level n, and Lemma 3.3 says

$$\langle \pi_\chi(a_m)\Phi_{K,\chi}, f'\rangle_G = q^{-m/2}\big(C_1(f')z^m + C_s(f')z^{-m}\big) \qquad (m \geq m_0),$$

with $C_w(f') = c_w(\chi)\langle \Phi^A_{w\chi}, \bar{R}_{A,w} f'\rangle$. This has the same two-term shape as Macdonald's formula (Example 2.3), but the coefficients depend on f' through its Jacquet module, and the formula holds only for m large compared with the level. For $f' = \Phi_{K,\chi^{-1}}$ it is Macdonald's formula and holds for all $m \geq 0$. In rank one, (C) is just this statement, which is Casselman's theorem.

3.2 Truncated inner products of matrix coefficients

Here we sum the expansions of §3.1 over the truncated set $\mathscr{A}_A(T)$ of dominant coweights. After defining J^T in (3.4), we prove the combinatorial results needed: Brion's theorem in a form that allows half-open polytopes and characters trivial on some edges (Lemma 3.5B), a positivity lemma for root systems (Lemma 3.5C), and the resulting vertex decomposition of exponential sums over $\mathscr{A}_A(T)$ (Lemma 3.5A). Proposition 3.6 then writes J^T as a sum over standard parabolics Q of terms $\psi(T_Q)R_Q$, where only the term for $Q = G$ is independent of T. Proposition 3.7 records the variant with a character κ, used in §5.

Let $T \in X_*(A)$ with $d(T) := \min_{\alpha\in\Delta}\langle T,\alpha\rangle > 0$. The set

$$\mathscr{A}_A(T) = \big\{\lambda \in X_*(A)/X_*(A_G) :\ \langle\lambda,\alpha\rangle \geq 0,\ \langle\lambda - T,\varpi_\alpha\rangle < 0\ \ \forall\alpha\in\Delta\big\} \tag{3.3}$$

is finite. Using (2.5), define

$$J^T(\chi,\chi',u,u') = \sum_{\lambda\in\mathscr{A}_A(T)} \int_{Ka_\lambda K} \langle\pi_\chi(g)\Phi_{K,\chi},u\rangle_G\langle\pi_{\chi'}(g)\Phi_{K,\chi'},u'\rangle_G\,dg, \tag{3.4}$$

and put $G(T) = \bigcup_{\lambda\in\mathscr{A}_A(T)} Ka_\lambda K$ as in (4.2).

Remark 2026.8 (the complement of the truncated region). One cannot choose T so that $\nu_A^{-1}(\mathscr{A}_A(T)^c) \subset A^-(\varepsilon)$, i.e. so that everything outside the truncated region is deep in every simple direction: in $SL(3)$, $\lambda_n = n(2\alpha_1^\vee + \alpha_2^\vee)$ lies outside $\mathscr{A}_A(T)$ for large n while $\langle\lambda_n,\alpha_2\rangle = 0$. The proof of Proposition 3.6 below avoids the claim by decomposing the dominant cone into regions on each of which some $A_P^-(\varepsilon)$-asymptotic applies.

The vertex decomposition. For a standard parabolic Q let T_Q denote the projection of T to $\mathfrak{a}_Q$. Let

$$L_0 = \{T \in X_*(A) :\ T_Q \in X_*(A) \text{ for every standard } Q\},$$

a sublattice of finite index in $X_*(A)$. If G is semisimple, $T_G = 0$ and $T_B = T$. For example, in rank one the only standard parabolics are B and G, so $L_0 = X_*(A)$.

Two general lemmas come first: a form of Brion's theorem that covers half-open polytopes and degenerate characters, and a positivity statement for root systems.

Lemma 3.5B (Brion's theorem, half-open and degenerate forms)**.** *Let Λ be a lattice in a real vector space V of dimension d, let $\ell_1,\dots,\ell_m$ be Λ-rational linear forms on V, and for $b\in\mathbb{R}^m$ let*

$$\Pi(b) = \{x\in V :\ \ell_i(x) \leq b_i\ (i\in I_\leq),\ \ \ell_i(x) < b_i\ (i\in I_<)\},\qquad I_\leq \sqcup I_< = \{1,\dots,m\}.$$

Let $\mathscr{C}\subset\mathbb{R}^m$ be an open set of parameters on which the closure $\bar\Pi(b)$ is a simple polytope of constant combinatorial type, with vertices $v(b)$ depending linearly on b, and let $B_0\subset\mathscr{C}$ be a set of parameters for which all $v(b)\in\Lambda$. For a vertex v let $C_v = \{x :\ \ell_i(x)\leq 0\ (i\in I_\leq \text{ active at } v),\ \ell_i(x) < 0\ (i\in I_< \text{ active at } v)\}$ be the half-open tangent cone; it does not depend on $b\in\mathscr{C}$.

(i) For $\psi\in\mathrm{Hom}(\Lambda,\mathbb{C}^\times)$ outside the polar divisors below, and $b\in B_0$,

$$\sum_{x\in\Pi(b)\cap\Lambda}\psi(x) = \sum_v \psi(v(b))\,\sigma_{C_v}(\psi),$$

where $\sigma_C(\psi)$ is the rational function continuing $\sum_{x\in C\cap\Lambda}\psi(x)$. Its denominator divides $\prod_e(1-\psi(\mu_e))$, the product over the d edges e of C with primitive vectors $\mu_e\in\Lambda$.

(ii) For every ψ (including points of the polar divisors) and $b\in B_0$, $\sum_{x\in\Pi(b)\cap\Lambda}\psi(x) = \sum_v\psi(v(b))\,p_v(b;\psi)$, where $p_v(\cdot;\psi)$ is a polynomial of degree $\leq d$. If ψ runs over an irreducible subvariety of $\mathrm{Hom}(\Lambda,\mathbb{C}^\times)$ on which the set $\{e :\ \psi(\mu_e)=1\}$ is constant, the coefficients of p_v are rational functions on that subvariety.

Proof. (i) For closed rational polyhedra this is Brion's theorem [Br88], in the form: the map $P \mapsto \sigma_P$ is a valuation on the algebra generated by indicator functions of rational polyhedra, vanishing on polyhedra that contain lines, and $[P] \equiv \sum_v [v + \mathrm{tcone}(P,v)]$ modulo polyhedra containing lines [Ba08]. For the half-open $\Pi(b)$, inclusion–exclusion over the strict constraints gives $\mathbf{1}_{\Pi(b)} = \sum_{J \subseteq I_<} (-1)^{|J|} \mathbf{1}_{F_J}$ with $F_J = \bar{\Pi}(b) \cap \{\ell_i = b_i,\ i \in J\}$, a closed face of $\bar{\Pi}(b)$ (possibly empty). Apply Brion's theorem to each F_J. The vertices of F_J are the vertices v of $\bar{\Pi}(b)$ at which all $i \in J$ are active, and $\mathrm{tcone}(F_J, v) = \mathrm{tcone}(\bar{\Pi}(b), v) \cap \{\ell_i = 0,\ i \in J\}$. Regrouping by v, the same inclusion–exclusion applied to the cone $\mathrm{tcone}(\bar{\Pi}(b), v)$ gives $\sum_J (-1)^{|J|} [\mathrm{tcone}(F_J, v)] = [C_v]$, the sum running over subsets J of the strict constraints active at v. Since $\bar{\Pi}(b)$ is simple, each C_v is a half-open simplicial cone, whose generating function has the stated denominator. Translating by $v(b) \in \Lambda$ multiplies by $\psi(v(b))$.

(ii) Choose $\xi \in \mathrm{Hom}(\Lambda, \mathbb{R})$ with $\xi(\mu_e) \neq 0$ for all edge vectors μ_e of all the C_v, and put $\psi_\varepsilon = \psi e^{\varepsilon \xi}$. For small $\varepsilon \neq 0$, ψ_ε avoids the polar divisors, so (i) applies. The left side is an entire function of ε (a finite sum). Each vertex term $\psi(v(b)) e^{\varepsilon \xi(v(b))} \sigma_{C_v}(\psi_\varepsilon)$ is meromorphic at $\varepsilon = 0$ with a pole of order $\leq d$. Expanding $e^{\varepsilon \xi(v(b))} = \sum_k \varepsilon^k \xi(v(b))^k / k!$, the coefficient of ε^0 is $\psi(v(b))$ times a polynomial of degree $\leq d$ in $\xi(v(b))$, hence in b. The sum of these coefficients equals the left side at $\varepsilon = 0$. The Laurent coefficients of $\sigma_{C_v}(\psi_\varepsilon)$ at $\varepsilon = 0$ are rational in ψ on any subvariety along which the set of vanishing factors $1 - \psi(\mu_e)$ is constant. □

Lemma 3.5C (positivity)**.** *Let $P = MN$ be a standard parabolic and let ϖ_γ^M and $\varpi_\gamma^{M\vee}$ ($\gamma \in \Delta^M$) be the fundamental weights and coweights of M, the latter in $\mathfrak{a}_0^P$.*

(a) The vectors $(\beta^\vee)_P$, $\beta \in \Delta \smallsetminus \Delta^M$, form the basis of $\mathfrak{a}_P$ dual to $\{\varpi_\beta|_{\mathfrak{a}_P}\}$. For $\alpha \in \Delta^M$ and $Y \in \mathfrak{a}_P$, $\varpi_\alpha(Y) = \sum_{\beta \notin \Delta^M} c_{\alpha\beta} \varpi_\beta(Y)$ with $c_{\alpha\beta} = \varpi_\alpha((\beta^\vee)_P) \geq 0$.

(b) If $X \in \mathfrak{a}_0$ is dominant, then $X^P = \sum_{\gamma \in \Delta^M} t_\gamma \gamma^\vee$ with $t_\gamma = \varpi_\gamma(X^P) \geq \kappa_M \min_{\delta \in \Delta^M} \langle X, \delta \rangle \geq 0$, where $\kappa_M > 0$ depends only on M. Moreover $\langle X_P, \beta \rangle \geq \langle X, \beta \rangle$ for $\beta \notin \Delta^M$, and $\langle X_P - X, \varpi_\alpha \rangle = -t_\alpha$ for $\alpha \in \Delta^M$.

Proof. (a) For $\beta \notin \Delta^M$, ϖ_β vanishes on $\mathfrak{a}_0^P$ (it pairs to 0 with $\Delta^{M\vee}$), so $\varpi_\beta((\beta'^\vee)_P) = \varpi_\beta(\beta'^\vee) = \delta_{\beta\beta'}$. This gives the dual basis and the expansion. Next, $(\beta^\vee)^P \in \mathfrak{a}_0^P$ is determined by $\langle (\beta^\vee)^P, \gamma \rangle = \langle \beta^\vee, \gamma \rangle$ for $\gamma \in \Delta^M$ (since γ vanishes on $\mathfrak{a}_P$), so $(\beta^\vee)^P = \sum_{\gamma \in \Delta^M} \langle \beta^\vee, \gamma \rangle \varpi_\gamma^{M\vee}$. As $\varpi_\alpha(\beta^\vee) = 0$ for $\alpha \neq \beta$,

$$c_{\alpha\beta} = -\varpi_\alpha((\beta^\vee)^P) = -\sum_{\gamma \in \Delta^M} \langle \beta^\vee, \gamma \rangle\, \varpi_\alpha(\varpi_\gamma^{M\vee}).$$

Here $\langle \beta^\vee, \gamma \rangle \leq 0$ (off-diagonal Cartan entries), and $\varpi_\alpha(\varpi_\gamma^{M\vee}) = \varpi_\alpha^M(\varpi_\gamma^{M\vee}) \geq 0$ is an entry of the inverse Cartan matrix of M (fundamental coweights are non-negative combinations of simple coroots). Hence $c_{\alpha\beta} \geq 0$.

(b) X^P is M-dominant, since $\langle X^P, \gamma \rangle = \langle X, \gamma \rangle \geq 0$. So

$$t_\gamma = \varpi_\gamma(X^P) = \sum_{\delta \in \Delta^M} \varpi_\gamma^M(\varpi_\delta^{M\vee})\, \langle X, \delta \rangle,$$

a combination with non-negative coefficients and positive diagonal coefficients; take $\kappa_M = \min_\gamma \varpi_\gamma^M(\varpi_\gamma^{M\vee})$. Then, for $\beta \notin \Delta^M$ and $\alpha \in \Delta^M$,

$$\langle X_P, \beta \rangle = \langle X, \beta \rangle - \sum_\gamma t_\gamma \langle \gamma^\vee, \beta \rangle \geq \langle X, \beta \rangle, \qquad \langle X_P - X, \varpi_\alpha \rangle = -\langle X^P, \varpi_\alpha \rangle = -t_\alpha.$$

□

Lemma 3.5A (vertex decomposition)**.** *Assume (H), so that $X_*(A_G)=0$. Let ψ be a character of $X_*(A)$, i.e. a point of $\widehat{T}$, and let $\mathsf{c}: X_*(A)^+ \to \mathbb{C}$ be a function that depends only on the set $\{\alpha\in\Delta : \langle\lambda,\alpha\rangle = 0\}$. For each standard parabolic Q there is a rational function $R_Q = R_Q^{\mathsf{c}}$ on $\widehat{T}$, independent of T, such that for all $T\in L_0$ with $d(T)>0$*

$$\sum_{\lambda\in\mathscr{A}_A(T)} \mathsf{c}(\lambda)\,\psi(\lambda) = \sum_{Q\supseteq B} \psi(T_Q)\,R_Q(\psi) \tag{3.5A}$$

as rational functions of ψ; for ψ on a polar divisor, $R_Q(\psi)$ is to be replaced by a polynomial $p_Q(T;\psi)$ in T as in Lemma 3.5B(ii). The denominator of R_Q is a product of factors $1-\psi(\mu)$, where $\mu\in X_*(A)\smallsetminus\{0\}$ runs over primitive vectors on the edges of the cone

$$C_Q = \{X\in\mathfrak{a}_0 :\ \langle X,\alpha\rangle\ge 0\ (\alpha\in\Delta^Q),\ \ \langle X,\varpi_\alpha\rangle\le 0\ (\alpha\in\Delta\smallsetminus\Delta^Q)\},$$

whose edges are spanned by the fundamental coweights of M_Q (in $\mathfrak{a}_0^Q$) and by the vectors $-(\alpha^\vee)_Q$, $\alpha\in\Delta\smallsetminus\Delta^Q$. The term for $Q=G$ is independent of T: $R_G(\psi)$ is the meromorphic continuation of $\sum_{\lambda\in X_*(A)^+}\mathsf{c}(\lambda)\psi(\lambda)$.

Proof. Let $\Pi_T = \{X\in\mathfrak{a}_0 : \langle X,\alpha\rangle\ge 0,\ \langle X-T,\varpi_\alpha\rangle\le 0\ \forall\alpha\}$; this is the dominant part of the convex hull of $W\cdot T$.

Vertices. For Q standard, T_Q satisfies $\langle T_Q,\alpha\rangle = 0$ for $\alpha\in\Delta^Q$ (since $T_Q\in\mathfrak{a}_Q$) and $\langle T_Q - T,\varpi_\alpha\rangle = 0$ for $\alpha\notin\Delta^Q$ (since such ϖ_α vanish on $\mathfrak{a}_0^Q$). By Lemma 3.5C(b) applied to $X=T$, $\langle T_Q,\alpha\rangle\ge\langle T,\alpha\rangle>0$ for $\alpha\notin\Delta^Q$ and $\langle T_Q-T,\varpi_\alpha\rangle = -t_\alpha<0$ for $\alpha\in\Delta^Q$. So T_Q is a vertex of Π_T at which exactly the constraints defining C_Q are active ($\ell=\dim\mathfrak{a}_0$ of them), and Π_T is simple there. Conversely, let v be a vertex and $\theta=\{\alpha:\langle v,\alpha\rangle=0\}$, so $v\in\mathfrak{a}_{P_\theta}$. By Lemma 3.5C(a), for $\alpha\in\theta$, $\varpi_\alpha(v) = \sum_{\beta\notin\theta} c_{\alpha\beta}\varpi_\beta(v) \le \sum_\beta c_{\alpha\beta}\varpi_\beta(T) = \varpi_\alpha(T_{P_\theta}) < \varpi_\alpha(T)$. So the constraints $\langle v-T,\varpi_\alpha\rangle\le 0$ with $\alpha\in\theta$ are not active, those with $\beta\notin\theta$ must all be active, and $v=T_{P_\theta}$. The combinatorial type is therefore the same for all regular T, and the vertices depend linearly on T.

Counting. Write $\mathsf{c}\cdot\mathbf{1}_{\mathscr{A}_A(T)} = \sum_{\theta\subseteq\Delta}\mathsf{c}_\theta\,\mathbf{1}_{\mathscr{A}_\theta(T)}$, where $\mathscr{A}_\theta(T) = \{\lambda\in\mathscr{A}_A(T) : \langle\lambda,\alpha\rangle = 0 \iff \alpha\in\theta\}$ and c_θ is the value of c there. Each $\mathscr{A}_\theta(T)$ is the set of lattice points of a half-open polytope in the subspace $\mathfrak{a}_{P_\theta}$ (with the lattice $X_*(A)\cap\mathfrak{a}_{P_\theta}$): the inequalities $\langle\lambda,\alpha\rangle>0$ ($\alpha\notin\theta$) and $\langle\lambda-T,\varpi_\alpha\rangle<0$ are strict. Its closure is the face $\Pi_T\cap\mathfrak{a}_{P_\theta}$ of Π_T, whose vertices are the T_Q with $Q\supseteq P_\theta$; its half-open tangent cones are the corresponding faces of the C_Q, independent of T. Apply Lemma 3.5B(i) to each $\mathscr{A}_\theta(T)$; for $T\in L_0$ the vertices T_Q are lattice points, so each vertex contributes $\psi(T_Q)$ times a T-independent rational function. Collecting by Q gives (3.5A), with $R_Q = \sum_{\theta:P_\theta\subseteq Q}\mathsf{c}_\theta\,\sigma_{C_{Q,\theta}}$, where $C_{Q,\theta}$ is the half-open tangent cone at T_Q of $\mathscr{A}_\theta(T)$. The statement for ψ on a polar divisor is Lemma 3.5B(ii). The edges of C_Q are as follows. An edge in $\mathfrak{a}_0^Q$ is an extremal ray of the dominant cone of M_Q, so it is spanned by a fundamental coweight of M_Q. An edge in $\mathfrak{a}_Q$ is dual to $\hat{\Delta}_Q$, so it is spanned by $-(\alpha^\vee)_Q$ for some $\alpha\notin\Delta^Q$ (Lemma 3.5C(a)). For $Q=G$ the only vertex is 0 and the cone is the closed dominant cone, so $R_G = \sum_\theta \mathsf{c}_\theta\sigma_{\text{relint face }\theta}$ continues $\sum_{\lambda\in X_*(A)^+}\mathsf{c}(\lambda)\psi(\lambda)$. □

Remark 2026.9 (Lemma 3.5A). Note that the vertex terms are *not* of the form $\prod_\alpha\chi(a_\alpha)^{\langle T,\alpha\rangle}$ times a T-independent function: already for $SL(2)$ the sum is $(z-z^T)/(1-z)$ plus a constant. Also, the

coordinates $\langle\lambda,\alpha\rangle$ on $X_*(A)$ are unimodular only for adjoint G, which is why Lemma 3.5B is stated for an arbitrary lattice. Lemma 3.5A was also checked by exact computation for $SL(3)$ and $PGL(3)$, with $\mathsf{c}=1$ and $\mathsf{c}(\lambda)=Q_{a_\lambda}$, including degenerate ψ (`examples/rank2/brion_check.py`). For $PGL(3)$ the dominant cone is not unimodular in root coordinates. The rank-one case is Example 3.5.

From [Ma71, Prop. 3.2.15] we have, for $a\in A^-$,

$$\mathrm{meas}(KaK)=\delta_B(a)^{-1}Q_a,\qquad Q_a=Q_G/Q_{W_a},\tag{3.7}$$

where $W_a\subset W$ is the stabilizer of $\nu_A(a)$ (generated by the w_α with $\langle\nu_A(a),\alpha\rangle=0$) and $Q_{W_a}=\sum_{w\in W_a}(IwI:I)^{-1}$. So Q_a is a rational number (for $SL(2)$ and $a\neq 1$, $Q_a=1+q^{-1}$), depending only on the face of the dominant cone containing $\nu_A(a)$.

The case of Lemma 3.5A needed for K-fixed vectors is this. For K-fixed $u=\Phi_{K,\chi^{-1}}$, $u'=\Phi_{K,\chi'^{-1}}$, Lemma 3.3 holds on all of A^-, and Lemma 3.4 with $P=B$ gives

$$J^T=\sum_{w,w'}C_{w,w'}\sum_{\lambda\in\mathscr{A}_A(T)}Q_{a_\lambda}\big((w\chi)(w'\chi')\big)(\lambda),$$

where

$$C_{w,w'}=c_w(\chi)c_{w'}(\chi')c(w,M_0,\chi)c(w',M_0,\chi')c_w(\chi^{-1})c_{w'}(\chi'^{-1}).$$

By Lemma 3.5A with $\mathsf{c}(\lambda)=Q_{a_\lambda}$ and $\psi=(w\chi)(w'\chi')$, this equals

$$\sum_{w,w'\in W}C_{w,w'}\sum_{Q\supseteq B}\big((w\chi)(w'\chi')\big)(T_Q)\,R_Q\big((w\chi)(w'\chi')\big).\tag{3.8}$$

The main proposition.

Proposition 3.6. *Assume (C) of §3.1 (automatic in rank one). Fix a level n. There are $c=c(n)\geq 1$ and $d_0=d_0(n)$ with the following property. Let χ,χ' be unramified regular characters and $u\in V_B(\chi^{-1})$, $u'\in V_B(\chi'^{-1})$ of level n. Then for $T\in L_0$ with $d(T)\geq d_0$,*

$$J^T(\chi,\chi',u,u')=\sum_{Q\supseteq B}J_Q^T(\chi,\chi',u,u'),\qquad J_Q^T=\sum_{P\subseteq Q}\sum_{w,w'\in[W_{M_P}\backslash W]}\psi_{w,w'}(T_Q)\,\mathscr{R}^{P,Q}_{w,w'}(\chi,\chi';u,u'),\tag{3.9}$$

where $\psi_{w,w'}=(w\chi)(w'\chi')$, and each $\mathscr{R}^{P,Q}_{w,w'}$ is bilinear in (u,u'), rational in (χ,χ') and independent of T. This holds for (χ,χ') off finitely many polar divisors. For all (χ,χ'), including $\chi'=\chi^{-1}$, (3.9) holds with each $\mathscr{R}^{P,Q}_{w,w'}$ replaced by a polynomial in T of degree $\leq\dim\mathfrak{a}_P$ (Lemma 3.5B(ii)). In particular $J_G^T(\chi,\chi',u,u')=:J_G(\chi,\chi',u,u')$ does not depend on T; it is the "T-independent remainder", and it contains the contribution of the compact core where no asymptotics are available. For K-fixed u,u', (3.9) is (3.8).

Proof. Let $\varepsilon=\varepsilon(n)$ be as in (C) and choose an integer $c\geq 1$ with $q^{-c}\leq\varepsilon$.

Step 1: regions. For a standard parabolic $P=MN$ let

$$S_P=\{\lambda\in X_*(A)^+:\ \langle\lambda,\alpha\rangle<c\iff\alpha\in\Delta^M\}.$$

These sets partition $X_*(A)^+$, and $a_\lambda \in A_P^-(\varepsilon)$ for $\lambda \in S_P$. The map $\lambda \mapsto \lambda^P$ identifies $X_*(A)/X_*(A_M)$ with a lattice in $\mathfrak{a}_0^P$, and for $\lambda \in S_P$, λ^P lies in the bounded set $\{X \in \mathfrak{a}_0^P : 0 \le \langle X, \alpha\rangle < c,\ \alpha \in \Delta^M\}$. So S_P meets only finitely many cosets of $X_*(A_M)$. Choose representatives $e \in S_P$ for them, a finite set E_P. Neither S_P nor E_P involves T: they depend only on $c = c(n)$, so the coefficients $C^P_{w,w',e}$ of Step 2 are independent of T. Then

$$S_P \cap (e + X_*(A_M)) = e + \{\nu \in X_*(A_M) :\ \langle e+\nu, \alpha\rangle \ge c\ \ (\alpha \in \Delta \smallsetminus \Delta^M)\}.$$

Step 2: exponential form on each region. Let $F(\lambda)$ denote the λ-th summand of (3.4). For $\lambda = e + \nu \in S_P$, apply Lemma 3.4. We have $a_\lambda = a_e a_\nu$ with $a_\nu \in A_M(F)$ central in M, so $i_{M,A}(w\chi)(a_\lambda) = (w\chi)(\nu)\, i_{M,A}(w\chi)(a_e)$. Further $\mathrm{meas}(Ka_\lambda K)\delta_P(a_\lambda) = \delta_{B\cap M}(a_\lambda)^{-1} Q_{a_\lambda} = \delta_{B\cap M}(a_e)^{-1} Q_{a_e}$, because $\delta_{B\cap M}$ is trivial on A_M and the stabilizer W_{a_λ} only involves roots in Δ^M, where $\langle \nu, \alpha\rangle = 0$. Hence

$$F(e+\nu) = \sum_{w,w' \in [W_M \backslash W]} \psi_{w,w'}(\nu)\, C^P_{w,w',e}(u,u'),$$

$$C^P_{w,w',e}(u,u') = \frac{Q_{a_e} c_w(\chi) c_{w'}(\chi')}{\delta_{B\cap M}(a_e)} \int_K \langle i_{M,A}(w\chi)(a_e)\Phi^M_{w\chi}, u(w,k)\rangle_M \times \langle i_{M,A}(w'\chi')(a_e)\Phi^M_{w'\chi'}, u'(w',k)\rangle_M\, dk.$$

Step 3: the truncated region on $S_P \cap (e + X_*(A_M))$. The conditions $\langle e+\nu-T, \varpi_\alpha\rangle < 0$ for $\alpha \in \Delta^M$ are implied by those for $\alpha \notin \Delta^M$ once $d(T)$ is large. Indeed, for $Y \in \mathfrak{a}_P$ and $\alpha \in \Delta^M$, $\varpi_\alpha(Y) = \sum_{\beta \notin \Delta^M} c_{\alpha\beta}\varpi_\beta(Y)$ with $c_{\alpha\beta} = \varpi_\alpha((\beta^\vee)_P) \ge 0$ (Lemma 3.5C(a)). So if $\varpi_\beta(e+\nu) < \varpi_\beta(T)$ for all $\beta \notin \Delta^M$, then

$$\varpi_\alpha(e+\nu) = \varpi_\alpha(e^P) + \textstyle\sum_\beta c_{\alpha\beta}\varpi_\beta(e+\nu) < \varpi_\alpha(e^P) + \varpi_\alpha(T_P) = \varpi_\alpha(T) - \varpi_\alpha(T^P) + \varpi_\alpha(e^P),$$

and $\varpi_\alpha(T^P) \ge \kappa_M d(T) \ge \varpi_\alpha(e^P)$ as soon as $d(T) \ge d_0$ (Lemma 3.5C(b); E_P is finite). (For $P = G$, $\nu = 0$ and the condition is $e \in \mathscr{A}_A(T)$, true for all $e \in E_G$ once $d(T) \ge d_0$.) Thus the ν that occur form the lattice points of the half-open polytope

$$\Pi_{P,e}(T) = \{\nu \in \mathfrak{a}_P :\ \langle \nu, \alpha\rangle \ge c - \langle e, \alpha\rangle\ (\alpha \in \Delta \smallsetminus \Delta^M),\ \ \langle \nu - (T-e), \varpi_\beta\rangle < 0\ (\beta \in \Delta \smallsetminus \Delta^M)\},$$

which has the shape of (3.3) for the datum $(\mathfrak{a}_P, \Delta_P, \hat{\Delta}_P)$, with the lower bounds shifted by the bounded amounts $c - \langle e, \alpha\rangle$. The argument for Lemma 3.5A carries over: it uses Lemma 3.5C for the parabolics $Q \supseteq P$, and the shifts are absorbed by enlarging d_0, since by Lemma 3.5C(b) the slack in every inactive constraint grows at least like $\kappa d(T)$. So for $d(T) \ge d_0$ the closure of $\Pi_{P,e}(T)$ is a simple polytope of constant combinatorial type whose vertices are the points $v_{Q,e}(T)$, $Q \supseteq P$, determined by $\langle v, \alpha\rangle = c - \langle e, \alpha\rangle$ for $\alpha \in \Delta^Q \smallsetminus \Delta^M$ and $\langle v - (T-e), \varpi_\beta\rangle = 0$ for $\beta \notin \Delta^Q$. Decomposing $v = v_Q + v^Q$ in $\mathfrak{a}_P = \mathfrak{a}_Q \oplus \mathfrak{a}_P^Q$ gives $v_Q = T_Q - e_Q$, and $v^Q \in \mathfrak{a}_P^Q$ is determined by T-independent equations. Thus $v_{Q,e}(T) = T_Q + x_{Q,e}$ with $x_{Q,e}$ independent of T. Since $T_Q \in X_*(A) \cap \mathfrak{a}_Q \subset X_*(A_M)$ for $T \in L_0$, the vertices move by lattice vectors as T varies in L_0, and Lemma 3.5B(i) (with $\Lambda = X_*(A_M)$; the vertex $T_Q + x_{Q,e}$ contributes $\psi(T_Q)\psi(x_{Q,e})\sigma$, where $x_{Q,e}$ is fixed and only its class modulo Λ matters) gives

$$\sum_{\nu \in \Pi_{P,e}(T) \cap X_*(A_M)} \psi(\nu) = \sum_{Q \supseteq P} \psi(T_Q)\, R_{P,Q,e}(\psi)$$

with T-independent rational $R_{P,Q,e}$; for degenerate ψ, Lemma 3.5B(ii) gives the polynomial version.

Step 4. Summing F over $\mathscr{A}_A(T) = \bigsqcup_P \bigsqcup_{e\in E_P} \big(\mathscr{A}_A(T) \cap (e + X_*(A_M))\big)$ gives (3.9) with $\mathscr{R}^{P,Q}_{w,w'} = \sum_{e\in E_P} C^P_{w,w',e} R_{P,Q,e}(\psi_{w,w'})$. For K-fixed vectors we may take $\varepsilon = 1$, and the computation reduces to (3.8). □

Example 3.5 (Brion in rank one, and $PGL(3)$). For $G = PGL(2)$ and $T = t\varpi^\vee$ we have $\mathscr{A}_A(T) = \{n\varpi^\vee : 0 \le n < t\}$, i.e. $G(T)$ is the ball of radius $t-1$ around the base vertex of the tree. With $z = \psi(\varpi^\vee)$ and $\mathsf{c} \equiv 1$, Lemma 3.5A is

$$\sum_{0\le n<t} z^n = \underbrace{\frac{1}{1-z}}_{Q=G,\ T_G=0} + \underbrace{z^t \cdot \Big(-\frac{1}{1-z}\Big)}_{Q=B,\ T_B=T},$$

so $R_G = \frac{1}{1-z}$ and $R_B = -\frac{1}{1-z}$. With Macdonald's weights ($\mathsf{c}(0) = 1$, $\mathsf{c}(n\varpi^\vee) = Q$ for $n \ge 1$) the sum is $1 + Q\frac{z}{1-z} - Q\frac{z^t}{1-z}$. For the degenerate character $\psi = 1$ the sum is the polynomial $1 + Q(t-1)$, as in Lemma 3.5B(ii). For $PGL(3)$, $\mathscr{A}_A(T)$ is the set of lattice points of a quadrilateral with vertices 0, T_{P_1}, T_{P_2} and T (Appendix A.1), and Lemma 3.5A has four vertex terms, one for each standard parabolic.

Remark 2026.10 (Proposition 3.6). The proof does not determine the remainder J_G, which is genuinely non-zero (in the $PGL(2)$ example it produces $|B_r| = O_\gamma(\mathbf{1}_K)$, §6.1). Steps 1–4 have been checked in rank one against §6. In higher rank, the combinatorial steps are now proved in full (Lemmas 3.5B, 3.5C) and were checked numerically for $SL(3)$ and $PGL(3)$. The representation-theoretic input (C) remains unverified for intermediate parabolics (Remark 2026.6), and the proposition is conditional on it.

Example 3.6 (the norm of the spherical function on a ball). Take $G = PGL(2)$, χ unitary and regular, $\chi' = \chi^{-1}$ and spherical u, u'. Then $J^T = \int_{G(T)} |\Gamma_\chi(g)|^2\, dg$. By Example 2.3, $\mathrm{vol}(Ka_nK)|\Gamma_\chi(a_n)|^2 = Q^{-1}|c_\alpha(\chi^{-1})z^n + c_\alpha(\chi)z^{-n}|^2$ for $n \ge 1$. Since $\overline{c_\alpha(\chi)} = c_\alpha(\chi^{-1})$ on the unitary circle, this is $Q^{-1}\big(2|c_\alpha(\chi)|^2 + 2\,\mathrm{Re}(c_\alpha(\chi^{-1})^2 y^n)\big)$. Summing over $1 \le n < t$,

$$J^T = 1 + \frac{t-1}{\rho(\chi)} + \frac{2}{Q}\,\mathrm{Re}\Big(c_\alpha(\chi^{-1})^2\,\frac{y - y^t}{1-y}\Big), \qquad \rho(\chi) = \frac{Q}{2}|c_\alpha(\chi)|^{-2}.$$

This is Proposition 3.6 in miniature. The degenerate pairs $w = w'$ give the polynomial term, whose slope is $1/\rho(\chi)$. The non-degenerate pairs give a T-independent term and a vertex term at T (the part with y^t). The growth rate of $\int_{G(T)} |\Gamma_\chi|^2$ is the reciprocal of the Plancherel density, as one expects from the Plancherel formula.

Proposition 3.7. *Let κ be an unramified character of $G(F)$. Then $w\kappa = \kappa$ for all $w \in W$ (represented in K), and the analogue $J^{T,\kappa}$ of (3.4) with an extra factor $\kappa(g)$ in the integrand has the decomposition (3.9) with $\psi_{w,w'}$ replaced by $\kappa|_{M_0(F)} \cdot \psi_{w,w'}$.*

Proof. By Remark 5 the extra factor is $\kappa(a_\lambda)$ on the λ-th summand, and $\lambda \mapsto \kappa(a_\lambda)$ is a character of $X_*(A)$; so the proof of Proposition 3.6 applies verbatim. □

4 Integrating the kernel

This section integrates the spherical function against an elliptic test function and proves the spectral identity (1.2); it is the analogue of [Wa89a, Prop. II.5]. §4.2 defines the truncated kernel $I^T(\chi,\phi)$, expresses it through the integrals J^T of §3, and defines $I(\chi,\phi)$ as its T-independent part (Proposition 4.2). §4.3 removes the truncation. The truncated identity (Theorem 4.5(a)) holds unconditionally, and the full identity (Theorem 4.5(b)) holds under the regularity Hypothesis (R). It ends with Conjecture 4.6, on the shape of the spectral measure when (R) fails, and with the geometric side of the identity.

(The numbering of results follows the original paper; its §4.1 is not needed and is omitted, see the revision log.)

4.2 The Fourier transform of a truncated orbital integral

We define the truncated kernel $I^T(\chi,\phi)$ by integrating $\Gamma_\chi(g^{-1}hg)$ over h against ϕ and over $g \in G(T)$. Expanding in a basis of vectors of fixed level expresses I^T through the integrals J^T of §3 (Proposition 4.2(a)), so I^T is an exponential polynomial in T, (4.7A). Its constant, non-oscillating part is the function $I(\chi,\phi)$ that appears on the spectral side of (1.2). Proposition 4.3 is the κ-twisted analogue needed in §5.

Let

$$G(T) := \bigcup_{\lambda \in \mathscr{A}_A(T)} K a_\lambda K, \tag{4.2}$$

and recall

$$\Gamma_\chi(g^{-1}hg) = \langle \pi_\chi(h)\pi_\chi(g)\Phi_{K,\chi},\ \pi_{\chi^{-1}}(g)\Phi_{K,\chi^{-1}}\rangle_G. \tag{4.3}$$

For $\phi \in C_c^\infty(G(F))$ we define the *truncated orbital integral* (the Fourier transform of the truncated orbital integral of ϕ)

$$I^T(\chi,\phi) = \int_{G(F)} \Gamma_\chi(h) \int_{G(T)} \phi(g^{-1}hg)\,dg\,dh = \int_{G(T)\times G(F)} \Gamma_\chi(g^{-1}hg)\phi(h)\,dh\,dg. \tag{4.4}$$

In §5 we also consider

$$I^{T,\kappa}(\chi,\phi) := \int_{G'(T)\times G(F)} \kappa(g)\Gamma_\chi(g^{-1}hg)\phi(h)\,dh\,dg, \tag{4.5}$$

where $G = SL(n)$, $G' = PGL(n)$, κ is an unramified character of $G'(F)$, and $G'(T)$ is the set (4.2) formed for G' (with $K' = PGL(n,\mathscr{O})$ and the same T, viewed in $X_*(A') \otimes \mathbb{Q}$).

The idea is to expand (4.3) using a basis and, term by term, to use the computations of §3.

Proposition 4.2. *Let $\phi \in C_c^\infty(G_{\mathrm{ell}})$ be bi-K_n-invariant and let χ be unramified and regular. Let $\{u_i\}$ be a basis of $V_B(\chi)^{K_n}$ and $\{u_i^*\}$ the dual basis of $V_B(\chi^{-1})^{K_n}$.*

(a) For every T, with finite sums,

$$I^T(\chi,\phi) = \sum_{i,j} J^T(\chi,\chi^{-1},u_i^*,u_j)\,\langle \pi_\chi(\phi)u_i, u_j^*\rangle_G. \tag{4.7}$$

(b) *Assume (C) (automatic in rank one). There is a finite set $\mathscr{M} = \mathscr{M}(G,n)$ of linear maps $\mu : L_0 \to X_*(A)$, containing 0 and independent of χ and ϕ, such that for $T \in L_0$ with $d(T) \geq d_0(n)$*

$$I^T(\chi,\phi) = \sum_{\mu\in\mathscr{M}} \chi(\mu(T))\, P_\mu(T;\chi,\phi), \tag{4.7A}$$

where $P_\mu(\cdot;\chi,\phi)$ is a polynomial function on L_0 whose coefficients are rational functions of χ, regular at generic regular χ, and linear in ϕ. The maps μ have the form $T \mapsto w^{-1}T_Q - w'^{-1}T_Q$ with Q standard and $w, w' \in W$.

(c) *Define*

$$I(\chi,\phi) := P_0(0;\chi,\phi),$$

the constant term of the non-oscillating part of $I^T(\chi,\phi)$. For generic regular χ this is intrinsic: it is the constant coefficient of the trivial character in the unique expansion of $T \mapsto I^T(\chi,\phi)$, on $\{T \in L_0 : d(T) \geq d_0\}$, as an exponential polynomial. The map $\phi \mapsto I(\chi,\phi)$ is linear and invariant under conjugation by K.

Proof. (a) Since ϕ is bi-K_n-invariant, $\pi_\chi(\phi)$ kills the complement of $V_B(\chi)^{K_n}$ and takes values in $V_B(\chi)^{K_n}$. So $\pi_\chi(\phi)v = \sum_{i,j}\langle v, u_i^*\rangle\langle \pi_\chi(\phi)u_i, u_j^*\rangle u_j$ for every v. Apply this to $v = \pi_\chi(g)\Phi_{K,\chi}$ in (4.3), integrate over h against ϕ and over $g \in G(T)$, and use $\langle u_j, \pi_{\chi^{-1}}(g)\Phi_{K,\chi^{-1}}\rangle = \langle \pi_{\chi^{-1}}(g)\Phi_{K,\chi^{-1}}, u_j\rangle$. Every vector involved has level n, independently of T. (Expanding $\pi(g)\Phi_{K,\chi}$ itself would not work, since its level grows with g.)

(b) Apply Proposition 3.6 directly with $\chi' = \chi^{-1}$ (which is regular), in its polynomial form, i.e. via Lemma 3.5B(ii) in Step 3. Each vertex $v_{Q,e}(T) = T_Q + x_{Q,e}$ contributes $\psi_{w,w'}(T_Q)\,\psi_{w,w'}(x_{Q,e})\,p(T)$ with p a polynomial, and

$$\psi_{w,w'}(T_Q) = (w\chi)(T_Q)\,(w'\chi^{-1})(T_Q) = \chi\big(w^{-1}T_Q - w'^{-1}T_Q\big) = \chi(\mu(T)).$$

Grouping the terms of (4.7) by the linear map μ gives (4.7A). The set $\mathscr{M}$ depends only on G and n (through the finitely many P, Q, w, w'), and it contains $\mu = 0$ (take $Q = G$, or $w = w'$).

Rationality. Let $\mathscr{Z} \subset \widehat{T}$ be the union of the finitely many proper subtori on which some factor $1 - \psi_{w,w'}(\mu_e)$ (with μ_e an edge vector of one of the cones in Step 3) vanishes without vanishing identically in χ. On $\widehat{T} \smallsetminus \mathscr{Z}$ the set of vanishing factors is constant, so by Lemma 3.5B(ii) the coefficients of the polynomials P_μ are rational functions of χ there. They are linear in ϕ by (4.7).

(c) Intrinsic: by uniqueness of the expansion. Linearity is clear from (4.7). For $k \in K$, conjugating ϕ by k amounts to translating $g \mapsto gk$ in (4.4), and $G(T)$ is right K-invariant. □

Remark 2026.11 (Proposition 4.2). $I^T(\chi,\phi)$ genuinely depends on T, and $I(\chi,\phi)$ is defined as its T-independent part. Part (b) follows from Proposition 3.6 and Lemma 3.5B(ii), and is therefore conditional on (C) in rank ≥ 2. That $\phi \mapsto I(\chi,\phi)$ is G-admissible in the sense of [HC78, §14] is plausible, since by (4.7) it is a finite combination of $\phi \mapsto \mathrm{tr}(A\,\pi_\chi(\phi))$ with finite-rank A, but it needs a bound uniform in n that we have not checked. Full G-invariance follows from Theorem 4.5(b) when Hypothesis (R) holds (Corollary 4.5A).

Proposition 4.3. *Let $G = SL(n)$, $G' = PGL(n)$, A' the diagonal torus of G', and κ an unramified character of $G'(F)$. For χ an unramified regular character of $M_0(F)$, the truncated κ-integral $I^{T,\kappa}(\chi,\phi)$ has an expansion (4.7A), obtained by applying Proposition 3.7 for G' to the restriction of Γ_χ to $G'(T)$-conjugates. We put $I^\kappa(\chi,\phi) := P_0^\kappa(0;\chi,\phi)$.*

Remark 9. We only prove Proposition 4.2; the proof of 4.3 is similar, using Remark 5 and Proposition 3.7, with $\kappa|_{A'(F)} \cdot \chi$ in place of χ in the frequencies.

Example 4.2 (the truncated kernel for $PGL(2)$). Let $G = PGL(2)$, and let γ be regular elliptic and unramified of depth 0, e.g. $\gamma = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ when $q \equiv 3 \bmod 4$. Then γ fixes only the base vertex v_0 of the tree, and $d(v, \gamma v) = 2d(v_0, v)$. There are $(q+1)q^{m-1}$ vertices at distance $m \geq 1$, and $(q+1)q^{m-1}\Gamma_\chi(a_{2m}) = c_\alpha(\chi^{-1})y^m + c_\alpha(\chi)y^{-m}$ by Example 2.3. So (4.4), with ϕ concentrated near γ, becomes

$$I^T(\chi,\gamma) = 1 + \sum_{m=1}^{t-1} \left(c_\alpha(\chi^{-1})y^m + c_\alpha(\chi)y^{-m}\right) = I(\chi,\gamma) - c_\alpha(\chi^{-1})\frac{y^t}{1-y} - c_\alpha(\chi)\frac{y^{-t}}{1-y^{-1}},$$

with

$$I(\chi,\gamma) = 1 + c_\alpha(\chi^{-1})\frac{y}{1-y} + c_\alpha(\chi)\frac{y^{-1}}{1-y^{-1}} = \frac{1-q^{-1}}{1-\cos\theta} \qquad (y = e^{i\theta}).$$

This is (4.7A) with three frequencies $\mu(T) \in \{0, \pm T\}$ (in the variable y). The T-independent part $I(\chi,\gamma)$ is a rational function with a double pole at $y = 1$. Details, and the other elliptic types, are in §6.1.

4.3 The spectral expansion

This subsection proves the spectral identity. After recalling that $f \mapsto I(f,\phi)$ has compactly generated support (Lemma 4.4), we show that for T large the identity holds exactly with I^T in place of I (Theorem 4.5(a)). To let $T \to \infty$ the oscillatory terms of (4.7A) must die out; Hypothesis (R) guarantees this (Theorem 4.5(b)). We then discuss what can happen without (R): the spectral side may acquire a singular part, as for $SL(2)$, and Conjecture 4.6 describes its expected shape. The subsection ends with the geometric side, obtained from the Weyl integration formula.

Let ϕ be a locally constant function with compact support in G_{ell}. By [Ka86, §4, Lemma 6] the distribution

$$f \longmapsto I(f,\phi) = \int_{G(F)\times G(F)} f(g^{-1}hg)\phi(h)\,dg\,dh = \int_{G(F)} f(h)\Phi_\phi(h)\,dh = \int_{G(F)} \Phi_f(h)\phi(h)\,dh, \tag{4.8}$$

$f \in C_c^\infty(G(F))$, has compactly generated support. Here, for $h \in G_{\mathrm{ell}}$, $\Phi_\phi(h) := \int_{G(F)} \phi(g^{-1}hg)\,dg$, and Φ_f is defined in the same way. (In (4.8) f may be any function in $C_c^\infty(G(F))$; from Lemma 4.4 on, $f \in \mathscr{H}(G,K)$.)

Lemma 4.4. *Let ϕ be locally constant with compact support in G_{ell}, and let $f \in \mathscr{H}(G,K)$. There is a compact set $C_{f,\phi} \subset G(F)$ such that $f(g^{-1}hg)\phi(h) \neq 0$ implies $g \in C_{f,\phi}$. If $G = SL(n)$ and $G' = PGL(n)$, there is likewise a compact $C'_{f,\phi} \subset G'(F)$ for the action of $G'(F)$ by conjugation.*

Proof. For each $h \in G_{\mathrm{ell}}$ there are a neighborhood $V_{h,f}$ of h and a compact set $C_{h,f} \subset G(F)$ such that $f(g^{-1}h'g) \neq 0$ and $h' \in V_{h,f}$ imply $g \in C_{h,f}$. This uses that the centralizer of an elliptic element is compact modulo A_G. Finitely many $V_{h,f}$ cover $\mathrm{supp}(\phi)$. The second part is similar. □

Choose $T = T(f,\phi) \in L_0$ with $d(T) \geq d_0(n)$ and $G(T) \supseteq C_{f,\phi}$, so that

$$\mathrm{supp}_g\left(\int_{G(F)} f(g^{-1}hg)\phi(h)\,dh\right) \subset G(T). \tag{4.9}$$

Then $I(f,\phi)=\int_{G(T)\times G(F)} f(g^{-1}hg)\phi(h)\,dg\,dh$. For $f\in\mathscr{H}(G,K)$, Corollary 2.3 gives $f(g^{-1}hg)=\int_{\widehat{T}^u} f^\vee(\chi)\Gamma_\chi(g^{-1}hg)\,d\mu(\chi)$. Write $d\mu(\chi)=\rho(\chi)\,d\chi$ with $\rho=\frac{Q_G}{|W|}|c(\chi)|^{-2}$.

Hypothesis (R) (regularity, for G and level n)**.** For every bi-K_n-invariant $\phi\in C_c^\infty(G_{\mathrm{ell}})$ and every $\mu\in\mathscr{M}(G,n)$, each coefficient of the polynomial $P_\mu(T;\chi,\phi)$ of (4.7A), multiplied by $\rho(\chi)$, extends to a C^∞ function of $\chi\in\widehat{T}^u$.

Hypothesis (R) is a strong condition with representation-theoretic content, not a routine regularity assumption. It asks that each oscillatory component separately be regular against the Plancherel density, and it fails for $SL(2)$ (§6.2) exactly at a reducibility point of the unitary principal series. For $PGL(2)$ it is proved in Lemma 6.1A. It also fails for $PGL(3)$, along non-root subtori where $\rho\neq 0$, but there the conclusion of Theorem 4.5(b) still holds (Appendix A). The weaker Hypothesis (R′) of Remark 2026.20 suffices for the proof of Theorem 4.5(b).

Remark 2026.12 (why (R) is needed). One might hope to remove the truncation by showing that the singularities of the individual vertex terms cancel in their sum, using $(G,A)^\sim$-families as in §2.5. This kind of cancellation does hold: in the $SL(2)$ example (§6.2) the sum of the vertex terms at the vertex T is a smooth function of χ for each T. But it is not enough for removing the truncation. The sum is smooth for each T but not uniformly in T: near the bad point it behaves like a Dirichlet kernel and tends to a point mass as $T\to\infty$. (See (2.13A) in §2.3A.) What is needed is regularity of each oscillatory component separately, which is Hypothesis (R). In particular, smoothness of the sum together with the Riemann–Lebesgue lemma does not suffice.

Theorem 4.5. *Let $f\in\mathscr{H}(G,K)$, let $\phi\in C_c^\infty(G_{\mathrm{ell}})$ be bi-K_n-invariant, and let $T(f,\phi)$ be as in* (4.9).

(a) For all $T\in L_0$ with $G(T)\supseteq G(T(f,\phi))$,

$$\begin{aligned} I(f,\phi) &= \sum_{\lambda\in\mathscr{A}_A(T)}\int_{Ka_\lambda K\times G(F)}\int_{\widehat{T}^u} f^\vee(\chi)\Gamma_\chi(g^{-1}hg)\,d\mu(\chi)\,\phi(h)\,dg\,dh \\ &= \int_{\widehat{T}^u} f^\vee(\chi)\,I^T(\chi,\phi)\,d\mu(\chi). \end{aligned}$$

(b) Assume (C) (automatic in rank one). If Hypothesis (R) holds for G and n, then

$$I(f,\phi)=\int_{\widehat{T}^u} f^\vee(\chi)\,I(\chi,\phi)\,d\mu(\chi),$$

the integral converging absolutely, with $I(\chi,\phi)$ as in Proposition 4.2(c).

Proof. (a) Fubini on the compact set $G(T)\times\operatorname{supp}\phi$, the integrand $f^\vee(\chi)\Gamma_\chi(g^{-1}hg)\phi(h)$ being bounded on $\widehat{T}^u\times G(T)\times\operatorname{supp}\phi$.

(b) By (a) and (4.7A), for $T\in L_0$ in the cone $\mathscr{C}=\{T: G(T)\supseteq G(T(f,\phi)),\ d(T)\geq d_0\}$,

$$I(f,\phi)=\sum_{\mu\in\mathscr{M}}\int_{\widehat{T}^u} f^\vee(\chi)\rho(\chi)\chi(\mu(T))P_\mu(T;\chi,\phi)\,d\chi.$$

Fix $T_2\in\mathscr{C}$ and a regular dominant $T_1\in L_0$ with $\mu(T_1)\neq 0$ for every $\mu\neq 0$ in $\mathscr{M}$ (possible, since $\mathscr{M}$ is finite). Put $T=T_2+tT_1$, $t\in\mathbb{Z}_{\geq 0}$. For $\mu\neq 0$ the μ-term is a finite sum of terms

$p(t)\,\hat{g}(-\mu(T_2)-t\mu(T_1))$, where p is a polynomial and $\hat{g}$ denotes Fourier coefficients of a function $g = f^\vee \rho\, a$ which is C^∞ by (R). Fourier coefficients of smooth functions on a compact torus decay faster than any power, so these terms tend to 0 as $t \to \infty$. Hence the $\mu = 0$ term, which is a polynomial in t, converges to $I(f,\phi)$ as $t \to \infty$, so it is constant in t. Since T_1 ranges over an open cone of directions, the polynomial $T \mapsto \int f^\vee \rho\, P_0(T;\chi,\phi)\,d\chi$ on L_0 is constant along a spanning set of directions from every point of $\mathscr{C}$, hence constant, and equal to its value at $T = 0$. That value is $\int f^\vee(\chi) I(\chi,\phi)\,d\mu(\chi)$. Absolute convergence holds because $I(\cdot,\phi)\rho$ is smooth by (R). □

Corollary 4.5A. *Assume (R) for G and all n. Then $\phi \mapsto I(\chi,\phi)\rho(\chi)$ is, for almost every χ, a G-invariant distribution on G_{ell}, and for each $\gamma \in G_{\mathrm{ell}}$ there is a function $I(\cdot,\gamma)$ with $I(\cdot,\gamma)\rho$ smooth, such that (1.5) holds in the form*

$$\int_{G(F)} f(g^{-1}\gamma g)\,dg = \int_{\widehat{T}^u} f^\vee(\chi) I(\chi,\gamma)\,d\mu(\chi) \qquad (f \in \mathscr{H}(G,K)).$$

Proof. $I(f,\phi)$ is invariant under conjugation of ϕ, and the W-invariant Laurent polynomials $f^\vee$ are dense in $C(\widehat{T}^u/W)$; since $I(\cdot,\phi)\rho$ is smooth and W-invariant, it is determined by the numbers $I(f,\phi)$. Taking ϕ to be normalized characteristic functions of small neighborhoods of γ, on which orbital integrals of f are constant, gives the second statement. □

Example 4.5 (removing the truncation in rank one). For $PGL(2)$ and γ as in Example 4.2, the oscillatory terms times ρ are

$$-c_\alpha(\chi^{-1})\frac{y^t}{1-y}\cdot\frac{Q}{2}|c_\alpha(\chi)|^{-2} = -\frac{Q}{2}\cdot\frac{y^t}{1-q^{-1}y},$$

and its mirror image. These are smooth on $|y| = 1$, so (R) holds. Better, their Fourier modes are y^k with $k \geq t$, so they pair to 0 with $f_n^\vee$ as soon as $2t > n$. Hence $\int f_n^\vee I\,d\mu = O_\gamma(f_n)$, which is $(q+1)q^{m-1}$ for $n = 2m \geq 2$, 1 for $n = 0$, and 0 for odd n.

For $SL(2)$ and an unramified elliptic γ of depth 0 the same computation gives the vertex terms

$$-c(\chi^{-1})\frac{z^{2t}}{1-z^2} - c(\chi)\frac{z^{-2t}}{1-z^{-2}}$$

(§6.2). Each has a simple pole at $z = -1$, where $\rho \neq 0$, so (R) fails. Near $z = -1$ both $c(\chi^{\pm 1})$ are close to $Q/2$, and the sum of the two terms is approximately $\frac{Q}{2}z^{2-2t}\sum_{m=0}^{2t-2} z^{2m}$, a two-sided Dirichlet kernel in z^2. By (2.13A) its integral against $f^\vee\,d\mu = f^\vee\rho\,d\chi$ tends to $\frac{Q}{2}\cdot\frac{1}{2}\rho(\chi_0)f^\vee(\chi_0) = \frac{1}{2}f^\vee(\chi_0)$. (The factor $\frac{1}{2}$ appears because $z \mapsto z^2$ is two-to-one; near $z = 1$ the double zero of ρ kills the contribution.) So the spectral measure of O_γ is $I\,d\mu + \frac{1}{2}\delta_{\chi_0}$, which is Proposition 6.4.

Remark 2026.13 (Theorem 4.5).

1. Part (b) is conditional on (R). Hypothesis (R) holds for $PGL(2)$ (§6.1). It fails for $SL(2)$ (§6.2), and there the conclusion of (b) is false: the spectral side needs an extra point mass. (R) can fail even when $Z(G)$ is connected: it fails for $PGL(3)$, although the conclusion of (b) holds there (Appendix A). The proof of (b) works verbatim under the weaker (R′) of Remark 2026.20. For $GL(N)$ the analogous statement is the content of [Wa89a], which we have not re-checked against the formulation here.

2. In rank one, the proof of (b) can be sharpened: the oscillatory terms pair to zero *exactly* with $f^\vee$ once T is large, by a Fourier-support argument (§6.1).

Functionals, distributions and measures. We distinguish three levels of structure on the spectral side. Let $\mathrm{PW} = \{f^\vee : f \in \mathscr{H}(G,K)\}$, the W-invariant Laurent polynomials on $\widehat{T}$, restricted to $\widehat{T}^u$.

(i) By Theorem 4.5(a), $\Lambda_\phi : f^\vee \mapsto I(f,\phi)$ is a well-defined *linear functional* on PW. This is all that is proved in general.

(ii) Λ_ϕ is a *distribution* on $\widehat{T}^u/W$ if it is continuous for the C^∞ topology, i.e. if $|\Lambda_\phi(f^\vee)| \le C\|f^\vee\|_{C^k}$ for some k. The bound $|I^T(\chi,\phi)| \le \mathrm{vol}(G(T))\|\phi\|_1$ from (4.4) grows exponentially with T, and T must grow with $\deg f$, so (i) does not imply (ii).

(iii) Λ_ϕ is given by a *finite complex measure* if $k = 0$ suffices.

Under (C) and (R), Λ_ϕ is the absolutely continuous measure $I(\chi,\phi)\,d\mu$ (Theorem 4.5(b)). In general, that Λ_ϕ is a measure is part of Conjecture 4.6. For $SL(2)$ it is a measure with an atom (Proposition 6.4).

When (R) fails. The singular behavior comes from the denominators $1-\psi(\mu_e)$ of Lemma 3.5A, with $\psi = w\chi/w'\chi$, at points where ρ does not vanish to compensate. On such a hypersurface a term $\chi(\mu(T))a(\chi)$ with $a\rho$ having a simple pole behaves, as $T\to\infty$, like a Dirichlet kernel, and its pairing with $f^\vee$ tends to a multiple of the restriction of $f^\vee$ to the hypersurface (the Sokhotski–Plemelj formula on the circle). The limit exists because $I(f,\phi)$ does not depend on T. In rank ≥ 2 the hypersurface can carry a further oscillation. In $PGL(3)$ the poles of two terms cancel in a pair whose frequencies agree on the hypersurface, and the common frequency still moves along the hypersurface as T grows. Then the Dirichlet kernel is multiplied by a character oscillating along the hypersurface, and the limit is 0 (Appendix A.6). This motivates:

Conjecture 4.6. *Let $\phi \in C_c^\infty(G_{\mathrm{ell}})$.*

(a) The linear functional $\Lambda_\phi : f^\vee \mapsto I(f,\phi)$ on Satake transforms is given by a finite complex measure dI_ϕ on $\widehat{T}^u/W$ of the form

$$dI_\phi = I(\chi,\phi)\,d\mu(\chi) + dI_\phi^{sing},$$

where $I(\cdot,\phi)$ is the function of Proposition 4.2(c), $I(\cdot,\phi)\rho \in L^1$, and dI_ϕ^{sing} is supported on the set of non-regular unitary χ at which ρ does not vanish (to the order forced by the stabilizer of χ). This set is, conjecturally, the set of χ for which the R-group of $i_{G,A}(\chi)$ is non-trivial.

(b) If $Z(G)$ is connected (equivalently, the derived group of $\hat{G}$ is simply connected), then the R-groups of the unitary unramified principal series are trivial [Ke82], and $dI_\phi^{sing} = 0$.

(c) For $G = SL(n)$, dI_ϕ^{sing} is the sum, over the non-trivial unramified κ (ramified κ do not contribute, by Lemma 5.0), of the contributions $(c_S|\mathfrak{D}(S)|)^{-1}\,dI_\phi^\kappa$ of (5.4A), which are push-forwards of spectral measures of the elliptic endoscopic groups (Proposition 5.2).

Remark 2026.14 (Conjecture 4.6). Verified for $PGL(2)$ (no singular part) and for $SL(2)$, where $dI_\gamma^{sing} = \frac{1}{2}\Delta\,\delta_{\chi_0}$ and χ_0 is the unramified quadratic character (§6.2). The precise support condition in (a) is a guess, and in (c) the normalizations (the size of $\mathfrak{D}$, transfer factors) have only been

checked for $SL(2)$. Part (b) relies on [Ke82] (Chevalley groups; the general unramified case via Steinberg's connectedness theorem for centralizers in $\hat{G}$) for the triviality of R-groups. Part (b) does not assert (R), which fails for $PGL(3)$ (Appendix A). The conjectural part of (b) is "trivial R-groups $\Rightarrow dI_\phi^{sing} = 0$", which we expect to go through (R′) (Remark 2026.20). For $PGL(3)$, part (b) holds for the depth-0 unramified elliptic elements of Appendix A, numerically and modulo Observation A.2.

The geometric side. The Weyl integration formula states that

$$\int_{G(F)} \psi(h)\,dh = \sum_S \frac{1}{|W(G,S)|} \int_{S(F)} |D(s)| \int_{S(F)\backslash G(F)} \psi(g^{-1}sg)\,d\dot{g}\,ds, \tag{4.10}$$

where S runs over a set of representatives of the $G(F)$-conjugacy classes of maximal F-tori, $W(G,S) = N_{G(F)}(S)/S(F)$, $D(s)$ is the Weyl discriminant, and $d\dot{g}$ is the quotient measure. If ϕ is supported on G_{ell}, only elliptic S contribute. Plugging $\psi = f\Phi_\phi$ into (4.10) gives the "geometric" expansion

$$I(f,\phi) = \sum_{S \text{ ell}} \frac{\text{vol}(S(F))}{|W(G,S)|} \int_{S(F)} |D(s)|\, O_s(f)\, O_s(\phi)\,ds, \tag{4.11}$$

where $O_s(f) = \int_{S(F)\backslash G(F)} f(g^{-1}sg)\,d\dot{g}$. The equality between Theorem 4.5 and (4.11) is a special case of Arthur's local trace formula (§7).

5 The κ-stable trace formula on $SL(n)$

This section treats a twisted version of (1.2) for $G = SL(n)$, in which conjugation is by $PGL(n,F)$, weighted by an unramified character κ. These κ-orbital integrals measure the difference between stable and ordinary conjugacy, and they are what the singular part of the spectral measure is expected to be built from (Conjecture 4.6(c)). §5.1 sets up the κ-orbital integrals and their spectral expansion. §5.2 uses the fundamental lemma, now a theorem, to identify the κ-spectral measures with push-forwards of stable spectral measures of the elliptic endoscopic groups (Propositions 5.1 and 5.2).

5.1 A twisted trace formula on $SL(n,F)$

We define the κ-orbital integral Φ_f^κ and show that only unramified κ matter for spherical f (Lemma 5.0). The Weyl integration formula gives the geometric side (5.4); Fourier inversion over a finite group $\mathfrak{D}(S)$ recovers ordinary orbital integrals from the κ-ones (5.4A); and the method of §4 gives the spectral side (5.7)–(5.9).

Let $G := SL(n)$, $G' := PGL(n)$, $m \mid n$, and let $\eta : F^\times \to \mathbb{C}^\times$ be an unramified character of order m. Then $\kappa'(g) := \eta(\det g)$ is a character of $GL(n,F)$, trivial on the center since $\eta(z^n) = 1$; let κ be the corresponding character of $G'(F)$. It is trivial on the image of $G(F)$, so for $f \in C_c^\infty(G(F))$ the *κ-orbital integral*

$$\Phi_f^\kappa(h) := \int_{G'(F)} \kappa(g) f(g^{-1}hg)\,dg \qquad (h \in G_{\text{ell}})$$

is invariant under conjugation by $G(F)$, and $\Phi_f^\kappa(x^{-1}hx) = \kappa(x)^{-1}\Phi_f^\kappa(h)$ for $x \in G'(F)$.

Let f and ϕ be as in Lemma 4.4. Then

$$I^\kappa(f,\phi) := \int_{G(F)}\int_{G'(F)} \kappa(g) f(g^{-1}hg)\phi(h)\,dg\,dh = \int_{G(F)} \Phi_f^\kappa(h)\phi(h)\,dh = \int_{G(F)} \Phi_\phi^{\kappa^{-1}}(h) f(h)\,dh. \tag{5.1}$$

The last equality follows from the substitution $h \mapsto ghg^{-1}$ followed by $g \mapsto g^{-1}$, which turns κ into κ^{-1}. The interchanges of integration are justified by Lemma 4.4. For h regular elliptic, the properties of Φ_f^κ near the singular set of the centralizer were investigated in [Ka84] (see also [He86]).

Let $S \subset G$ be an elliptic maximal torus and $S' \subset G'$ the centralizer in G' of S (the image of the corresponding maximal torus of $GL(n)$). The map $G(F) = SL(n,F) \to G'(F) = PGL(n,F)$ has kernel $\mu_n(F)$; write $\bar G(F)$ for its image, a normal subgroup with abelian quotient. The $G'(F)$-conjugacy class of a regular $s \in S(F)$ is the union of the $G(F)$-classes of the $x^{-1}sx$, $x \in \mathfrak{D}(S) := G'(F)/\bar G(F)S'(F)$, a finite abelian group. We give $\bar G(F)$ the push-forward of the Haar measure of $G(F)$ (so $\mathrm{vol}(\bar K) = 1$ for the image $\bar K$ of K), and we put $c_S = \mathrm{vol}(S'(F))/\mathrm{vol}(\bar S(F))$, where $\bar S(F)$ is the image of $S(F)$. Hence

$$\Phi_f^\kappa(s) = c_S \sum_{x \in \mathfrak{D}(S)} \kappa(x)\, O_{x^{-1}sx}(f) \quad \text{if } \kappa|_{S'(F)} = 1, \qquad \Phi_f^\kappa(s) = 0 \quad \text{otherwise.} \tag{5.2}$$

Lemma 5.0. *Let $f \in \mathscr{H}(G,K)$. If κ is ramified, i.e. η is non-trivial on $\mathscr{O}^\times$, then $\Phi_f^\kappa \equiv 0$. Hence only unramified κ contribute to κ-orbital integrals of spherical functions.*

Proof. Choose $u \in \mathscr{O}^\times$ with $\eta(u) \neq 1$ and let x be the image in $G'(F)$ of $\mathrm{diag}(u,1,\dots,1) \in GL(n,\mathscr{O})$. Conjugation by x preserves $K = SL(n,\mathscr{O})$ and fixes the diagonal elements a_λ, so it preserves every double coset $Ka_\lambda K$, and $f(x^{-1}gx) = f(g)$. Substituting $g \mapsto gx$ in the definition of Φ_f^κ gives $\Phi_f^\kappa(h) = \kappa(x)\Phi_f^\kappa(h) = \eta(u)\Phi_f^\kappa(h)$, so $\Phi_f^\kappa(h) = 0$. □

Since Φ_f^κ is $G(F)$-invariant, the Weyl integration formula (4.10) for $G(F)$ gives

$$I^\kappa(f,\phi) = \sum_{S \text{ ell}} \frac{1}{|W(G,S)|} \int_{S(F)} |D(s)|\, \Phi_f^\kappa(s)\, O_s(\phi)\,ds. \tag{5.4}$$

The finite group $\mathfrak{D}(S)$ in (5.2) is the source of the κ-weights. Fourier inversion on $\mathfrak{D}(S)$ recovers ordinary orbital integrals:

$$O_s(f) = \frac{1}{c_S|\mathfrak{D}(S)|} \sum_{\kappa \in \mathfrak{D}(S)^\wedge} \Phi_f^\kappa(s). \tag{5.4A}$$

The method of proof of Theorem 4.5 applies to I^κ (Proposition 4.3). It gives an expression

$$I^\kappa(f,\phi) = \int_{\widehat{T}^u} f^\vee(\chi)\, dI_\phi^\kappa(\chi), \tag{5.7}$$

where A is the maximal split torus of G. Under (R) for $I^{T,\kappa}$, $dI_\phi^\kappa = I^\kappa(\chi,\phi)\,d\mu(\chi)$. Combining (5.7) and (5.4),

$$\int_{\widehat{T}^u} f^\vee(\chi)\, dI_\phi^\kappa(\chi) = \sum_{S \text{ ell}} \frac{1}{|W(G,S)|} \int_{S(F)} |D(s)|\, \Phi_f^\kappa(s)\, O_s(\phi)\,ds, \tag{5.8}$$

a special case of a κ-stable form of Arthur's local trace formula. For $\gamma \in G_{\text{ell}}$ we write (5.7) as

$$\Phi_f^\kappa(\gamma) = \int_{\widehat{T}^u} f^\vee(\chi)\, dI_\gamma^\kappa(\chi), \qquad f \in \mathscr{H}(G,K). \tag{5.9}$$

5.2 The fundamental lemma and the spectral measures dI_γ^κ

Here the fundamental lemma enters. For κ of order $m \mid n$ we introduce the elliptic endoscopic group H_m and the map ι from its unramified characters to those of $SL(n)$. When κ has order n the group H is compact, and the κ-spectral measure is a point mass (Proposition 5.1); for $n=2$ this is the atom of the $SL(2)$ example in §6.2. For general m, the κ-spectral measure is the push-forward of a stable spectral measure on H_m, assuming a stable form of Theorem 4.5 for H_m (Proposition 5.2).

For κ of order $m \mid n$ the corresponding elliptic endoscopic group of $SL(n)$ is

$$H = H_m = \ker\big(N_{E/F} \circ \det :\ \mathrm{Res}_{E/F}\, GL(n/m) \to \mathbb{G}_m\big), \qquad E/F \text{ unramified},\ [E:F] = m.$$

It is unramified over F and split over E. For $m = n$, H is the norm-one torus $\ker N_{E/F}$, an elliptic maximal torus of $SL(n)$. Fix an L-embedding $\iota : {}^L H \hookrightarrow {}^L G$ and let $\iota^* : \mathscr{H}(G,K) \to \mathscr{H}(H,K_H)$ be the induced homomorphism, characterized by

$$(\iota^* f)^\vee(\chi_H) = f^\vee(\iota(\chi_H)),$$

where ι also denotes the induced map $\widehat{T}_H^u/W_H \to \widehat{T}^u/W$ on unramified unitary characters.

Normalizations. Orbital integrals $O_\gamma(f) = \int_{G_\gamma(F)\backslash G(F)} f(g^{-1}\gamma g)\, d\dot{g}$ use $\mathrm{vol}(K) = 1$ on $G(F)$. Stable orbital integrals on $H(F)$ use $\mathrm{vol}(K_H) = 1$ for a hyperspecial K_H. The Haar measures on the centralizers $G_\gamma = S$ and $H_{\gamma_H} = S_H$ correspond under the admissible isomorphism $S_H \cong S$ attached to the matching $\gamma_H \leftrightarrow \gamma$. In the form of Langlands and Shelstad [LS87, (1.4)], the *fundamental lemma* for the spherical Hecke algebra asserts that for every (G,H)-regular semisimple $\gamma_H \in H(F)$,

$$\sum_{\gamma'} \Delta_{H,\kappa}(\gamma_H, \gamma')\, O_{\gamma'}(f) = S\Phi^H_{\iota^* f}(\gamma_H), \tag{5.10}$$

where γ' runs over the $G(F)$-conjugacy classes of regular elements of $G(F)$ whose stable class corresponds to γ_H (the sum is 0 if there are none), $S\Phi^H$ is the stable orbital integral on $H(F)$, and $\Delta_{H,\kappa}$ is the transfer factor. For $SL(n)$ the classes γ' in the stable class of γ are the $x^{-1}\gamma x$, $x \in \mathfrak{D}(S)$, and $\Delta_{H,\kappa}(\gamma_H, x^{-1}\gamma x) = \kappa(x)\Delta_{H,\kappa}(\gamma_H,\gamma)$ (the κ-dependence of transfer factors on the invariant $\mathrm{inv}(\gamma,\gamma')$). Comparing with (5.2), (5.10) becomes

$$\Phi_f^\kappa(\gamma) = c_S\, \Delta_{H,\kappa}(\gamma_H,\gamma)^{-1}\, S\Phi^H_{\iota^* f}(\gamma_H) \qquad (\gamma \text{ corresponding to } \gamma_H), \tag{5.11}$$

and $\Phi_f^\kappa(\gamma) = 0$ for γ not coming from H. (Transfer factors here have absolute value 1, and with the normalizations we use in §6.2, $\Delta = 1$.) This is now a theorem. For the unit element and $SL(n)$ it is due to Kazhdan [Ka84] (κ of order n) and Waldspurger [Wa91]. The extension from the unit element to the whole Hecke algebra is due to Hales [Ha95]. The general case is due to Ngô [Ng10], combined with Waldspurger's reduction from groups to Lie algebras.

Proposition 5.1. *Let κ have order n, so $H = \ker N_{E/F}$ is compact. Let $\chi_\kappa \in \widehat{T}^u/W$ be the image under ι of the trivial character of $H(F)$; its Satake parameter is the class of* $\mathrm{diag}(1,\zeta,\dots,\zeta^{n-1})$ *in $PGL(n,\mathbb{C})$, ζ a primitive n-th root of unity. For $\gamma \in G_{\mathrm{ell}}$ coming from $\gamma_H \in H(F)$,*

$$\Phi_f^\kappa(\gamma) = c_S\, \Delta_{H,\kappa}(\gamma_H,\gamma)^{-1}\, f^\vee(\chi_\kappa) \qquad (f \in \mathscr{H}(G,K)),$$

that is, $dI_\gamma^\kappa = c_S \Delta_{H,\kappa}(\gamma_H,\gamma)^{-1}\, \delta_{\chi_\kappa}$ is a point mass. This is unconditional: it uses only the fundamental lemma (5.10).

Proof. $H(F)$ is compact, so $\mathscr{H}(H,H(F))$ consists of constants, $S\Phi^H_{\iota^* f}(\gamma_H) = (\iota^* f)(\gamma_H)$ (with $\mathrm{vol}\, H(F) = 1$), and $\iota^* f$ is the constant $(\iota^* f)^\vee(1) = f^\vee(\chi_\kappa)$. Now apply (5.11). Under ι, Frobenius is sent to an n-cycle in the Weyl group of $\hat{G} = PGL(n,\mathbb{C})$, whose semisimple class has eigenvalues the n-th roots of unity. □

Proposition 5.2. *Let $m \mid n$ and suppose that Theorem 4.5 holds for $H = H_m$ in its stable form: for $\gamma_H \in H_{\mathrm{ell}}$ there is a measure $dI^{H,st}_{\gamma_H}$ on $\widehat{T}^u_H/W_H$ with $S\Phi^H_{f_H}(\gamma_H) = \int f_H^\vee \, dI^{H,st}_{\gamma_H}$ for $f_H \in \mathscr{H}(H,K_H)$. Then for γ coming from γ_H,*

$$dI^\kappa_\gamma = c_S \Delta_{H,\kappa}(\gamma_H,\gamma)^{-1} \cdot \iota_* \big(dI^{H,st}_{\gamma_H} \big).$$

In words: the κ-spectral measure of G is the transfer factor times the push-forward, along the dual embedding, of the stable spectral measure of H.

Proof. Combine (5.11) with $(\iota^* f)^\vee = f^\vee \circ \iota$: with $\Delta' = c_S \Delta_{H,\kappa}(\gamma_H,\gamma)^{-1}$, $\Phi^\kappa_f(\gamma) = \Delta' \int (f^\vee \circ \iota)\, dI^{H,st}_{\gamma_H} = \Delta' \int f^\vee \, d\iota_*(dI^{H,st}_{\gamma_H})$. The measure dI^κ_γ is determined by its values on the dense set of Satake transforms. □

Conditionally on its hypothesis, Proposition 5.2 says that the distribution $\phi \mapsto I^\kappa(\iota(\chi_H),\phi)$ on G "transfers" to $\phi_H \mapsto I^{H,st}(\chi_H,\phi_H)$. By (5.4A) and Lemma 5.0, the ordinary spectral functional of $SL(n)$ is $(c_S|\mathfrak{D}(S)|)^{-1}\sum_\kappa \Lambda^\kappa_\gamma$, summed over the unramified characters κ of $\mathfrak{D}(S)$. The $\kappa = 1$ term is the stable part. If Proposition 5.2 applies to the $\kappa \neq 1$ terms, they are push-forwards from endoscopic groups supported on the proper subvarieties $\iota(\widehat{T}^u_H/W_H)$ of $\widehat{T}^u/W$; this is the expected source of the singular part in Conjecture 4.6(c). For κ of order n no hypothesis is needed (Proposition 5.1), and for $n = 2$ this is the atom found in §6.2.

Remark 2026.15 (§5). The structure of §5 (signs of κ, $\mathfrak{D}(S)$, the endoscopic groups H_m, the fundamental lemma as a theorem) is standard. The normalizations (c_S, the measures on $S(F)$, $S'(F)$ and $\bar{G}(F)$, the transfer factor $\Delta_{H,\kappa}$ and the choice of ι) have been checked only for $SL(2)$ and a depth-zero unramified elliptic γ, where $c_S = 1$, $\Delta = 1$ and $\Phi^\kappa_f(\gamma) = O_\gamma(f) - O_{\gamma'}(f) = f^\vee(\chi_0)$ (§6.2). The relation $\Delta(\gamma_H, x^{-1}\gamma x) = \kappa(x)\Delta(\gamma_H,\gamma)$ is quoted (sign conventions vary between κ and κ^{-1}). The description of χ_κ via the n-cycle is correct up to the choice of L-embedding, which may twist it by a central element. Proposition 5.2 assumes a stable version of Theorem 4.5 for H, which is not proved here (for $m < n$, H is not a torus).

Example 5.1 ($n = 2$). Let $G = SL(2)$, $G' = PGL(2)$, and let $\kappa = \eta \circ \det$ with η the unramified quadratic character, so $m = n = 2$. Let γ be unramified elliptic of depth 0 (§6.2), with elliptic torus S. Then $\mathfrak{D}(S)$ has two elements, represented by 1 and $\mathrm{diag}(\pi,1)$, and the $G'(F)$-class of γ is the union of the $G(F)$-classes of γ and $\gamma' = \mathrm{diag}(\pi,1)^{-1}\gamma\,\mathrm{diag}(\pi,1)$. With $c_S = 1$, (5.2) reads $\Phi^\kappa_f(\gamma) = O_\gamma(f) - O_{\gamma'}(f)$ and $\Phi^1_f(\gamma) = O_\gamma(f) + O_{\gamma'}(f)$, the stable orbital integral. The endoscopic group is $H = \ker(N_{E/F})$, E/F the unramified quadratic extension, which is compact. The image χ_κ of the trivial character of $H(F)$ has Satake parameter $\mathrm{diag}(1,-1)$ in $PGL(2,\mathbb{C})$, i.e. $\chi_\kappa = \chi_0$. Proposition 5.1 then gives $\Phi^\kappa_f(\gamma) = f^\vee(\chi_0)$, and (5.4A) gives

$$O_\gamma(f) = \tfrac{1}{2}\big(\Phi^1_f(\gamma) + \Phi^\kappa_f(\gamma)\big) = \tfrac{1}{2}\, SO_\gamma(f) + \tfrac{1}{2} f^\vee(\chi_0).$$

So the atom found in Example 4.5 is half the κ-orbital integral. For a ramified elliptic γ the relevant κ is ramified, and by Lemma 5.0 there is no atom (Remark 2026.16).

6 Worked examples: $PGL(2)$ and $SL(2)$

This section works out the identity (1.2) completely in rank one, where everything can be computed on the Bruhat–Tits tree. The two examples show the two behaviors anticipated by Conjecture 4.6. For $PGL(2)$ (§6.1), Hypothesis (R) holds, the spectral measure is absolutely continuous, and $I(\chi,\gamma)$ is an explicit rational function for every type of elliptic element. For $SL(2)$ (§6.2), (R) fails and the spectral measure has an atom at the unramified quadratic character, given by the κ-orbital integral of §5. A rank-two example, $PGL(3)$, is in Appendix A. §6.0 collects in one place the rank-one data used in the examples of §§2–5.

In this section $F=\mathbb{Q}_p$ is allowed in the numerical checks ($q=p=3$), but all formulas hold for any p-adic F. The geometric side is computed on the Bruhat–Tits tree $\mathscr{X}$ (vertices $PGL(2,F)/PGL(2,\mathscr{O})$, regular of degree $q+1$, metric d). The spectral side uses Lemma 2.1 and Lemma 2.2. Every formula below was checked numerically by the scripts `examples/pgl2/pgl2_check.py` and `examples/sl2/sl2_check.py`: orbital integrals by exact counts of vertices, and spectral integrals by the midpoint rule with 4×10^5 nodes on the unit circle.

6.0 The rank-one dictionary

This subsection gathers the rank-one formulas of Examples 2.1–5.1, so that §§6.1–6.2 can be read on their own. Throughout, $Q=1+q^{-1}$ and $d\chi$ is the Haar probability measure on the unit circle.

	$PGL(2)$	$SL(2)$
$X_*(A)$, a_n	$\mathbb{Z}\varpi^\vee$, $\mathrm{diag}(\pi^n,1)$	$\mathbb{Z}\alpha^\vee$, $\mathrm{diag}(\pi^n,\pi^{-n})$
character, root value	$z=\chi(a_1)$, $y=\chi(a_\alpha)=z^2$	$z=\chi(a_1)=\chi(a_\alpha)$
$\delta_B^{1/2}(a_n)$, $\mathrm{vol}(Ka_nK)$ $(n\ge 1)$	$q^{-n/2}$, $(q+1)q^{n-1}$	q^{-n}, $(q+1)q^{2n-1}$
c-function	$c_\alpha(\chi)=\frac{1-q^{-1}y}{1-y}$	$c(\chi)=\frac{1-q^{-1}z}{1-z}$
$\Gamma_\chi(a_n)$	$Q^{-1}q^{-n/2}(c_\alpha(\chi^{-1})z^n+c_\alpha(\chi)z^{-n})$	$Q^{-1}q^{-n}(c(\chi^{-1})z^n+c(\chi)z^{-n})$
Plancherel density ρ	$\frac{Q}{2}\frac{\lvert 1-y\rvert^2}{\lvert 1-q^{-1}y\rvert^2}$, zeros at $z=\pm1$	$\frac{Q}{2}\frac{\lvert 1-z\rvert^2}{\lvert 1-q^{-1}z\rvert^2}$, zero only at $z=1$
non-regular unitary χ	$z=\pm1$ (on the root wall)	$z=1$; $z=-1$ (χ_0, off the wall)
$V_B(\chi)$, χ unitary	always irreducible	$V_B(\chi_0)=\pi^+\oplus\pi^-$
$G(T)$, $T=t\varpi^\vee$ resp. $t\alpha^\vee$	ball of radius $t-1$ in the tree	even vertices at distance $<2t$

Representations (Examples 2.1A, 2.4). For $PGL(2)$ the compact picture of $V_B(\chi)$ is a space of functions on $\mathbb{P}^1(\mathscr{O})$; the spherical vector is the constant function, and the level-one vectors are the functions on $\mathbb{P}^1(\mathbb{F}_q)$. For regular χ the Jacquet module along B is $\chi\oplus s\chi$, which is why every matrix coefficient is eventually a two-term exponential sum $q^{-n/2}(C_1z^n+C_sz^{-n})$ (Example 3.1).

Fourier analysis (Example 2.3A). The Satake transforms are the Laurent polynomials in $z+z^{-1}$, with $f_n^\vee=\mathrm{vol}(Ka_nK)\Gamma_\chi(a_n)$; for instance $f_1^\vee=q^{1/2}(z+z^{-1})$ for $PGL(2)$. Inversion is $f(a_n)=\int f^\vee\Gamma_\chi(a_n)\,d\mu$. The scripts of §6 check $\int f_n^\vee\Gamma_\chi(a_m)\,d\mu=\delta_{nm}$ numerically.

Truncation and Brion (Examples 3.5, 3.6). In rank one, Lemma 3.5A is $\sum_{0\le n<t} z^n = \frac{1}{1-z} - \frac{z^t}{1-z}$: one vertex term at $T_G = 0$ and one at $T_B = T$. For spherical vectors, $\int_{G(T)} |\Gamma_\chi|^2 = 1 + (t-1)/\rho(\chi) +$ (bounded oscillatory terms), so the slope of the norm growth is $1/\rho$.

The truncated kernel and the limit (Examples 4.2, 4.5, 5.1). For an unramified elliptic γ of depth 0 the truncated kernel is a sum of two geometric series with ratio $y^{\pm 1}$ ($PGL(2)$) or $z^{\pm 2}$ ($SL(2)$). Its T-independent part $I(\chi,\gamma)$ is $\frac{1-q^{-1}}{1-\cos\theta}$ in both cases ($y = e^{i\theta}$, resp. $z = e^{i\theta}$, with I halved for $SL(2)$, Proposition 6.4). For $PGL(2)$ the oscillatory terms times ρ are smooth and the spectral measure is $I\,d\mu$. For $SL(2)$ they form a Dirichlet kernel at $z = -1$, where $\rho \neq 0$, and leave the atom $\frac{1}{2} f^\vee(\chi_0) = \frac{1}{2}\Phi_f^\kappa(\gamma)$, the κ-orbital integral of Example 5.1.

6.1 $PGL(2)$

We classify the regular elliptic elements of $PGL(2)$ by how they act on the tree, count the vertices that each moves a given distance (which gives the orbital integrals of spherical functions), and compute the truncated kernel by summing geometric series. The results are a closed form for $I(\chi,\gamma)$ for each type (Theorem 6.1), a proof of Hypothesis (R) for $PGL(2)$ (Lemma 6.1A), and a numerical comparison with counts on the tree.

Setup. Let $G = PGL(2)$, $K = PGL(2,\mathcal{O})$, A the diagonal torus. Then $X_*(A) = \mathbb{Z}\varpi^\vee$ with $a_n := \varpi^\vee(\pi)^n = \mathrm{diag}(\pi^n, 1)$, and $\alpha^\vee = 2\varpi^\vee$, so $a_\alpha = a_2$. An unramified character is determined by $z = \chi(a_1) \in \mathbb{C}^\times$, unitary iff $|z| = 1$. Then $\chi(a_\alpha) = z^2 =: y$, $W = \{1, s\}$ acts by $z \mapsto z^{-1}$, $A^- = \{a_n : n \ge 0\}$ and $\delta_B^{1/2}(a_n) = q^{-n/2}$. The double cosets are Ka_nK with $\mathrm{vol}(Ka_nK) = (q+1)q^{n-1}$ for $n \ge 1$; let $f_n = \mathbf{1}_{Ka_nK}$. For γ regular elliptic its centralizer S_γ is compact and we give it volume 1, so $O_\gamma(f) = \int_G f(g^{-1}\gamma g)\,dg$.

Spectral side. With $c_\alpha(\chi) = (1 - q^{-1}y)/(1-y)$, Lemma 2.1 reads

$$\Gamma_\chi(a_n) = Q^{-1} q^{-n/2}\big(c_\alpha(\chi^{-1}) z^n + c_\alpha(\chi) z^{-n}\big), \qquad Q = 1 + q^{-1},\ n \ge 0, \tag{6.1}$$

and $f_n^\vee(\chi) = \mathrm{vol}(Ka_nK)\Gamma_\chi(a_n)$. On $|z| = 1$, with $z = e^{i\theta}$ and $d\chi = d\theta/2\pi$, Lemma 2.2 gives $d\mu = \rho\, d\chi$ with

$$\rho(z) = \frac{Q}{2} |c_\alpha(\chi)|^{-2} = \frac{Q}{2} \frac{|1-y|^2}{|1 - q^{-1}y|^2}. \tag{6.2}$$

Plancherel inversion $\int f_n^\vee \Gamma_\chi(a_m)\, d\mu = \delta_{nm}$ holds (checked for $0 \le n, m \le 3$).

Geometric side. Since f_n is bi-K-invariant, $g^{-1}\gamma g \in Ka_nK$ iff $d(gK, \gamma gK) = n$, so

$$O_\gamma(f_n) = \#\{v \in \mathscr{X} :\ d(v, \gamma v) = n\}. \tag{6.3}$$

Up to conjugacy and the center, a regular elliptic γ generates a quadratic extension $E = F[\gamma]$, and there are three types:

($\mathbf{U}_r$) E/F unramified, $\gamma \in \mathcal{O}_E^\times$ of depth $r \ge 0$. The fixed set of γ is a ball B_r of radius r, and $d(v, \gamma v) = 2m$ at distance $m \ge 1$ from it.

($\mathbf{R}_r$) E/F ramified, $\gamma \in \mathcal{O}_E^\times$ of depth $r \geq 0$. The fixed set is the r-neighborhood of an edge, and $d(v,\gamma v) = 2m$ at distance $m \geq 1$ from it.

($\mathbf{R}'$) E/F ramified and $v_E(\gamma)$ odd. Then γ reverses an edge e, and $d(v,\gamma v) = 2d(v,e)+1$.

Counting vertices on spheres gives, for $m \geq 1$:

type	$O_\gamma(f_0)$	$O_\gamma(f_{2m})$	$O_\gamma(f_{2m+1})$
(U_r)	$\lvert B_r\rvert = 1 + \frac{(q+1)(q^r-1)}{q-1}$	$(q+1)q^{r+m-1}$	0
(R_r)	$\frac{2(q^{r+1}-1)}{q-1}$	$2q^{r+m}$	0
(R')	0	0	$2q^m$ (and $O_\gamma(f_1) = 2$)

These counts were verified for $q = 3$ by enumerating the 4373 vertices of the ball of radius 7, for $\gamma = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ (U_0; this element is elliptic of type U_0 exactly when -1 is a non-square in the residue field, e.g. $q \equiv 3 \bmod 4$, as for $q = 3$), $\left(\begin{smallmatrix} 1 & -3 \\ 3 & 1 \end{smallmatrix}\right)$ (U_1), $\left(\begin{smallmatrix} 1 & 3 \\ 1 & 1 \end{smallmatrix}\right)$ (R_0) and $\left(\begin{smallmatrix} 0 & 3 \\ 1 & 0 \end{smallmatrix}\right)$ (R').

The truncated kernel. If the fixed ball of γ of type (U_r) is centered at the base vertex, then $G(T)$ consists of the vertices at distance $< T$ from it, and (4.4) becomes

$$I^T(\chi,\gamma) = \lvert B_r\rvert + q^r \sum_{m=1}^{M} \left(c_\alpha(\chi^{-1})y^m + c_\alpha(\chi)y^{-m}\right), \qquad M = T-r-1, \tag{6.4}$$

because $O_\gamma(f_{2m})Q^{-1}q^{-m} = q^r$. (Equivalently, $I^T = \sum_{n \leq 2M} O_\gamma(f_n)\Gamma_\chi(a_n)$; this is the truncation used in the separate note `examples/pgl2/pgl2_worked_example.pdf`.) Summing the geometric series gives the vertex decomposition of Proposition 4.2:

$$I^T = I - q^r\left(c_\alpha(\chi^{-1})\frac{y^{M+1}}{1-y} + c_\alpha(\chi)\frac{y^{-M-1}}{1-y^{-1}}\right), \tag{6.5}$$

where $I = I(\chi,\gamma)$ is the T-independent part.

Theorem 6.1. *For $G = PGL(2)$ and regular unitary χ ($z \neq \pm 1$), $I(\chi,\gamma)$ is the rational function*

$$(U_r): \quad I(\chi,\gamma) = \lvert B_r\rvert + q^r \sum_{w \in W} c_\alpha\big((w\chi)^{-1}\big)\frac{(w\chi)(a_\alpha)}{\tilde\theta(w\chi)} = \lvert B_r\rvert + q^r\,\frac{\cos\theta - q^{-1}}{1-\cos\theta} \qquad (y = e^{i\theta}),$$

$$(R_r): \quad I(\chi,\gamma) = \frac{2(q^{r+1}-1)}{q-1} + \frac{2q^{r+1}}{q+1}\cdot\frac{\cos\theta - q^{-1}}{1-\cos\theta},$$

$$(R'): \quad I(\chi,\gamma) = \frac{q^{1/2}(1-q^{-1})}{q+1}\cdot\frac{\cos\psi}{\sin^2\psi} \qquad (z = e^{i\psi}).$$

Hypothesis (R) holds (Lemma 6.1A), and $O_\gamma(f) = \int f^\vee(\chi)I(\chi,\gamma)\,d\mu(\chi)$ for all $f \in \mathscr{H}(G,K)$, the integral converging absolutely.

Proof. Type (U_r); the others are identical. Multiply (6.5) by $\rho = \frac{Q}{2}(c_\alpha(\chi)c_\alpha(\chi^{-1}))^{-1}$:

$$(I^T - I)\rho = -\frac{Q}{2}\,q^r\left(\frac{y^{M+1}}{1-q^{-1}y} + \frac{y^{-M-1}}{1-q^{-1}y^{-1}}\right).$$

The coefficient of each oscillatory term, times ρ, is smooth on $|z|=1$, which is (R). Better, the first term has Fourier modes y^k only for $k \geq M+1$ and the second only for $k \leq -M-1$. Since $f^\vee$ is a Laurent polynomial in z of degree $\leq \deg f$ (the largest n with $f(a_n) \neq 0$), $\int f^\vee (I^T - I)\, d\mu = 0$ *exactly* once $2(M+1) > \deg f$. Combine with Theorem 4.5(a). The closed forms follow by summing the geometric series. $I\rho$ is smooth: see the next remark. □

Lemma 6.1A ((R) for $PGL(2)$)**.** *Hypothesis (R) holds for $G = PGL(2)$ and every level n.*

Proof. For $\phi \in C_c^\infty(G_{\mathrm{ell}})$, (4.4) gives $I^T(\chi,\phi) = \int \phi(h) I^T(\chi,h)\, dh$ with

$$I^T(\chi,h) = \int_{G(T)} \Gamma_\chi(g^{-1}hg)\, dg = \sum_v \Gamma_\chi(a_{d(v,hv)}),$$

the sum over the vertices v with $d(v_0,v) < T$. For elliptic h, let F_h be its fixed-point set (a ball or the neighborhood of an edge), or the reversed edge in type (R′). For $v \notin F_h$ we have $d(v,hv) = 2d(v,F_h)$, or $2d(v,F_h)+1$ in type (R′). Sort the vertices at distance $m \geq 1$ from F_h by the branch of the tree through which they leave F_h, and, on the branch towards v_0 (if $v_0 \notin F_h$), by the point where the geodesic to v leaves the geodesic to v_0. There are finitely many classes j, depending only on h. In each class the number of vertices at distance m is $A_j q^m$ for $m \geq m_j$, and $d(v_0,v) = m + k_j$ with $k_j \in \mathbb{Z}$ fixed. Hence, for T large,

$$I^T(\chi,h) = C(h,\chi) + \sum_j A_j \sum_{m=m_j}^{T-1-k_j} q^m \Gamma_\chi(a_{2m+\varepsilon}) \qquad (\varepsilon \in \{0,1\}),$$

where $C(h,\chi)$ collects finitely many terms independent of T. By (6.1), $q^m\Gamma_\chi(a_{2m+\varepsilon})$ is a combination of $c_\alpha(\chi^{-1}) z^\varepsilon y^m$ and $c_\alpha(\chi) z^{-\varepsilon} y^{-m}$, with constant coefficients. Summing the geometric series, the oscillatory terms are fixed multiples of

$$c_\alpha(\chi^{-1}) \frac{y^{T-k_j}}{1-y}, \qquad c_\alpha(\chi) \frac{y^{-(T-k_j)}}{1-y^{-1}}$$

(times $z^{\pm\varepsilon}$), and the T-independent part is $C(h,\chi)$ plus terms of the same shape with T removed. Multiplying by $\rho = \frac{Q}{2}(c_\alpha(\chi)c_\alpha(\chi^{-1}))^{-1}$ gives $\frac{Q}{2}(1-q^{-1}y)^{-1}$ and $\frac{Q}{2}(1-q^{-1}y^{-1})^{-1}$ up to monomials, which are smooth on $|y|=1$. The terms in $C(h,\chi)$ are finite combinations of $\Gamma_\chi(a_k)$, and $\Gamma_\chi\rho$ is smooth. All the data (F_h, the classes j, A_j, k_j, m_j) are locally constant in h, so on $\operatorname{supp}\phi$ they take finitely many values. Integrating against ϕ therefore gives coefficients of the required form. □

Remark 6.2 (poles and the Plancherel density). $I(\chi,\gamma)$ has a *double* pole at the non-regular characters $z = \pm 1$ (i.e. $y = 1$), and no other poles on $|z| = 1$. The density ρ has a double zero there, and

$$I(\chi,\gamma)\rho(z) = \frac{Q}{2} \cdot \frac{|B_r|(2-2\cos\theta) + q^r(2\cos\theta - 2q^{-1})}{1 - 2q^{-1}\cos\theta + q^{-2}} \qquad (\text{type } \mathrm{U}_r).$$

The cancellation is structural: $c_\alpha((w\chi)^{-1})\rho/\tilde\theta(w\chi) = \frac{Q}{2}(1-q^{-1}(w\chi)(a_\alpha))^{-1}$, because $\rho \propto |c|^{-2}$ supplies the factor $1-(w\chi)(a_\alpha)$ that cancels $\tilde\theta$. On its own $I(\cdot,\gamma)$ is not locally integrable at $z=\pm1$ against $d\chi$; only $I\rho$ is.

Remark 6.3 (the core term, and sign). In type (U_r) the constant $|B_r| = O_\gamma(\mathbf{1}_K)$ comes from the vertices fixed by γ. There no Casselman asymptotic is available, and this is the T-independent remainder J_G of Proposition 3.6. The second term is a sum over W of c-functions divided by $\tilde{\theta}$. In types (U_r) and (R_r), I is positive on regular unitary χ (its minimum, at $\cos\theta = -1$, is $|B_r| - q^r\frac{1+q^{-1}}{2} > 0$). In type ($R'$) it changes sign with $\cos\psi$. So $I(\cdot,\gamma)\,d\mu$ is a signed measure in general.

	(U_0)		(U_1)		(R_0)		(R')	
n	tree	spectral	tree	spectral	tree	spectral	tree	spectral
0	1	1.000000	5	5.000000	2	2.000000	0	0.000000
1	0	0.000000	0	0.000000	0	0.000000	2	2.000000
2	4	4.000000	12	12.000000	6	6.000000	0	0.000000
3	0	0.000000	0	0.000000	0	0.000000	6	6.000000
4	12	12.000000	36	36.000000	18	18.000000	0	0.000000
5	0	0.000000	0	0.000000	0	0.000000	18	18.000000
6	36	36.000000	108	108.000000	54	54.000000	0	0.000000
7	0	0.000000	0	0.000000	0	0.000000	54	54.000000

Table 1: $PGL(2)$, $q=3$: $O_\gamma(f_n)$ counted on the tree versus $\int f_n^\vee I(\chi,\gamma)\,d\mu$ with I from Theorem 6.1. For $f = f_4$ and type (U_0), $\int f^\vee I^T\,d\mu$ equals 0 for $2M < 4$ and exactly 12 for $2M \geq 4$.

6.2 $SL(2)$: an atom

For $SL(2)$ the same computation has a different outcome. At the unramified quadratic character χ_0, a non-regular unitary character where the Plancherel density does not vanish, the vertex terms have poles that nothing cancels, and Hypothesis (R) fails. Proposition 6.4 shows that the spectral measure of an elliptic orbital integral then has an atom at χ_0, equal to half the κ-orbital integral, as §5 predicts.

Setup. Let $G = SL(2)$, $K = SL(2,\mathcal{O})$. Now $X_*(A) = \mathbb{Z}\alpha^\vee$, $a_n = \mathrm{diag}(\pi^n,\pi^{-n})$, $z = \chi(a_1) = \chi(a_\alpha)$, $\delta_B^{1/2}(a_n) = q^{-n}$ and $\mathrm{vol}(Ka_nK) = (q+1)q^{2n-1}$ for $n \geq 1$. Lemma 2.1 reads $\Gamma_\chi(a_n) = Q^{-1}q^{-n}(c(\chi^{-1})z^n + c(\chi)z^{-n})$ with $c(\chi) = (1-q^{-1}z)/(1-z)$, and $\rho = \frac{Q}{2}|1-z|^2/|1-q^{-1}z|^2$. The vertices of G/K are the vertices of $\mathscr{X}$ at even distance from the base vertex v_0. So

$$O_\gamma(f_n) = \#\{v:\ d(v_0,v) \text{ even},\ d(v,\gamma v) = 2n\}.$$

The non-regular unitary characters are $z=1$ and $z=-1$. At $z=1$, ρ has a double zero, as for $PGL(2)$. At $z=-1$ (the unramified quadratic character χ_0), $s\chi_0 = \chi_0$ but $\chi_0(a_\alpha) = -1 \neq 1$, so $\rho(\chi_0) = \frac{Q}{2}\cdot\frac{4}{(1+q^{-1})^2} \neq 0$. Here $i_{G,A}(\chi_0) = \pi^+ \oplus \pi^-$ is reducible, with R-group of order 2.

An unramified elliptic element. Let $\gamma \in SL(2,\mathcal{O})$ have characteristic polynomial $x^2 - tx + 1$ irreducible modulo π. Then $E = F[\gamma]$ is the unramified quadratic extension, $\gamma \in \mathcal{O}_E^\times$ has depth 0

(type U_0), and γ fixes exactly the base vertex v_0. For example, $\gamma = \left(\begin{smallmatrix} 0 & -1 \\ 1 & t \end{smallmatrix}\right)$ for any such t; when $q \equiv 3 \bmod 4$ one may take $\gamma = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, as in the numerical checks ($q = 3$). (For $q \equiv 1 \bmod 4$ that element is split, since -1 is a square.) Let γ' be its conjugate by $\mathrm{diag}(\pi, 1) \in GL(2,F)$. Then γ' is stably conjugate but not $SL(2,F)$-conjugate to γ; it fixes only the neighbor $\mathrm{diag}(\pi,1)v_0$, which has odd type. Counting gives

$$O_\gamma(f_n) = (q+1)q^{n-1} \cdot \tfrac{1+(-1)^n}{2}, \qquad O_{\gamma'}(f_n) = (q+1)q^{n-1} \cdot \tfrac{1-(-1)^n}{2} \qquad (n \geq 1),$$

with $O_\gamma(f_0) = 1$ and $O_{\gamma'}(f_0) = 0$. The paper's truncated kernel (4.4) is

$$I^T(\chi, \gamma) = 1 + \sum_{k=1}^{T-1} \left(c(\chi^{-1}) z^{2k} + c(\chi) z^{-2k} \right),$$

since $G(T)$ consists of the vertices at distance $2k < 2T$ from v_0, and $d(v, \gamma v) = 2d(v_0, v)$. Its T-independent part is

$$I(\chi, \gamma) = 1 + c(\chi^{-1}) \frac{z^2}{1 - z^2} + c(\chi) \frac{z^{-2}}{1 - z^{-2}}, \tag{6.6}$$

which is regular on $|z| = 1$ away from $z = 1$. In fact, since $z\,c(\chi^{-1}) + c(\chi) = 1 + z$, a short computation gives the closed form

$$I(\chi, \gamma) = \frac{1 - q^{-1}}{2(1 - \cos\theta)} \qquad (z = e^{i\theta}),$$

which has the same shape as the $PGL(2)$ kernel of type (U_0) in Theorem 6.1. The vertex terms

$$-c(\chi^{-1}) \frac{z^{2T}}{1 - z^2} - c(\chi) \frac{z^{-2T}}{1 - z^{-2}}$$

each have a simple pole at $z = -1$ where $\rho \neq 0$. So Hypothesis (R) *fails*, although their sum is smooth for each T (Remark 2026.12).

Proposition 6.4. *For $G = SL(2)$, γ as above and all $f \in \mathscr{H}(G,K)$, let $I_{st}(\chi) = 1 + c(\chi^{-1}) \frac{z}{1-z} + c(\chi) \frac{z^{-1}}{1-z^{-1}}$ (the kernel (6.6) for $SO_\gamma = O_\gamma + O_{\gamma'}$). Then $I(\chi,\gamma) = \frac{1}{2} I_{st}(\chi)$, and*

$$O_\gamma(f) = \int f^\vee(\chi)\, I(\chi,\gamma)\, d\mu(\chi) + \tfrac{1}{2} f^\vee(\chi_0) = \tfrac{1}{2} \int f^\vee(\chi)\, I_{st}(\chi)\, d\mu(\chi) + \tfrac{1}{2} f^\vee(\chi_0),$$

$$O_{\gamma'}(f) = \tfrac{1}{2} \int f^\vee(\chi)\, I_{st}(\chi)\, d\mu(\chi) - \tfrac{1}{2} f^\vee(\chi_0).$$

So $dI_\gamma = I(\chi,\gamma)\, d\mu + \frac{1}{2}\delta_{\chi_0}$: the spectral measure of O_γ has an atom at χ_0, and the κ-orbital integral is

$$\Phi^\kappa_f(\gamma) = O_\gamma(f) - O_{\gamma'}(f) = f^\vee(\chi_0),$$

in agreement with Proposition 5.1 ($n = 2$, $\chi_\kappa = \chi_0$, $\Delta = 1$).

Proof. The stable orbital integral has coefficients $SO_\gamma(f_n) = (q+1)q^{n-1}$ for $n \geq 1$, exactly as for $PGL(2)$ type (U_0), and the same argument as for Theorem 6.1 gives $SO_\gamma(f) = \int f^\vee I_{st}\, d\mu$. The poles of I_{st} are at $z = 1$ only, where ρ has a double zero. For the κ-part, $f_n^\vee(\chi_0) = \mathrm{vol}(Ka_nK)Q^{-1}q^{-n}(-1)^n \cdot 2c(\chi_0) = (q+1)q^{n-1}(-1)^n$ for $n \geq 1$, since $c(\chi_0) = (1+q^{-1})/2 = Q/2$; and $f_0^\vee(\chi_0) = 1$. This equals $O_\gamma(f_n) - O_{\gamma'}(f_n)$. Adding and subtracting gives the formulas for O_γ and $O_{\gamma'}$. Finally, $2I - I_{st} = 1 - \frac{zc(\chi^{-1})+c(\chi)}{1+z} = 0$ by the identity $zc(\chi^{-1}) + c(\chi) = 1+z$, so $I = \frac{1}{2}I_{st}$. □

Remark 2026.16 (§6.2). Proposition 6.4 is proved above, and was also verified numerically for $q = 3$ and $f = f_0, \ldots, f_3$ (`sl2_check.py`, checks (2), (3) and (5)); the identity $I = \frac{1}{2}I_{st}$ was checked symbolically. For a ramified elliptic $\gamma = \left(\begin{smallmatrix} 2 & 3 \\ 1 & 2 \end{smallmatrix}\right)$ ($E = \mathbb{Q}_3(\sqrt{3})$) there is no atom (check (4)), as the fundamental lemma predicts: the relevant κ is ramified, and κ-orbital integrals of spherical functions vanish.

n	$O_\gamma(f_n)$	$O_{\gamma'}(f_n)$	$\frac{1}{2}\int f_n^\vee I_{st}\, d\mu$	$\frac{1}{2}f_n^\vee(\chi_0)$	$\int f_n^\vee I(\chi,\gamma)\, d\mu$
0	1	0	0.5	0.5	0.5
1	0	4	2.0	−2.0	2.0
2	12	0	6.0	6.0	6.0
3	0	36	18.0	−18.0	18.0

Table 2: $SL(2)$, $q = 3$, $\gamma = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$. The atom-free integral in the last column misses exactly $\frac{1}{2}f_n^\vee(\chi_0)$.

7 Route B: the identity from Arthur's local trace formula

This section explains how (1.2) follows from Arthur's local trace formula [Ar91], and serves as an introduction to Arthur's work through this special case. It is a sketch.

Arthur's formula expresses the same distribution T_ϕ in much greater generality, as a sum of weighted orbital integrals on one side and of weighted characters on the other. §7.1 states it. §7.2 and §7.3 specialize its two sides to f_1 spherical and f_2 elliptic, where most terms vanish. §7.4 lists what would be needed to turn this into a proof of Conjecture 4.6, and compares the two routes.

Remark 2026.17 (§7). Nothing in this section has been checked against [Ar91, Ar93] at the level of normalizations: the constants $|W_0^M||W_0^G|^{-1}$, the signs $(-1)^{\dim(A_M/A_G)}$, the measures on the spectral sets, and the identification of $I(\chi,\phi)$ with a weighted character (Conjecture 7.3). The structural statements (which terms survive) use only the definitions and the splitting formulas.

7.1 Arthur's formula

We recall Arthur's truncated kernel and the two expansions of the constant term of its integral: over Levi subgroups M, as weighted orbital integrals and as weighted characters.

Let $f = f_1 \otimes f_2 \in C_c^\infty(G(F) \times G(F))$ act on $L^2(G(F))$ by R as in §1, with kernel (1.4). Arthur truncates the diagonal integral with a function $u(x,T)$, the characteristic function of a large compact

set modulo A_G built from the building, and shows that $J^T(f) = \int_{G(F)/A_G(F)} K_f(x,x)u(x,T)\,dx$ is asymptotic to a polynomial in T, whose constant term $J(f)$ has two expansions [Ar91]:

$$\begin{aligned} J(f) &= \sum_M \frac{|W_0^M|}{|W_0^G|}(-1)^{\dim(A_M/A_G)} \int_{\Gamma_{\mathrm{ell}}(M)} J_M(\gamma,f)\,d\gamma \\ &= \sum_M \frac{|W_0^M|}{|W_0^G|}(-1)^{\dim(A_M/A_G)} \int_{\Pi(M)} J_M(\pi,f)\,d\pi. \end{aligned}$$

Both sums run over Levi subgroups $M \supseteq M_0$. In the first (geometric) expansion, $\Gamma_{\mathrm{ell}}(M)$ is the set of elliptic conjugacy classes of M, and $J_M(\gamma,f)$ is the weighted orbital integral of $f_1 \otimes f_2$ over γ, with weight $v_M(x_1,x_2)$ given by the (G,M)-family $e^{-\lambda(H_P(x_1)-H_{\bar{P}}(x_2))}$. In the second (spectral) expansion, $\Pi(M)$ is a space of tempered representations of M; in [Ar93] it is reorganized in terms of the virtual elliptic tempered characters $T_{\mathrm{ell}}(M)$. $J_M(\pi,f)$ is the weighted character, built from the (G,M)-family of normalized intertwining operators $\lambda \mapsto R_{P'|P}(\pi)^{-1}R_{P'|P}(\pi_\lambda)$. For $M = G$ the weights are 1 and $J_G(\gamma,f) = |D(\gamma)|O_\gamma(f_1)O_\gamma(f_2)$, $J_G(\pi,f) = \operatorname{tr}\pi(f_1)\operatorname{tr}\pi^\vee(f_2)$.

7.2 Specialization: the geometric side

On the geometric side, the elliptic support of ϕ kills every term except $M = G$, and no truncation is needed.

Take $f_1 = f \in \mathscr{H}(G,K)$ and $f_2 = \phi$ supported on G_{ell}. By Lemma 4.4, $x \mapsto K_f(x,x)$ has compact support modulo A_G, so $J^T(f) = I(f,\phi)$ for all large T and $J(f) = I(f,\phi)$. For $M \neq G$ and $\gamma \in \Gamma_{\mathrm{ell}}(M)$ regular in G, γ is not elliptic in G, so ϕ vanishes on its G-orbit and $J_M(\gamma,f) = 0$. (Singular γ contribute a set of measure zero.) The geometric side therefore reduces to the $M = G$ term, which is (4.11).

7.3 Specialization: the spectral side

On the spectral side, the splitting formula for weighted characters and the special form of f_1 and f_2 leave only unramified principal series of Levi subgroups, paired with weighted characters of ϕ. This leads to Conjecture 7.3, which identifies $I(\chi,\phi)$ with a weighted character.

The weighted characters satisfy a splitting formula [Ar88]: $J_M(\pi, f_1 \otimes f_2)$ is a sum over pairs (L_1,L_2) of Levi subgroups containing M, with coefficients $d_M^G(L_1,L_2)$, of products of weighted characters of L_1 applied to (a constant term of) f_1 and of L_2 applied to f_2. Two facts simplify it here.

- f_1 *spherical.* Normalized intertwining operators fix the spherical vector of an unramified principal series. So the weighted characters of f_1 for $L_1 \supsetneq M$ vanish, forcing $L_1 = M$ and $L_2 = G$ ($d_M^G(M,G) = 1$). Moreover $\operatorname{tr}\pi(f_1)$ is non-zero only if π has a $K \cap M$-fixed vector, i.e. π is a constituent of an unramified principal series of M.
- ϕ *elliptic.* ϕ is cuspidal: $\operatorname{tr}\sigma(\phi) = 0$ for every properly induced σ. So only the elliptic part of the spectrum survives in the $L_2 = G$ factor, and the factor is the weighted character $J_M(\pi^\vee,\phi)$.

The $M = M_0$ term is therefore

$$\frac{(-1)^{\dim(A_0/A_G)}}{|W|}\int_{\widehat{T}^u} f^\vee(\chi)\, J_{M_0}(\chi^{-1},\phi)\, d\chi,$$

an absolutely continuous measure in χ. The terms with $M \neq M_0$ come from elliptic tempered virtual characters τ of M with $K\cap M$-fixed vectors. These exist only at unramified χ whose R-group (for M) is non-trivial, and they contribute $f^\vee(\chi)$ times $J_M(\tau^\vee,\phi)$, summed over a set of χ of lower dimension. For $SL(2)$ and $M = G$ this is $\tau = \pi^+ - \pi^-$ at χ_0, with $\operatorname{tr}\tau(f) = f^\vee(\chi_0)$ (the spherical vector lies in π^+) and $\operatorname{tr}\tau^\vee(\phi)$ given by the κ-character. This is the atom of Proposition 6.4.

Conjecture 7.3 (expectation)**.** *For regular unitary χ,*

$$I(\chi,\phi)\rho(\chi) = (-1)^{\dim(A_0/A_G)}|W|^{-1} J_{M_0}(\chi^{-1},\phi).$$

The singular part dI_ϕ^{sing} of Conjecture 4.6 is the sum of the contributions of the elliptic tempered virtual characters with K-fixed vectors of the Levi subgroups $M \supsetneq M_0$.

Example 7.3 (rank one, in outline). For $PGL(2)$ the discrete series have no K-fixed vectors, so for f_1 spherical only the $M = A$ term survives. It is an integral over $\widehat{T}^u$ of $f^\vee$ against a weighted character of ϕ, which is the shape of $\int f^\vee I\, d\mu$. For $SL(2)$, Arthur's elliptic set for $M = G$ also contains the virtual character $\pi^+ - \pi^-$ attached to χ_0 (the R-group has order 2; Example 2.1A). Since only π^+ has a K-fixed vector, its trace against f_1 is $f^\vee(\chi_0)$, and this term is the point mass at χ_0 found in Example 4.5. The constants have not been checked against [Ar93] (Remark 2026.17).

7.4 A proposed route to Conjecture 4.6, and comparison with Route A

We list the steps needed to make Route B a proof, and say what each route contributes that the other does not.

The argument of §§7.2–7.3 is a proposed route to Conjecture 4.6(a), not a proof. To complete it one would have to (i) justify the splitting and vanishing statements of §7.3 with Arthur's actual normalizations; (ii) show that the spectral side, for f_1 spherical and f_2 elliptic, is given by a finite complex measure in the unramified variable (Arthur's spectral side is a priori an expansion of a linear form, and its convergence in this form is part of what must be checked); and (iii) identify the singular part with the elliptic tempered characters having K-fixed vectors. If this is carried out, the result would be an existence statement for dI_ϕ as in Conjecture 4.6(a), with absolutely continuous part a weighted character and singular part supported where R-groups are non-trivial.

What Route A adds is an explicit and elementary description of the absolutely continuous part, as the T-independent part of the truncated kernel (Proposition 4.2), together with closed forms in rank one (§6). The two truncations are related: Arthur's $u(x,T)$ and our $G(T)$ are both defined by the convex hull of $W\cdot T$ in $\mathfrak{a}_0$, and our vertex terms $\psi(T_Q)R_Q$ (Lemma 3.5A) are periodic counterparts of Arthur's $e^{\lambda(T_Q)}c_Q(\lambda)/\theta_Q(\lambda)$. (We do not claim a general dictionary between the two: Lemma 2.8, which would provide one, holds only when the coroot lattice is saturated; Remark 2026.4.) In rank one, the constant term of $\sum_Q e^{\lambda(T_Q)}c_Q(\lambda)/\theta_Q(\lambda)$ at $\lambda = 0$ is a logarithmic derivative of an intertwining operator, which is how weighted characters arise; that is the structure visible in (6.5).

8 Summary of results and open questions

This section collects the main results, with the status that the text gives them, and lists the questions that remain open. Nothing in it is new. Each entry repeats a statement that is made, and proved or qualified, earlier in the paper.

Results. The table gives each main statement and its status.

Statement	Status
Truncated identity, Theorem 4.5(a)	proved, for G satisfying (H)
Brion decomposition, Lemmas 3.5A–3.5C	proved; checked for $SL(3)$, $PGL(3)$
Vertex decomposition of J^T, I^T (Props. 3.6, 4.2)	proved in rank one; in rank ≥ 2 conditional on Casselman's input (C); (4.7A) confirmed for one γ in $PGL(3)$ (App. A)
Spectral identity, Theorem 4.5(b)	proved under (C) and (R); holds numerically for one γ in $PGL(3)$, where (R) fails (App. A)
Hypothesis (R)	proved for $PGL(2)$ (Lemma 6.1A); fails for $SL(2)$ (§6.2) and for $PGL(3)$ (App. A)
Weaker hypothesis (R′), Remark 2026.20	proposed; consistent with the $PGL(3)$ data
$SL(2)$ atom, Proposition 6.4	proved, all q
Shape of the spectral measure, Conjecture 4.6	verified for $PGL(2)$, $SL(2)$; numerically for one γ in $PGL(3)$; open in general
κ-spectral measure, κ of order n, Prop. 5.1	proved (from the fundamental lemma); normalizations checked for $SL(2)$
κ-spectral measure, general $m \mid n$, Prop. 5.2	conditional on a stable Theorem 4.5 for H_m
Lemma 2.8	proved when $\mathbb{Z}\Delta^\vee$ is saturated; false for $PGL(2)$
Route B (Section 7)	proposed route; normalizations not checked

The numerical checks are `examples/pgl2/pgl2_check.py`, `examples/sl2/sl2_check.py`, `examples/rank2/brion_check.py` and `examples/pgl3/pgl3_check.py` (run all with `python3 run_checks.py` at the top of the repository).

Open questions. Each of the following is raised in the text at the place indicated.

1. Prove the input (C) of §3.1 for intermediate parabolic subgroups (Remark 2026.6). This would make Propositions 3.6 and 4.2 unconditional in rank ≥ 2.

2. Prove the weaker hypothesis (R′) of Remark 2026.20, or find the right replacement for (R). Remark 2026.20 conjectures that (R′), rather than (R), is the condition behind Conjecture 4.6(b) (see also Remark 2026.14).

3. Prove Conjecture 4.6. Part (a) asks that the spectral functional be a finite complex measure whose singular part is supported on the non-regular unitary characters at which ρ does not vanish. Part (c) asks that for $SL(n)$ the singular part be given by endoscopy.

4. Prove the stable form of Theorem 4.5 for the endoscopic groups H_m that Proposition 5.2 assumes (Remark 2026.15).

5. Show that $\phi \mapsto I(\chi,\phi)$ is G-admissible (Remark 2026.11).

6. Prove Observation A.2, the displacement table for $PGL(3)$ (Remark 2026.19). Treat the other conjugacy classes of $PGL(3)$: positive depth, ramified E, and tori split by a quadratic extension (§A.7).

7. Complete Route B. This means checking Arthur's normalizations and proving Conjecture 7.3. The steps are listed in §7.4.

8. Two extensions are not attempted here: general reductive G, with central characters (Remark 2026.1), and a form of Lemma 2.8 without the saturation hypothesis (Remark 2026.4).

A $PGL(3)$: a rank-two test of Hypothesis (R)

This appendix reports a computation on the Bruhat–Tits building of $PGL(3)$. It was done to test Hypothesis (R) in rank two. In brief:

- for an unramified elliptic element γ of depth 0, the T-independent part $I(\chi,\gamma)$ has the closed form (A.4), with double poles only on the root walls;
- the spectral identity $O_\gamma(f) = \int f^\vee(\chi) I(\chi,\gamma)\, d\mu(\chi)$ holds, with no singular part (Table A.1);
- yet Hypothesis (R) *fails*: some oscillatory coefficients, times ρ, have simple poles along non-root subtori where $\rho \neq 0$.

So (R) is sufficient for Theorem 4.5(b) but not necessary. A weaker hypothesis (R′), which separates $PGL(3)$ from $SL(2)$, is proposed in Remark 2026.20. The computations are in `examples/pgl3/` (`pgl3_check.py` runs them in a few seconds).

A.1 Setup

We fix the group and the element γ, recall how vertices of the building correspond to double cosets, and write the orbital integral and the truncated kernel as sums over vertices.

Let $G = PGL(3)$ over F with residue field of order q, $K = PGL(3,\mathcal{O})$, and $\mathrm{vol}(K) = 1$. The vertices of the building $\mathscr{X}$ are the homothety classes of $\mathcal{O}$-lattices in F^3, and $G(F)/K$ is the set of vertices; v_0 is the class of $\mathcal{O}^3$. For vertices v, v' the *vector distance* $d(v,v')$ is the dominant coweight $\lambda = (l_1, l_2, 0)$, $l_1 \geq l_2 \geq 0$, such that $v' = g a_\lambda v_0$ when $v = g v_0$, where $a_\lambda = \mathrm{diag}(\pi^{l_1}, \pi^{l_2}, 1)$. Concretely, if $L' \subseteq L$ represent v', v, then π^{l_1}, π^{l_2}, 1 are the elementary divisors of L' in L after

scaling. We write $a = \langle \lambda, \alpha_1 \rangle = l_1 - l_2$, $b = \langle \lambda, \alpha_2 \rangle = l_2$, and use (a,b) and (l_1,l_2) interchangeably. Then

$$\mathrm{vol}(Ka_\lambda K) = \begin{cases} 1 & \lambda = 0, \\ \frac{Q}{1+q^{-1}}\, q^{2l_1} & \lambda \neq 0,\ ab = 0, \\ Q\, q^{2l_1} & ab \neq 0, \end{cases} \qquad Q = Q_G = (1+q^{-1})(1+q^{-1}+q^{-2}), \tag{A.1}$$

and $\langle 2\rho, \lambda \rangle = 2l_1$.

Let $\gamma \in GL(3, \mathscr{O})$ be the companion matrix of a monic cubic whose reduction mod π is irreducible. Then $E = F[\gamma]$ is the unramified cubic extension, $\gamma \in \mathscr{O}_E^\times$ has depth 0, and γ is regular elliptic in $G(F)$. Its only fixed vertex is v_0, since an invariant lattice strictly between $\mathscr{O}^3$ and $\pi\mathscr{O}^3$ would give an invariant line or plane mod π. The orbital integral and the truncated kernel (4.4) are

$$O_\gamma(f) = \int_{G(F)} f(g^{-1}\gamma g)\, dg = \sum_{v \in \mathscr{X}^0} f\big(a_{d(v,\gamma v)}\big), \qquad I^T(\chi, \gamma) = \sum_{v:\ d(v_0,v) \in \mathscr{A}_A(T)} \Gamma_\chi\big(a_{d(v,\gamma v)}\big),$$

where $\mathscr{X}^0$ is the set of vertices. In the coordinates $t_1 = \langle T, \alpha_1 \rangle$, $t_2 = \langle T, \alpha_2 \rangle$, the truncation set $\mathscr{A}_A(T)$ of §3 is $\{(a,b) \in \mathbb{Z}^2_{\geq 0} :\ 2a+b < 2t_1+t_2,\ a+2b < t_1+2t_2\}$, because $\langle \lambda, \varpi_1 \rangle = (2a+b)/3$ and $\langle \lambda, \varpi_2 \rangle = (a+2b)/3$. We take γ to be the companion matrix of x^3+x+1 for $q=2$, and of x^3-x+1 for $q=3$.

A.2 The displacement table

The geometric input is a table of how far γ moves each vertex, sorted by the vertex's distance from v_0. We compute it by enumerating the building and find a simple closed form, which we state as an observation.

Write the vertices in canonical (Hermite normal form) bases, and enumerate them by breadth-first search from v_0. Each vertex has $2(q^2+q+1)$ neighbors, corresponding to the lines and planes of $\mathbb{F}_q^3$. The number of vertices at vector distance λ from v_0 agrees with (A.1). Tabulating

$$N(\lambda, \mu) = \#\{v :\ d(v_0, v) = \lambda,\ d(v, \gamma v) = \mu\}$$

for all v with $l_1 \leq 6$ when $q=2$ (86 017 vertices) and all v with $l_1 \leq 4$ when $q=3$ (62 193 vertices) gives the following.

Observation A.2. Let $k = l_1 = a+b$ and $s = \min(a,b)$. Then $d(v, \gamma v) = \lambda + \lambda^* - j\alpha^\vee$ for some $\alpha \in \{\alpha_1, \alpha_2\}$ and some $0 \leq j \leq s$, where $\lambda + \lambda^* = k\rho^\vee$, i.e. $\mu = (2k,k)$, $(2k-j, k+j)$ or $(2k-j, k-2j)$ in the coordinates (m_1, m_2). The proportions are as follows.

$$\frac{N(\lambda,\mu)}{\mathrm{vol}(Ka_\lambda K)} = \begin{cases} 1, & s = 0,\ \mu = k\rho^\vee; \\ \dfrac{q-1}{q+1}, & s \geq 1,\ \mu = k\rho^\vee; \\ \dfrac{q-1}{q+1}\, q^{-j} \ \text{ for each } \alpha, & s \geq 1,\ \mu = k\rho^\vee - j\alpha^\vee,\ 1 \leq j \leq s-1; \\ \dfrac{q^{1-s}}{q+1} \ \text{ for each } \alpha, & s \geq 1,\ \mu = k\rho^\vee - s\alpha^\vee. \end{cases} \tag{A.2}$$

The table is the same for three other elements of γK_1 that were tried, so for ϕ the normalized characteristic function of γK_1 one has $I^T(\chi,\phi) = I^T(\chi,\gamma)$.

Remark 2026.19 (Observation A.2). (A.2) is an observation, checked on every vertex in the ranges stated, not a theorem. On the walls ($s = 0$) it is easy: γ moves every line and every plane of $\mathbb{F}_q^3$, and $d(v,\gamma v) = k\rho^\vee$. In the interior the parameter j measures how far the flag attached to v is from being γ-stable modulo π^j. We have not written out a proof. Everything below that uses (A.2) for l_1 beyond the checked range is conditional on it.

A.3 The spectral side

We combine the table with Macdonald's formula. The sum over the truncated cone becomes a sum of geometric series, and its T-independent part gives the closed form (A.4) for $I(\chi,\gamma)$.

Write $z = (z_1,z_2,z_3)$, $z_1z_2z_3 = 1$, for the unramified character χ with $\chi(\mathrm{diag}(x_1,x_2,x_3)) = \prod z_i^{v(x_i)}$, $z^\alpha = z_i/z_j$ for $\alpha = e_i - e_j$, and $c(\chi) = \prod_{\alpha>0}(1-q^{-1}z^\alpha)/(1-z^\alpha)$. For dominant $\mu = (m_1,m_2,0)$, Macdonald's formula (Lemma 2.1, written for the dominant representative of $Ka_\mu K$) is

$$\Gamma_\chi(a_\mu) = Q^{-1}q^{-m_1}\sum_{w\in W} c\big((w\chi)^{-1}\big)(w\chi)(a_\mu), \qquad \rho(\chi) = \frac{Q}{6}\,|c(\chi)|^{-2}. \tag{A.3}$$

Passing to the dominant representative may replace χ by χ^{-1}. This does not matter here, because every function below is invariant under $\chi \mapsto \chi^{-1}$.

Put $u_w = (w\chi)(a_{\alpha_1^\vee})$, $v_w = (w\chi)(a_{\alpha_2^\vee})$ and $X_w = u_wv_w$. Insert (A.2) into $\sum_\mu N(\lambda,\mu)\Gamma_\chi(a_\mu)$. The volume (A.1), the factor $q^{-m_1} = q^{-(2k-j)}$ and Q^{-1} combine to the weight $\frac{q-1}{q+1}$ for $j<s$ and $\frac{q}{q+1}$ for $j=s$ and on the walls. Since $\mu = k\rho^\vee - j\alpha^\vee$, one gets $(w\chi)(a_\mu) = X_w^k u_w^{-j}$ or $X_w^k v_w^{-j}$. Hence

$$I^T(\chi,\gamma) = 1 + \sum_{w\in W} c\big((w\chi)^{-1}\big)\sum_{0\neq\lambda\in\mathscr{A}_A(T)} S_\lambda(u_w,v_w),$$

$$S_\lambda(u,v) = \begin{cases} \dfrac{q}{q+1}X^k, & s=0,\\[2mm] X^k\Big[\dfrac{q-1}{q+1}\Big(1+\sum_{j=1}^{s-1}(u^{-j}+v^{-j})\Big) + \dfrac{q}{q+1}\big(u^{-s}+v^{-s}\big)\Big], & s\geq 1, \end{cases} \qquad X = uv.$$

The T-independent part of the inner sum is its sum over the whole dominant cone, summed as geometric series (the origin vertex term of Lemma 3.5A):

$$F_0(u,v) = \frac{X\big[(q-1)(X+X^3) - (q+1)X^2 + q\big(2+(u+v)(1-X)\big)\big]}{(q+1)(1-X)(1-u^2v)(1-uv^2)}.$$

The denominators $1-u_w^2v_w$ and $1-u_wv_w^2$ vanish on the subtori $H_i = \{z_i^3 = 1\}$, which are not root hypersurfaces. On these subtori $\rho \neq 0$. These poles cancel in the sum over W, and an exact (computer-algebra) computation gives

$$\begin{aligned} I(\chi,\gamma) &= \frac{3(q-1)^2}{q^3(q+1)}\cdot\frac{2(q^2+q+1) - q\sum_{\alpha\in\Phi} z^\alpha}{\prod_{\alpha>0}|1-z^\alpha|^2},\\ I(\chi,\gamma)\rho(\chi) &= \frac{(q-1)^2(q^2+q+1)}{2q^6}\cdot\frac{2(q^2+q+1) - q\sum_{\alpha\in\Phi} z^\alpha}{\prod_{\alpha\in\Phi}(1-q^{-1}z^\alpha)}. \end{aligned} \tag{A.4}$$

Thus $I(\cdot,\gamma)$ has double poles exactly on the root walls, like the $PGL(2)$ kernel of Theorem 6.1, and $I\rho$ is smooth on $\widehat{T}^u$. Since $\sum_\Phi z^\alpha \le 6$, the numerator is at least $2(q-1)^2$, so $I>0$. Compare type (U_0) for $PGL(2)$: $I = 2(1-q^{-1})/|1-y|^2$.

A.4 The spectral identity

We test the spectral identity by computing both sides independently for several spherical functions.

$O_\gamma(f_v)$ was counted on the building and compared with $\int f_v^\vee I\,d\mu$, where

$$f_v^\vee = \mathrm{vol}(Ka_vK)\Gamma_\chi(a_v)$$

is computed as an explicit Laurent polynomial. The integral was evaluated on a 160×160 grid. The integrand is smooth, so this is accurate to rounding error.

$v=(n_1,n_2)$	$(0,0)$	$(2,1)$	$(3,0)$	$(4,2)$	$(5,1)$	$(5,4)$	$(6,3)$	$(6,0)$
$O_\gamma(f_v)$, $q=2$	1	14	14	70	112	112	336	112
$O_\gamma(f_v)$, $q=3$	1	26	39	312	702	702	3510	1053

TABLE A.1. Elliptic orbital integrals of $PGL(3)$ for γ of depth 0. In every case $\int f_v^\vee I(\chi,\gamma)\,d\mu$ agrees to within 10^{-12}, and also for $v=(1,0),(2,0),(3,1),(4,1)$, where both sides are 0.

So $dI_\gamma = I(\chi,\gamma)\,d\mu$, with no singular part. This is what Conjecture 4.6(b) predicts, since $Z(G)$ is connected.

A.5 The expansion in T

To test Hypothesis (R) we need the whole exponential-polynomial expansion (4.7A) of I^T in T, not only its constant term. We find it numerically.

At a fixed generic χ, $I^T(\chi,\gamma)$ was computed for 66 values of T with t_1,t_2 even and $t_1\equiv t_2 \bmod 3$, so that all the vertices of the truncated region are lattice points. Every value is fitted exactly (residual 10^{-14}) by an exponential polynomial with constant coefficients:

$$I^T(\chi,\gamma) = \sum_\mu a_\mu(\chi)\,\chi(\mu(T)),$$

with 19 non-zero terms. The maps μ all have the form $T\mapsto w^{-1}T - w'^{-1}T$, as in (4.7A) with $Q=B$. Here $a_0 = I(\chi,\gamma)$ is (A.4), and a fit on the cone $t_1\le t_2$ predicts the values on $t_1>t_2$. This is a rank-two check of Proposition 4.2(b) for this γ. (The individual Weyl terms in §A.3 are only *piecewise* exponential polynomials, switching along $t_1=t_2$ because of $\min(a,b)$ in (A.2). The switching cancels in the sum over W.)

A.6 Hypothesis (R) fails

We examine the coefficients of the oscillatory terms near the singular set, and explain why their poles nevertheless leave no singular part in the limit.

Let χ approach a point of $H_3 = \{z_3^3 = 1\}$ that is not on a root wall, at distance ε. Then 8 of the coefficients $a_\mu(\chi)\rho(\chi)$ grow like ε^{-1} (measured for $\varepsilon = 10^{-2}, 10^{-3}, 10^{-4}$). So they have simple poles along H_3, where $\rho \neq 0$, and Hypothesis (R) fails for $PGL(3)$. By symmetry the same holds along H_1 and H_2.

The poles cancel in pairs. For example, take

$$\mu_1(T) = (t_1 + t_2,\, -t_1,\, -t_2), \qquad \mu_2(T) = (t_1,\, -t_1 - t_2,\, t_2)$$

(exponents of z_1, z_2, z_3). Then $\chi(\mu_1(T) - \mu_2(T)) = y^{t_2}$ with $y = z_3^{-3}$, which is 1 on H_3, and $a_{\mu_1} + a_{\mu_2}$ is regular there. Writing $a_{\mu_1} = A/(1-y) + (\text{regular})$, the pair is

$$a_{\mu_1}\chi(\mu_1(T)) + a_{\mu_2}\chi(\mu_2(T)) = -A\,\chi(\mu_2(T)) \sum_{m=0}^{t_2-1} y^m + (\text{regular coefficients}).$$

That is a Dirichlet-type kernel transverse to H_3, times $\chi(\mu_2(T))$. On H_3 (say $z_3 = 1$, $z_2 = z_1^{-1}$) the factor $\chi(\mu_2(T))$ restricts to $z_1^{2t_1+t_2}$, which oscillates faster and faster along H_3. Every frequency $\mu_2(T) + m(1,1,-2)$, $0 \le m < t_2$, has component $2t_1 + t_2$ along H_3. So the pairing with a smooth $f^\vee \rho A$ is a sum of t_2 Fourier coefficients, each $O(|T|^{-N})$, and it tends to 0. In $SL(2)$ the corresponding geometric sum is $z^{-2T}\sum_{m=0}^{2T-1} z^{2m}$, near $z = -1$. Its frequencies run through 0, and the limit is the atom of Proposition 6.4.

Remark 2026.20 (a weaker regularity hypothesis). The computation suggests replacing (R) by:

(R′) The oscillatory part $I^T(\chi,\phi) - I(\chi,\phi)$ is, on a cone of T, a finite sum of terms

$$b(\chi)\, p(T)\, \chi(\mu(T)) \sum_{m=0}^{M(T)-1} \chi(m\nu),$$

where $b\rho$ is smooth on $\widehat{T}^u$, p is a polynomial, $\nu \in X_*(A)$, M is affine with $M \ge 1$, and $|\mu(T) + m\nu| \ge c|T|$ for $0 \le m < M(T)$, with some $c > 0$.

Hypothesis (R) is the case $M \equiv 1$. Under (R′) the proof of Theorem 4.5(b) goes through unchanged: each term pairs with $f^\vee$ to at most $M(T)$ Fourier coefficients of the smooth function $f^\vee \rho b$, at frequencies of size $\ge c|T|$. For the $PGL(3)$ element above, the numerical structure of §A.6 is that of (R′) with ν transverse to H_i. This has not been verified for all 19 terms analytically, or for other γ. For $SL(2)$, (R′) fails, because $\mu(T) + m\nu$ passes through 0. We conjecture that (R′), rather than (R), is the condition behind Conjecture 4.6(b).

A.7 What this shows

We summarize what is proved, what is observed numerically, and what remains open.

For $PGL(3)$, an unramified elliptic γ of depth 0 and ϕ supported on γK_1:

1. (A.2) is observed on the building for $l_1 \le 6$ ($q = 2$) and $l_1 \le 4$ ($q = 3$), but not proved;
2. given (A.2), the closed form (A.4) is exact, and $I\rho$ is smooth;
3. the spectral identity of Theorem 4.5(b) holds, numerically to 10^{-12}, for all f_ν with $n_1 \le 6$, $q = 2, 3$, with no singular part;

4. (4.7A) holds, with frequencies of the predicted form;
5. Hypothesis (R) fails. Its conclusion survives through the mechanism of §A.6, which suggests the weaker (R′) of Remark 2026.20.

Other conjugacy classes (positive depth, ramified E, split-by-quadratic tori) have not been treated.

Glossary

This glossary gives short definitions of the main terms used in the paper, with pointers to where they are used. It is meant for readers who are not specialists in the harmonic analysis of p-adic groups.

Unramified group
A reductive group over F that is quasi-split over F and split over an unramified extension. Such a group has hyperspecial maximal compact subgroups.

Hyperspecial maximal compact subgroup K
The stabilizer of a hyperspecial vertex of the building; for G split, a conjugate of $G(\mathscr{O})$ for a smooth reductive $\mathscr{O}$-model. Then $G(F) = P(F)K$ for every parabolic P.

Iwahori subgroup I
The stabilizer of a chamber of the building contained in K; $K = \bigsqcup_w IwI$.

Relative root system; reduced roots
The roots Σ of the maximal split torus A acting on $\mathrm{Lie}\, G$; reduced means $\alpha/2 \notin \Sigma$.

Coroots, fundamental weights and coweights
$\alpha^\vee \in X_*(A) \otimes \mathbb{Q}$ with $\langle \alpha^\vee, \alpha \rangle = 2$; ϖ_α is the basis dual to $\Delta^\vee$; the fundamental coweights are the basis dual to Δ.

$\mathfrak{a}_P$, $\mathfrak{a}_P^Q$, H_M, $G(F)^1$
$\mathfrak{a}_P = \mathrm{Hom}(X^*(M_P)_F, \mathbb{R})$; $\mathfrak{a}_P^Q$ is the kernel of $\mathfrak{a}_P \to \mathfrak{a}_Q$; $H_M : M(F) \to \mathfrak{a}_M$ is given by $e^{\langle H_M(m), \chi \rangle} = |\chi(m)|$; $G(F)^1 = \ker H_G$.

τ_P^Q, $\hat{\tau}_P^Q$
Characteristic functions of the cones in $\mathfrak{a}_0$ cut out by positivity of the simple roots, resp. of the fundamental weights, of $(M_Q \cap P, A_P)$.

Unramified character; tori $\widehat{T}$, $\widehat{T}^u$
A character of $M_0(F)$ trivial on $M_0(F) \cap K$; equivalently a character of $X_*(A)$. All of them form the complex torus $\widehat{T}$, the unitary ones the compact torus $\widehat{T}^u$, which supports the Plancherel measure.

Regular character
χ with $w\chi \neq \chi$ for $w \neq 1$.

Unramified principal series $i_{G,A}(\chi)$
Normalized induction of χ from B; it has a one-dimensional space of K-fixed vectors, spanned by $\Phi_{K,\chi}$.

Normalized Jacquet functor $r_{M,G}$
$\pi \mapsto \delta_P^{-1/2} \otimes \pi/\langle \pi(n)v - v \rangle$; left adjoint of $i_{G,M}$ (Frobenius reciprocity).

Geometric lemma
The Bernstein–Zelevinsky filtration of $r_{M,G} \circ i_{G,A}(\chi)$ with subquotients $i_{M,A}(w\chi)$, $w \in [W_M \backslash W]$ (Lemma 2.4).

Intertwining operator T_w; c-functions
$T_w : i_{G,A}(\chi) \to i_{G,A}(w\chi)$, given by an integral over $N \cap wN^-w^{-1}$ (convergent in a cone and continued meromorphically). $T_w \Phi_{K,\chi} = c_w(\chi)\Phi_{K,w\chi}$ with c_w a product of rank-one factors c_α (2.8); $c(\chi) = c_{w_\ell}(\chi)$.

Zonal spherical function Γ_χ
The bi-K-invariant matrix coefficient $\langle \pi_\chi(g)\Phi_{K,\chi}, \Phi_{K,\chi^{-1}} \rangle$; normalized by $\Gamma_\chi(1) = 1$.

Macdonald's formula
The explicit formula for Γ_χ on A^- (Lemma 2.1).

Satake (Fourier) transform $f^\vee$
$f^\vee(\chi) = \int f(g)\Gamma_\chi(g^{-1})\,dg$, a W-invariant Laurent polynomial on $\widehat{T}$ (2.9).

Spherical Plancherel measure
$d\mu = \frac{Q_G}{|W|}|c(\chi)|^{-2}\,d\chi$ on $\widehat{T}^u$ (Lemma 2.2); $Q_G = \sum_w (IwI : I)^{-1}$.

Casselman's canonical pairing
The M-invariant pairing between $r_{M,G}(\pi)$ and $r_{M,\bar{P}}(\tilde{\pi})$ that computes matrix coefficients deep in the chamber (3.1).

Level
A vector has level n if it is fixed by the principal congruence subgroup K_n.

Regular elliptic element
A regular semisimple element whose centralizer is an elliptic torus (anisotropic modulo A_G).

Orbital integral
$O_\gamma(f) = \int_{G_\gamma(F)\backslash G(F)} f(g^{-1}\gamma g)\,d\dot{g}$.

Stable conjugacy; stable orbital integral
Elements conjugate over $\bar{F}$ (for $SL(n)$: under $GL(n,F)$); $SO_\gamma(f)$ is the sum of the orbital integrals over the classes in a stable class.

κ-orbital integral
$\sum_x \kappa(x) O_{x^{-1}\gamma x}(f)$ over the classes in a stable class, weighted by a character κ of the group $\mathfrak{D}(S)$ classifying them (5.2).

Weyl discriminant $D(s)$
$\det(1 - \mathrm{Ad}(s))|_{\mathrm{Lie}\, G/\mathrm{Lie}\, S}$.

(G,M)-family; $(G,A)^\sim$-family
Families of functions indexed by $\mathscr{P}(M)$ that agree on common walls; sums $\sum_P c_P/\theta_P$ are then smooth (Lemmas 2.6–2.7). The $\sim$-version is the periodic analogue on $\widehat{T}^u$.

Truncation parameter T; $\mathscr{A}_A(T)$; $G(T)$
$T \in X_*(A)$ regular dominant; $\mathscr{A}_A(T)$ is the set of dominant λ below T in the sense of (3.3), and $G(T) = \bigcup K a_\lambda K$.

Vertex decomposition (Brion)
The generating function of lattice points of a rational polytope is the sum, over its vertices, of the generating functions of the tangent cones (Lemma 3.5A).

Langlands' combinatorial lemma; Γ_M^G
The identity $\sum_{Q \supseteq P} (-1)^{\dim \mathfrak{a}_Q^G} \tau_P^Q \hat{\tau}_Q = \delta_{PG}$ and its relatives [Ar78, CLL84].

Hypothesis (R)
Smoothness of the oscillatory coefficients of the truncated kernel times the Plancherel density (§4.3).

***R*-group**
For unramified χ, the quotient of the stabilizer W_χ by its subgroup generated by the reflections s_α with $\chi(a_\alpha) = 1$. It governs the reducibility of $i_{G,A}(\chi)$ for unitary χ.

Elliptic tempered (virtual) character
A virtual tempered character not in the span of properly induced ones (Arthur [Ar93]); for $SL(2)$, $\pi^+ - \pi^-$ at χ_0.

***G*-admissible distribution**
An invariant distribution satisfying Harish-Chandra's finiteness condition [HC78, §14], and hence represented by a locally integrable function on G_{reg}.

Cuspidal function
ϕ with $\operatorname{tr} \sigma(\phi) = 0$ for all properly induced σ; e.g. any ϕ supported on G_{ell}.

Weighted orbital integral; weighted character
Orbital integrals and characters weighted by the (G,M)-families $v_M(x)$, resp. normalized intertwining operators (§7).

Endoscopic group
For $SL(n)$ and κ of order m, $H_m = \ker(N_{E/F} \circ \det) \subset \mathrm{Res}_{E/F} GL(n/m)$, with an L-embedding ${}^L H \to {}^L G$ (§5).

Transfer factor $\Delta_{H,\kappa}$
The Langlands–Shelstad factor relating κ-orbital integrals on G to stable orbital integrals on H [LS87].

Fundamental lemma
The identity (5.11) for spherical functions, $f_H = \iota^* f$; now a theorem.

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ChatGPT (OpenAI), an AI model, acted as an independent referee. It wrote the second referee report, on the first revised draft of November 2026. Several hypotheses, remarks and examples of the present draft, and Appendix A, were added or corrected in response to that report.

Review by AI systems is not an independent certification of the results. The errors that remain are the authors'. Further details on the AI usage, including the referee reports, the responses to them and a list of every change, can be found in the revision log (`revision_log.md`) that accompanies the source.

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